



Final Step - B | Mathematics

Functions

CHOOSE THE CORRECT ALTERNATIVE. ONLY ONE CHOICE IS CORRECT. HOWEVER, QUESTIONS MARKED '**' MAY HAVE MORE THAN ONE CORRECT OPTION.

- Let $f(x) = |x-2| + |x-3| + |x-4|$ and $g(x) = f(x+1)$. Then :
 (A) $g(x)$ is an even function (B) $g(x)$ is an odd function
 (C) $g(x)$ is neither even nor odd (D) $g(x)$ is periodic
- Period of $f(x) = x - [x+a] - b$, where $a, b \in \mathbb{R}^+$ and $[.]$ denotes the greatest integer function is :
 (A) a (B) b (C) $|a-b|$ (D) 1
- The domain of the function $f(x) = \log \left[\frac{1}{x+\frac{1}{2}} \right] |x^2 - 5x + 6|$ is : (where $[.]$ denotes the greatest integer function)
 (A) $x \in \left[\frac{3}{2}, 2 \right) \cup (2, 3) \cup (3, \infty)$ (B) $x \in \left[\frac{3}{2}, \infty \right)$
 (C) $x \in \left[\frac{1}{2}, \infty \right)$ (D) None of these
- The function $f(x) = \sec \left[\log \left(x + \sqrt{1+x^2} \right) \right]$ is :
 (A) Even (B) Odd (C) Constant (D) None of these
- If $f(x) = \log \left(\frac{1+x}{1-x} \right)$, $-1 < x < 1$, then $f \left(\frac{3x+x^3}{1+3x^2} \right) - f \left(\frac{2x}{1+x^2} \right)$ is :
 (A) $[f(x)]^3$ (B) $[f(x)]^2$ (C) $-f(x)$ (D) $f(x)$
- The range of the function $f(x) = x^2 + \frac{1}{x^2+1}$ is :
 (A) $[1, \infty)$ (B) $[2, \infty)$ (C) $\left[\frac{3}{2}, \infty \right)$ (D) None of these
- If $f(x) = \frac{4^x}{4^x+2}$, then $f\left(\frac{1}{97}\right) + f\left(\frac{2}{97}\right) + \dots + f\left(\frac{96}{97}\right)$ is equal to :
 (A) 1 (B) 48 (C) -48 (D) -1
- Let $f(x+y) + f(x-y) = 2f(x)f(y)$, $\forall y \in \mathbb{R}$ and $f(0) = k$, then :
 I $f(x)$ is even, if $k = 1$ II $f(x)$ is odd, if $k = 0$
 III $f(x)$ is always odd IV $f(x)$ is neither even nor odd for any value of k
 The correct choice is :
 (A) I, III (B) II, III (C) I, II (D) III, IV

9. The period of the function $f(x)$ which satisfies the relation $f(x) + f(x+4) = f(x+2) + f(x+6)$ is :
 (A) 6 (B) 7 (C) 8 (D) None of these

10. Let $x = \left(\frac{81^{\frac{1}{\log_5 9}} + 3^{\frac{3}{\log_{\sqrt{6}} 3}}}{409} \right) \cdot \left((\sqrt{7})^{\frac{2}{\log_{25} 7}} - 125^{\log_{25} 6} \right)$, then value of $\log_2 x$ is equal to :
 (A) 0 (B) 1 (C) -1 (D) None of these

11. Which of the following when simplified reduces to unity ?

I. $\log_{1.5} \log_4 \log_{\sqrt{3}} 81$ II. $\log_2 \sqrt{6} + \log_2 \sqrt{\frac{2}{3}}$
 III. $-\frac{1}{6} \log_{\frac{\sqrt{3}}{2}} \left(\frac{64}{27} \right)$ IV. $\log_{3.5} (1 + 2 + 3 \div 6)$

The correct choice is :

- (A) I only (B) II and IV only (C) I and III only (D) All the above
12. Given, $\log_a x = \alpha$; $\log_b x = \beta$; $\log_c x = \gamma$ & $\log_d x = \delta$ ($x \neq 1$), $a, b, c, d \in \mathbb{R}^+ - \{1\}$ then $\log_{abcd} x$ has the value equal to :

(A) $\frac{1}{\alpha\beta\gamma\delta}$ (B) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} + \frac{1}{\delta}$ (C) $\frac{1}{\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} + \frac{1}{\delta}}$ (D) None of these

13. The number of integral solutions of the equation $4\log_{x/2}(\sqrt{x}) + 2\log_{4x}(x^2) = 3\log_{2x}(x^3)$ is :

(A) 0 (B) 1 (C) 2 (D) None of these

14. If $f(x) = -\frac{x|x|}{1+x^2}$, then $f^{-1}(x)$ equals :

(A) $\sqrt{\frac{|x|}{1-|x|}}$ (B) $(\operatorname{sgn}(-x))\sqrt{\frac{|x|}{1-|x|}}$ (C) $-\sqrt{\frac{x}{1-x}}$ (D) $(\operatorname{sgn}(x))\sqrt{\frac{|x|}{1+|x|}}$

15. If $f(x) = \sin\left\{[x+5] + \left\{x - \{x - \{x\}\}\right\}\right\}$ for $x \in \left(0, \frac{\pi}{4}\right)$ is invertible, where $\{.\}$ and $[.]$ represent fractional part and greatest integer functions respectively, then $f^{-1}(x)$ is :

I. $\sin^{-1} x$ II. $\frac{\pi}{2} - \cos^{-1} x$ III. $\sin^{-1}\{x\}$ IV. $\cos^{-1}\{x\}$

The correct choice is :

- (A) I, II, III (B) II, III (C) III, IV (D) None of these
16. Let $G(x) = \left(\frac{1}{a^x - 1} + \frac{1}{2}\right)F(x)$, where 'a' is a positive real number not equal to 1 and $F(x)$ is an odd function. Which

of the following statements is true ?

- (A) $G(x)$ is an odd function
 (B) $G(x)$ is an even function
 (C) $G(x)$ is neither even nor odd function
 (D) Whether $G(x)$ is an odd or even function depends on the value of 'a'.

17. If $f\left(2x + \frac{y}{8}, 2x - \frac{y}{8}\right) = xy$, then $f(m, n) + f(n, m)$ is :
- (A) Depends over m and n both (B) Periodic and odd function
(C) Constant number (D) None of these
18. If $\sum_{r=0}^{21} f\left(\frac{r}{11} + 2x\right) = \text{constant } \forall x \in R$ and $f(x)$ is periodic, then period of $f(x)$ is :
- (A) 1 (B) $1/11$ (C) 2 (D) 4
19. Suppose f is a real function $f(x + f(x)) = 4f(x)$ and $f(1) = 4$. Then the value of $f(21)$ is :
- (A) 16 (B) 21 (C) 64 (D) 105
20. Let f be a real valued function such that for any real x
 $f(15 + x) = f(15 - x)$ and $f(30 + x) = -f(30 - x)$
 Then which of the following statements is true ?
- (A) f is odd and periodic (B) f is odd but not periodic
(C) f is even and periodic (D) f is even but not periodic
21. For $x \in R, x \neq 0, x \neq 1$, let $f_0(x) = \frac{1}{1-x}$ and $f_{n+1}(x) = f_0(f_n(x)), n = 0, 1, 2, \dots$
 Then the value of $f_{100}(3) + f_1\left(\frac{2}{3}\right) + f_2\left(\frac{3}{2}\right)$ is equal to :
- (A) $\frac{8}{3}$ (B) $\frac{5}{3}$ (C) $\frac{4}{3}$ (D) $\frac{1}{3}$
22. Let $f : R \rightarrow R$ be defined by $f(x) = \frac{|x|-1}{|x|+1}$ then f is :
- (A) both one-one and onto (B) one-one but not onto
(C) onto but not one-one (D) neither one-one nor onto
23. Let $f(n) = \left[\frac{1}{3} + \frac{3n}{100}\right]n$, where $[n]$ denotes the greatest integer less than or equal to n . Then $\sum_{n=1}^{56} f(n)$ is equal to :
- (A) 56 (B) 689 (C) 1287 (D) 1399
- *24. The given function $f(x) = \frac{a^x - 1}{x^n(a^x + 1)}$ is :
- (A) Odd, if n is even (B) Even, if n is odd (C) Odd, if n is odd (D) Even, if n is even
25. The domain of $f(x) = \log_2\left(-\log_{1/2}\left(1 + \frac{1}{x^{1/4}}\right)\right)$ is :
- (A) $(0, 1)$ (B) $(0, 1]$ (C) $[1, \infty)$ (D) $(1, \infty)$

Integer Answer Type Questions

The Answer to the following questions are positive integers of 1/2/3 digits and zero

26. Let f be a real valued function such that $f(x) + 2f\left(\frac{2002}{x}\right) = 3x, \forall x > 0$. The value of $\left(\frac{f(2)}{2000}\right)$ is _____.
27. Let a and b be constants and $f(x) = a \sin x + bx \cos x + 2x^2$. If $f(2) = 15$, then the value of $f(-2)$ is _____.
28. The range of $f(x) = \log_{\sqrt{2}}\left(2 - \log_2(16 \sin^2 x + 1)\right)$ is $(-\infty, a]$, then the value of a is _____.
29. The period of $f(x)$ satisfying $f(x-1) + f(x+3) = f(x+1) + f(x+5)$, is _____.
30. The period of $f(x) = \sin\left(\frac{\pi}{2}[x]\right)$ (where, $[.]$ denotes greatest integer function), is _____.
31. If $f(x) = \left(\frac{4a-7}{3}\right)x^3 + (a-3)x^2 + x + 5$ is a one-one function, where $a \in [\lambda, \mu]$, then the value of $|\lambda - \mu|$ is ____.
32. If $f(x)$ be a polynomial of degree 4 with leading coefficient 1 satisfying $f(1) = 10, f(2) = 20, f(3) = 30$, where $f(12) + f(-8) = (3968)\lambda$, then the value of λ is _____.
33. Let $f: \left[\frac{\pi}{6}, \frac{2\pi}{3}\right] \rightarrow B$. Defined by $f(x) = 2\cos^2 x + \sqrt{3}\sin 2x + 1$, where $B \in [a, b]$ such that $f^{-1}(x)$ exists, then $(a+b)$ is _____.
34. If $f(x)$ is real valued function satisfying $f(x+y) = f(x) + f(y) - xy - 1, \forall x, y \in \mathbb{R}$ such that $f(1) = 1$, then the number of solutions, of $f(n) = n, n \in \mathbb{N}$, is _____.
35. If $[.]$ denotes the greatest integer function and $n \in \mathbb{N}$, then period of $f(x) = nx + n - [nx + n] + \tan \frac{\pi x}{2}$, is ____.
36. Let for $a \neq a_1 \neq 0, f(x) = ax^2 + bx + c, g(x) = a_1x^2 + b_1x + c_1$ and $p(x) = f(x) - g(x)$. If $p(x) = 0$ only for $x = -1$ and $p(-2) = 2$, then the value of $p(2)$ is _____.
37. Let $f(x)$ be a polynomial function satisfying the relation $f(x) \cdot f\left(\frac{1}{x}\right) = f(x) + f\left(\frac{1}{x}\right), \forall x \in \mathbb{R} - \{0\}$ and $f(3) = -26$, then $|f(2)|$ is ____.
38. If $f(x+y) = f(x) \cdot f(y), \forall x, y \in \mathbb{N}$ and $f(1) = 2$, then $\left|\sum_{n=1}^{10} f(n) - 2050\right|$ is ____.
39. Let $f(x)$ be an invertible function such that the graph $y = f(x)$ intersects the coordinate axes in points A and B respectively, such that Area of $\Delta OAB = 16$ square units. If the graph of $y = f^{-1}(x)$ intersects the coordinate axes in points A' and B' respectively, find the minimum length of diagonal OC of rectangle $OA'CB'$.
40. If $f(x) = \frac{\alpha x}{x+1}, x \neq -1$, then for what value of $|\alpha|$ is $f\{f(x)\} = x$?

Differential Calculus-1

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- If $\lim_{x \rightarrow \infty} \left(1 + \frac{a}{x} - \frac{4}{x^2}\right)^{2x} = e^3$, then ‘a’ is equal to :

(A) 2 (B) $\frac{3}{2}$ (C) $\frac{2}{3}$ (D) $\frac{1}{2}$
- If the function $f(x) = \begin{cases} -x, & x < 1 \\ a + \cos^{-1}(x+b), & 1 \leq x \leq 2 \end{cases}$ is differentiable at $x = 1$, then $\frac{a}{b}$ is equal to :

(A) $\frac{\pi-2}{2}$ (B) $\frac{-\pi-2}{2}$ (C) $\frac{\pi+2}{2}$ (D) $-1 - \cos^{-1}(2)$
- If $f(x)$ is differentiable function in the interval $(0, \infty)$ such that $f(1) = 1$ and $\lim_{t \rightarrow x} \frac{t^2 f(x) - x^2 f(t)}{t - x} = 1$, for each $x > 0$, then $f(3/2)$ is equal to :

(A) $\frac{13}{6}$ (B) $\frac{23}{18}$ (C) $\frac{25}{9}$ (D) $\frac{31}{18}$
- $\lim_{x \rightarrow 0} \frac{(1 - \cos 2x)^2}{2x \tan x - x \tan 2x}$ is :

(A) -2 (B) $-\frac{1}{2}$ (C) $\frac{1}{2}$ (D) 2
- Let $a, b \in \mathbb{R}$, ($a \neq 0$). If the function f defined as $f(x) = \begin{cases} \frac{2x^2}{a}, & 0 \leq x < 1 \\ a, & 1 \leq x < \sqrt{2} \\ \frac{2b^2 - 4b}{x^3}, & \sqrt{2} \leq x < \infty \end{cases}$ is continuous in the interval $[0, \infty)$, then an ordered pair (a, b) is :

(A) $(\sqrt{2}, 1 - \sqrt{3})$ (B) $(-\sqrt{2}, 1 + \sqrt{3})$ (C) $(\sqrt{2}, -1 + \sqrt{3})$ (D) $(-\sqrt{2}, 1 - \sqrt{3})$
- $\lim_{x \rightarrow 0} \frac{e^{x^2} - \cos x}{\sin^2 x}$ is equal to :

(A) 3 (B) $\frac{3}{2}$ (C) $\frac{5}{4}$ (D) 2
- Let k be a non-zero real number. If $f(x) = \begin{cases} \frac{(e^x - 1)^2}{\sin\left(\frac{x}{k}\right) \log\left(1 + \frac{x}{4}\right)}, & x \neq 0 \\ 12, & x = 0 \end{cases}$ is a continuous function, then the value of k is :

(A) 1 (B) 2 (C) 3 (D) 4

8. If $f(x)$ is continuous and $f(9/2) = 2/9$, then $\lim_{x \rightarrow 0} f\left(\frac{1 - \cos 3x}{x^2}\right)$ is equal to :
 (A) $9/2$ (B) $2/9$ (C) 0 (D) $8/9$
9. If $y = e^{nx}$, then $\left(\frac{d^2y}{dx^2}\right)\left(\frac{d^2x}{dy^2}\right)$ is equal to :
 (A) ne^{nx} (B) ne^{-nx} (C) 1 (D) $-ne^{-nx}$
10. If $\lim_{x \rightarrow 2} \frac{\tan(x-2)\{x^2 + (k-2)x - 2k\}}{x^2 - 4x + 4} = 5$, then k is equal to :
 (A) 0 (B) 1 (C) 2 (D) 3
11. Let $f(x) = x|x|$, $g(x) = \sin x$ and $h(x) = (g \circ f)(x)$. Then :
 (A) $h(x)$ is not differentiable at $x = 0$.
 (B) $h(x)$ is differentiable at $x = 0$, but $h'(x)$ is not continuous at $x = 0$.
 (C) $h'(x)$ is continuous at $x = 0$ but it is not differentiable at $x = 0$.
 (D) $h'(x)$ is differentiable at $x = 0$.
12. If $f(x) = x^2 - x + 5$, $x > \frac{1}{2}$, and $g(x)$ is its inverse function, then $g'(7)$ equals :
 (A) $-\frac{1}{13}$ (B) $\frac{1}{13}$ (C) $\frac{1}{3}$ (D) $-\frac{1}{3}$
13. Let $f, g : \mathbb{R} \rightarrow \mathbb{R}$ be two function defined by $f(x) = \begin{cases} x \sin\left(\frac{1}{x}\right), & x \neq 0 \\ 0, & x = 0 \end{cases}$, and $g(x) = xf(x)$
Statement I : f is a continuous function at $x = 0$
Statement II : g is a differentiable function at $x = 0$
 (A) Statement I is true, statement II is false (B) Both statements I and II are true
 (C) Statement I is false, statement II is true (D) Both statements I and II are false
14. If the function $f(x) = \begin{cases} \frac{\sqrt{2 + \cos x} - 1}{(\pi - x)^2}, & x \neq \pi \\ k, & x = \pi \end{cases}$ is continuous at $x = \pi$, then k equals :
 (A) 0 (B) $\frac{1}{2}$ (C) 2 (D) $\frac{1}{4}$
15. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function such that $|f(x)| \leq x^2$, for all $x \in \mathbb{R}$. Then, at $x = 0$, f is :
 (A) continuous but not differentiable (B) continuous as well as differentiable
 (C) neither continuous nor differentiable (D) differentiable but not continuous
16. The value of $\lim_{x \rightarrow 0} \frac{1}{x} \left[\tan^{-1}\left(\frac{x+1}{2x+1}\right) - \frac{\pi}{4} \right]$ is :
 (A) 1 (B) $-\frac{1}{2}$ (C) 2 (D) 0

17. For $a > 0$, $t \in \left(0, \frac{\pi}{2}\right)$, let $x = \sqrt{a^{\sin^{-1}t}}$ and $y = \sqrt{a^{\cos^{-1}t}}$. Then, $1 + \left(\frac{dy}{dx}\right)^2$ equals :
- (A) $\frac{x^2}{y^2}$ (B) $\frac{y^2}{x^2}$ (C) $\frac{x^2 + y^2}{y^2}$ (D) $\frac{x^2 + y^2}{x^2}$
18. Let $f(x) = -1 + |x - 2|$, and $g(x) = 1 - |x|$; then the set of all points where $f \circ g$ is discontinuous is :
- (A) $\{0, 2\}$ (B) $\{0, 1, 2\}$ (C) $\{0\}$ (D) an empty set
19. If $f(x) = \sin(\sin x)$ and $f''(x) + \tan x f'(x) + g(x) = 0$, then $g(x)$ is :
- (A) $\cos^2 x \cos(\sin x)$ (B) $\sin^2 x \cos(\cos x)$ (C) $\sin^2 x \sin(\cos x)$ (D) $\cos^2 x \sin(\sin x)$
20. Let f be a composite function of x defined by $f(u) = \frac{1}{u^2 + u - 2}$, $u(x) = \frac{1}{x - 1}$. Then the number of points x where f is discontinuous is :
- (A) 4 (B) 3 (C) 2 (D) 1
21. The value of $\lim_{x \rightarrow 1^+} \left(\{x\}\right)^{\frac{1}{\ln x}}$, is : (where $\{.\}$ denotes the fractional part function)
- (A) 1 (B) -1 (C) 0 (D) None of these
22. The value of $\lim_{x \rightarrow \infty} \cos(\sqrt{x+1}) - \cos(\sqrt{x})$ is :
- (A) 1 (B) -1 (C) 0 (D) None of these
23. The value of $\lim_{x \rightarrow \infty} \left((x+1)^{2/3} - (x-1)^{2/3}\right)$ is :
- (A) 1 (B) -1 (C) 0 (D) None of these
24. The value of $\lim_{n \rightarrow \infty} \frac{\sqrt{n^3 - 2n^2 + 1} + \sqrt[3]{n^4 + 1}}{\sqrt[4]{n^6 + 6n^5 + 2} - \sqrt[5]{n^7 + 3n^3 + 1}}$, $n \in N$ is :
- (A) 1 (B) -1 (C) 0 (D) ∞
25. The value of $\lim_{x \rightarrow \infty} \left(\left((x+1)(x+2)(x+3)(x+4)\right)^{1/4} - x\right)$ is :
- (A) $\frac{1}{2}$ (B) $\frac{5}{2}$ (C) $\frac{3}{2}$ (D) $-\frac{1}{2}$
26. The value of $\lim_{x \rightarrow 1^-} \left([x]\right)^{1-x}$, is : (where $[.]$ denotes greatest integer function)
- (A) 0 (B) 1 (C) Not defined (D) None of these
27. The value of $\lim_{x \rightarrow 2} \frac{(x+2)^{1/2} - (15x+2)^{1/5}}{(7x+2)^{1/4} - x}$ is :
- (A) $-\frac{1}{5}$ (B) $-\frac{1}{2}$ (C) $-\frac{2}{25}$ (D) None of these
28. Suppose that $f(x) = x^3 - 3x^2 - 4x + 12$ and $h(x) = \begin{cases} \frac{f(x)}{x-3}, & x \neq 3 \\ K, & x = 3 \end{cases}$, then :
- For that value of K that makes $h(x)$ continuous at $x = 3$ then $h(x)$ is :
- (A) Even (B) Odd (C) $f(0) = 1$ (D) Neither even nor odd

29. If $f(x) = \begin{cases} ax^2 - b & \text{if } |x| < 1 \\ -\frac{1}{|x|} & \text{if } |x| \geq 1 \end{cases}$ is derivable at $x = 1$. Then the values of a and b are :
- (A) $a = \frac{1}{2}, b = \frac{3}{2}$ (B) $a = \frac{1}{2}, b = \frac{-1}{2}$ (C) $a = \frac{-1}{2}, b = \frac{-3}{2}$ (D) None of these
30. Let a function $f : R \rightarrow R$ be given by $f(x+y) = f(x)f(y)$ for all $x, y \in R$ and $f(x) \neq 0$ for any $x \in R$. If the function $f(x)$ is differentiable at $x = 0$.
- (A) $e^{x \cdot f'(0)}$ (B) $x \cdot f'(0)$ (C) $\frac{x^2}{2} f'(x)$ (D) None of these
31. The value of $\lim_{x \rightarrow 0} \frac{(4^x - 1)^3}{\sin\left(\frac{x}{p}\right) \ln\left(1 + \frac{x^2}{3}\right)}$ is :
- (A) $9p(\ln 4)$ (B) $3p(\ln 4)^3$ (C) $12p(\ln 4)^3$ (D) $27p(\ln 4)^2$
32. $\lim_{x \rightarrow 1} \frac{\left(\sum_{k=1}^{100} x^k\right) - 100}{x - 1}$ is :
- (A) 0 (B) 5050 (C) 4550 (D) -5050
33. The value of $\lim_{x \rightarrow \frac{\pi}{2}} \tan^2 x \left(\sqrt{2 \sin^2 x + 3 \sin x + 4} - \sqrt{\sin^2 x + 6 \sin x + 2} \right)$ is :
- (A) 1/10 (B) 1/11 (C) 1/12 (D) 1/8
34. $\lim_{n \rightarrow \infty} \frac{5^{n+1} + 3^n - 2^{2n}}{5^n + 2^n + 3^{2n+3}}, n \in N$ is :
- (A) 5 (B) 3 (C) 1 (D) Zero
- *35. If $f(x) = \frac{\sqrt{x^2 + 2}}{3x - 6}$, then :
- (A) $\lim_{x \rightarrow -\infty} f(x) = -\frac{1}{3}$ (B) $\lim_{x \rightarrow \infty} f(x) = \frac{1}{3}$
- (C) $\lim_{x \rightarrow -\infty} f(x) = \frac{1}{3}$ (D) $\lim_{x \rightarrow \infty} f(x) = -\frac{1}{3}$
36. $\lim_{x \rightarrow \infty} \left(x - x^2 \ln \left(1 + \frac{1}{x} \right) \right)$ is equal to :
- (A) 1/2 (B) 3/2 (C) 1/3 (D) 1
37. $\lim_{x \rightarrow 0} \frac{e^{-x^2/2} - \cos x}{x^3 \sin x}$ is equal to :
- (A) 1/4 (B) 1/6 (C) 1/12 (D) 1/8

38. Consider the following statements :

$S_1 :$ $\lim_{x \rightarrow 0^-} \frac{[x]}{x}$ is an indeterminate form (where $[.]$ denotes greatest integer function).

$S_2 :$ $\lim_{x \rightarrow \infty} \frac{\sin(3^x)}{3^x} = 0$

$S_3 :$ $\lim_{x \rightarrow \infty} \sqrt{\frac{x - \sin x}{x + \cos^2 x}}$ does not exist.

$S_4 :$ $\lim_{n \rightarrow \infty} \frac{(n+2)! + (n+1)!}{(n+3)!} (n \in N) = 0$

Which of the statements S_1, S_2, S_3, S_4 are true or false :

(A) FTFT (B) FTTT (C) FTFF (D) TTFT

*39. If $\ell = \lim_{x \rightarrow 0} \frac{x(1 + a \cos x) - b \sin x}{x^3} = \lim_{x \rightarrow 0} \frac{1 + a \cos x}{x^2} - \lim_{x \rightarrow 0} \frac{b \sin x}{x^3}$, where $\ell \in R$, then :

(A) $(a, b) = (-1, 0)$ (B) a and b are any real numbers

(C) $\ell = 0$ (D) $\ell = \frac{1}{2}$

40. The value of $\lim_{x \rightarrow \frac{\pi}{4}} \frac{1}{(1 + [x])^{\ln(\tan x)}}$ is equal to : (where $[.]$ denotes the integer function)

(A) 0 (B) 1 (C) e (D) e^{-1}

41. The value of $\lim_{n \rightarrow \infty} \frac{1}{n^4} ([1^3 x] + [2^3 x] + \dots + [n^3 x])$ is : [where $[x]$ denotes greatest integer less than or equal to x]

(A) $\frac{x}{2}$ (B) $\frac{x}{3}$ (C) $\frac{x}{6}$ (D) $\frac{x}{4}$

42. The value of $\lim_{x \rightarrow \infty} \left(\frac{x^2 - 2x + 1}{x^2 - 4x + 2} \right)^x$ is equal to :

(A) 1 (B) 2 (C) e^2 (D) e

43. Let $f(x) = \left[\left(x + \frac{1}{2} \right) [x] \right]$, when $-2 \leq x \leq 2$ where $[.]$ represents greatest integer function. Then :

(A) $f(x)$ is continuous at $x = 2$ (B) $f(x)$ is continuous at $x = 1$
(C) $f(x)$ is continuous at $x = -1$ (D) $f(x)$ is discontinuous at $x = 0$

44. Let $f(x) = \text{sgn}(x)$ and $g(x) = x(x^2 - 5x + 6)$. Then function $f(g(x))$ is discontinuous at :

(A) Infinitely many points (B) Exactly one point
(C) Exactly three points (D) No point

45. If $f(x) = \begin{cases} x(3e^{1/x} + 4) & , x \neq 0 \\ 2 - e^{1/x} & , x = 0 \end{cases}$, then $f(x)$ is :

(A) Continuous as well differentiable at $x = 0$ (B) Continuous but not differentiable at $x = 0$
(C) Neither differentiable at $x = 0$ nor continuous at $x = 0$ (D) None of these

46. If $f(x) = \begin{cases} x + \{x\} + x \sin \{x\} & \text{for } x \neq 0 \\ 0 & \text{for } x = 0 \end{cases}$, (where $\{.\}$ denotes the fractional part function), then :
- (A) f is continuous and differentiable at $x = 0$ (B) f is continuous but not differentiable at $x = 0$
 (C) f is continuous and differentiable at $x = 2$ (D) None of these

47. Given $f(x) = \begin{cases} \log_a \left(a^{\left| [x] + [-x] \right|} \right)^x \frac{a^{\left(\frac{[x] + [-x]}{|x|} \right)^{-5}}}{3 + a^{1/|x|}} & \text{for } |x| \neq 0, a > 1 \\ 0 & \text{for } x = 0 \end{cases}$

where $[.]$ represents the integral part function, then :

- (A) f is continuous but not differentiable at $x = 0$
 (B) f is continuous and differentiable at $x = 0$
 (C) The differentiability of ' f ' at $x = 0$ depends on the value of a
 (D) f is continuous and differentiable at $x = 0$ and for $a = e$ only
- *48. The function $f(x) = \begin{cases} |x-3| & , x \geq 1 \\ \left(\frac{x^2}{4} \right) - \left(\frac{3x}{2} \right) + \left(\frac{13}{4} \right) & , x < 1 \end{cases}$ is :
- (A) Continuous at $x = 1$ (B) Differentiable at $x = 1$
 (C) Continuous at $x = 3$ (D) Differentiable at $x = 3$

49. If $f(x) = \begin{cases} \frac{x^2 - 1}{x^2 + 1} & , 0 < x \leq 2 \\ \frac{1}{4}(x^3 - x^2) & , 2 < x \leq 3 \\ \frac{9}{4}(|x-4| + |2-x|) & , 3 < x < 4 \end{cases}$, then :
- (A) $f(x)$ is differentiable at $x = 2$ and $x = 3$ (B) $f(x)$ is non-differentiable at $x = 2$ and $x = 3$
 (C) $f(x)$ is differentiable at $x = 3$ but not at $x = 2$ (D) $f(x)$ is differentiable at $x = 2$ but not at $x = 3$

50. If $f(x)$ is differentiable everywhere, then :
- (A) $|f|$ is differentiable everywhere (B) $|f|^2$ is differentiable everywhere
 (C) $f|f|$ is not differentiable at some point (D) $f + |f|$ is differentiable everywhere

- *51. $f(x) = \begin{cases} \left(\sin^{-1} x \right)^2 \cdot \cos \left(\frac{1}{x} \right) & , x \neq 0 \\ 0 & , x = 0 \end{cases}$ then :
- (A) Continuous no where in $-1 \leq x \leq 1$ (B) Continuous everywhere in $-1 \leq x \leq 1$
 (C) Differentiable no where in $-1 \leq x \leq 1$ (D) Differentiable everywhere in $-1 < x < 1$

52. Let $f(x)$ be defined in $[-2, 2]$ by $f(x) = \begin{cases} \max(\sqrt{4-x^2}, \sqrt{1+x^2}) & , -2 \leq x \leq 0 \\ \min(\sqrt{4-x^2}, \sqrt{1+x^2}) & , 0 < x \leq 2 \end{cases}$, then :
- (A) is continuous at all points (B) is not continuous at more than one point
 (C) is not differentiable only at one point (D) is not differentiable at more than one points

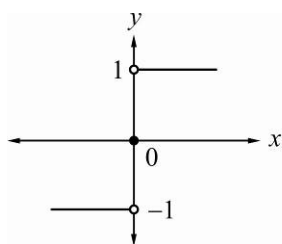
53. The number of points at which the function $f(x) = \max\{a-x, a+x, b\}$, $-\infty < x < \infty$, $0 < a < b$ can not be differentiable is :
- (A) 1 (B) 2 (C) 3 (D) None of these
54. Let $f(x) = x - x^2$ and $g(x) = \begin{cases} \max f(t) & , 0 \leq t \leq x, 0 \leq x \leq 1 \\ \sin \pi x & , x > 1 \end{cases}$, then in the interval $[0, \infty)$
- (A) $g(x)$ is everywhere continuous except at two points
 (B) $g(x)$ is everywhere differentiable except at two points
 (C) $g(x)$ is everywhere differentiable except at $x = 1$ (D) None of these
55. If $f: R \rightarrow R$ be a differentiable function, such that $f(x+2y) = f(x) + f(2y) + 4xy \forall x, y \in R$, then :
- (A) $f'(1) = f'(0) + 1$ (B) $f'(1) = f'(0) - 1$
 (C) $f'(0) = f'(1) + 2$ (D) $f'(0) = f'(1) - 2$
56. Consider the following statements :
- S_1 : Number of points where $f(x) = |x \operatorname{sgn}(1-x^2)|$ is non-differentiable is 3.
- S_2 : Defined $f(x) = \begin{cases} a \sin \frac{\pi}{2}(x+1) & , x \leq 0 \\ \frac{\tan x - \sin x}{x^3} & , x > 0 \end{cases}$, if $f(x)$ be continuous at $x = 0$, then $a = \frac{1}{2}$
- S_3 : The set of all points, where the function $\sqrt[3]{x^2|x|}$ is differentiable is $(-\infty, 0) \cup (0, \infty)$
- S_4 : Number of number where $f(x) = \frac{1}{\sin^{-1}(\sin x)}$ is non-differentiable in the interval $(0, 3\pi)$ is 3.
- Which of the statements S_1, S_2, S_3, S_4 are true or false :
- (A) TTTF (B) TTTT (C) FTTF (D) TFFT

For Questions 57 - 59

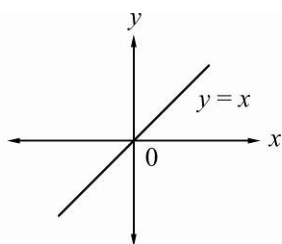
- (A) Statement-1 is True, Statement-2 is True and Statement-2 is a correct explanation for Statement-1
 (B) Statement-1 is True, Statement-2 is True and Statement-2 is NOT a correct explanation for Statement-1
 (C) Statement-1 is True, Statement-2 is False
 (D) Statement-1 is False, Statement-2 is True
57. **Statement 1 :** $\lim_{x \rightarrow 0} \frac{\sqrt{1 - \cos 2x}}{x}$ does not exist **Statement 2 :** $\lim_{x \rightarrow \frac{\pi}{2}} \frac{\sin x - (\sin x)^{\sin x}}{1 - \sin x + \ln \sin x} = 2$
58. **Statement 1 :** $\lim_{x \rightarrow \infty} \frac{2x^4 + 3x^3 + 7x}{3x^4 + 2x^2 + 3x} = \frac{2}{3}$.
- Statement 2 :** If $P(x)$ and $Q(x)$ are two polynomials with rational coefficients, then :
- $$\lim_{x \rightarrow \infty} \frac{P(x)}{Q(x)} = \frac{\text{coefficient of highest power of } x \text{ in } P(x)}{\text{coefficient of highest power of } x \text{ in } Q(x)}$$
59. **Statement 1 :** $f(x) = |x-2| + \frac{x^2 - 5x + 6}{x-1} + \tan x$ is continuous function within the domain of $f(x)$.
- Statement 2 :** All absolute valued polynomial function, rational polynomial function, trigonometric functions are continuous within their domain.

60. The value of a and b so that $\lim_{x \rightarrow \infty} \left(\sqrt{x^4 + ax^3 + 3x^2 + bx + 2} - \sqrt{x^4 + 2x^3 - cx^2 + 3x - d} \right) = 4$
- (A) $a = 2, b \in R, c = 5, d \in R$ (B) $a = 0, b \in R, c \in R, d \in R$
 (C) $a = 2, b = 2, c = 3, d \in R$ (D) None of these
61. The value of a and b so that $\lim_{x \rightarrow 0} \frac{axe^x - b \ln(1+x) + cxe^{-x}}{x^2 \sin x} = 2$ is :
- (A) $a = 3, b = 12, c = -9$ (B) $a = 2, b = 12, c = 9$
 (C) $a = 0, b = 1, c = 9$ (D) None of these
62. The value of $\lim_{x \rightarrow 0} \left[\sin \left(\frac{\pi}{2 - ax} \right) \right]^{\sec^2 \left(\frac{\pi}{2 - bx} \right)}$ is :
- (A) $e^{a/b}$ (B) e^{a^2/b^2} (C) e^{-a^2/b^2} (D) None of these
63. The function $f(x) = \begin{cases} b+2 & , \quad x = \frac{\pi}{2} \\ (1 + |\cos x|) \left(\frac{a |\tan x|}{b} \right) & , \quad \frac{\pi}{2} < x < \pi \end{cases}$. Determine the values of ' a ' and ' b ', if f is continuous at $x = \frac{\pi}{2}$.
- (A) $a = 0, b = -1$ (B) $a = 0, b = 0$ (C) $a = 1, b = -1$ (D) None of these
64. Let $f(x) = \frac{\sin^{-1}(1 - \{x\}) \cdot \cos^{-1}(1 - \{x\})}{\sqrt{2\{x\}(1 - \{x\})}}$, then the $\lim_{x \rightarrow 0^+} f(x)$ and $\lim_{x \rightarrow 0^-} f(x)$ is :
 (where $\{.\}$ denotes the fractional part function)
- (A) $\pi, \frac{\pi}{2}$ (B) $\frac{\pi}{2}, \frac{\pi}{2\sqrt{2}}$ (C) $\frac{-\pi}{2}, \frac{-\pi}{2\sqrt{2}}$ (D) $\frac{\pi}{2}$, does not exist
65. If $f(x) = \begin{cases} \frac{\sin[x^2]\pi}{x^2 - 3x + 8} + ax^3 + b & , \quad 0 \leq x \leq 1 \\ 2 \cos \pi x + \tan^{-1} x & , \quad 1 < x \leq 2 \end{cases}$ is differentiable in $[0, 2]$, then ' a ' and ' b ' are : (where $[.]$ stands greatest integer function)
- (A) $a = 1, b = \frac{\pi}{4}$ (B) $a = \frac{1}{6}, b = \frac{\pi}{4} - \frac{13}{6}$
 (C) $a = 6, b = \frac{-13}{6}$ (D) None of these
66. If $y = \sec(\tan^{-1} x)$, then $\frac{dy}{dx}$ at $x = 1$ is equal to :
- (A) $\frac{1}{\sqrt{2}}$ (B) $\frac{1}{2}$ (C) 1 (D) $\sqrt{2}$
67. If $f(x) = \begin{cases} x-1 & , \quad x \geq 1 \\ 2x^2 - 2 & , \quad x < 1 \end{cases}$, $g(x) = \begin{cases} x+1 & , \quad x > 0 \\ -x^2 + 1 & , \quad x \leq 0 \end{cases}$ and $h(x) = |x|$ then $\lim_{x \rightarrow 0} f(g(h(x)))$ is equal to :
- (A) 1 (B) 0 (C) -1 (D) Does not exist

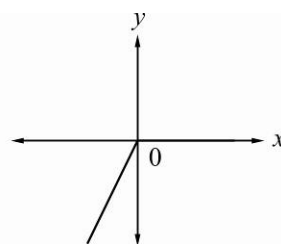
68. Let $f(x) = \begin{cases} x \sin\left(\frac{1}{x}\right) + \sin\left(\frac{1}{x^2}\right), & x \neq 0 \\ 0, & x = 0 \end{cases}$, then $\lim_{x \rightarrow \infty} f(x)$ is equal to :
- (A) 0 (B) $-\frac{1}{2}$ (C) 1 (D) None of these
69. $\lim_{x \rightarrow a^-} \left(\frac{|x|^3}{a} - \left[\frac{x}{a} \right]^3 \right)$ ($a > 0$), where $[x]$ denotes the greatest integer less than or equal to x , is equal to :
- (A) $a^2 + 1$ (B) $-a^2 - 1$ (C) a^2 (D) $-a^2$
70. $\lim_{x \rightarrow 0} \left[(1 - e^x) \frac{\sin x}{|x|} \right]$, where $[.]$ represents greatest integer function, is equal to :
- (A) -1 (B) 1 (C) 0 (D) Does not exist
71. If $\ell = \lim_{x \rightarrow \infty} (\sin \sqrt{x+1} - \sin \sqrt{x})$ and $m = \lim_{x \rightarrow -\infty} [\sin \sqrt{x+1} - \sin \sqrt{x}]$, where $[.]$ denotes the greatest integer function, then :
- (A) $\ell = m = 0$ (B) $\ell = 0, m$ is undefined
(C) ℓ, m both do not exist (D) $\ell = 0, m \neq 0$ (although m exist)
72. If $f(x) = \sum_{\lambda=1}^n \left(x - \frac{1}{\lambda} \right) \left(x - \frac{1}{\lambda+1} \right)$, then $\lim_{n \rightarrow \infty} f(0)$ is equal to :
- (A) 1 (B) -1 (C) 2 (D) None of these
73. Given a real valued function f such that $f(x) = \begin{cases} \frac{\tan^2[x]}{(x^2 - [x]^2)}, & x > 0 \\ 1, & x = 0 \\ \sqrt{\{x\}} \cot \{x\}, & x < 0 \end{cases}$
- where $[.]$ represents greatest integer function and $\{.\}$ represents fractional part function, then :
- (A) $\lim_{x \rightarrow 0} f(x) = 1$ (B) $\lim_{x \rightarrow 0^-} f(x) = \cot 1$
(C) $\cot^{-1} \left(\lim_{x \rightarrow 0^-} f(x) \right)^2 = 1$ (D) $\lim_{x \rightarrow 0^+} f(x) = 1$
74. The graph of the function $f(x) = \lim_{t \rightarrow 0} \left(\frac{2x}{\pi} \cot^{-1} \frac{x}{t^2} \right)$ is :



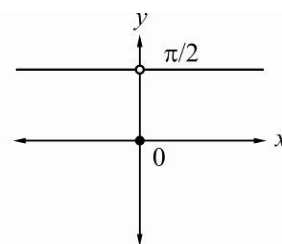
(A)



(B)



(C)



(D)

75. If $\lim_{x \rightarrow \infty} f(x)$ exists and is finite and non-zero and $\lim_{x \rightarrow \infty} \left(f(x) + \frac{3f(x)-1}{f^2(x)} \right) = 3$, then the value of $\lim_{x \rightarrow \infty} f(x)$ is equal to:

- (A) 1 (B) -1 (C) 2 (D) None of these

76. $\lim_{x \rightarrow \infty} \frac{e^x \left(\left(2^{x^n} \right)^{\frac{1}{e^x}} - \left(3^{x^n} \right)^{\frac{1}{e^x}} \right)}{x^n}$, $n \in N$, is equal to :

- (A) 0 (B) $\ln\left(\frac{2}{3}\right)$ (C) $\ln\left(\frac{3}{2}\right)$ (D) None of these

77. $\lim_{y \rightarrow 0} \left(\lim_{x \rightarrow \infty} \frac{\exp\left(x \ln\left(1 + \frac{ay}{x}\right) - \exp\left(x \ln\left(1 + \frac{by}{x}\right)\right)\right)}{y} \right)$ is equal to :

- (A) $a+b$ (B) $a-b$ (C) $b-a$ (D) $-(a+b)$

78. The function $f(x)$ is defined $f(x) = \begin{cases} \log_{(4x-3)}(x^2 - 2x + 5), & \text{if } \frac{3}{4} < x < 1 \text{ or } x > 1 \\ 4, & \text{if } x = 1 \end{cases}$

- (A) is continuous at $x = 1$
 (B) is discontinuous at $x = 1$ since $f(1^+)$ does not exist though $f(1^-)$ exists
 (C) is discontinuous at $x = 1$ since $f(1^-)$ does not exist though $f(1^+)$ exists
 (D) is discontinuous since neither $f(1^-)$ nor $f(1^+)$ exists

79. If $f(x) = \begin{cases} \sqrt{x} \left(1 + x \sin \frac{1}{x} \right), & x > 0 \\ -\sqrt{-x} \left(1 + x \sin \frac{1}{x} \right), & x < 0 \\ 0, & x = 0 \end{cases}$, then $f(x)$ is :

- (A) Continuous as well as diff. at $x = 0$ (B) Continuous at $x = 0$, but not diff. at $x = 0$
 (C) Neither continuous at $x = 0$ nor diff. at $x = 0$ (D) None of these

80. Given $f(x) = \begin{cases} x^2 e^{2(x-1)} & \text{for } 0 \leq x \leq 1 \\ a \operatorname{sgn}(x+1) \cos(2x-2) + bx^2 & \text{for } 1 < x \leq 2 \end{cases}$. $f(x)$ is differentiable at $x = 1$ provided :

- (A) $a = -1, b = 2$ (B) $a = 1, b = -2$ (C) $a = -3, b = 4$ (D) $a = 3, b = -4$

*81. Let $f(x) = \frac{|x+\pi|}{\sin x}$, then :

- (A) $f(-\pi^+) = -1$ (B) $f(-\pi^-) = 1$
 (C) $\lim_{x \rightarrow -\pi} f(x)$ does not exist (D) $\lim_{x \rightarrow \pi} f(x)$ does not exist

- *82. $\lim_{x \rightarrow \infty} \frac{(ax+1)^n}{x^n + A}$ is equal to :
- (A) a^n if $n \in \mathbb{N}$ (B) ∞ if $n \in \mathbb{Z}^-$ and $a = A = 0$
 (C) $\frac{1}{1+A}$ if $n = 0$ (D) a^n if $n \in \mathbb{Z}^-$, $A = 0$ and $a \neq 0$
83. Let $L = \lim_{x \rightarrow 0} \frac{a - \sqrt{a^2 - x^2} - \frac{x^2}{4}}{x^4}$, $a > 0$. If L is finite, then :
- (A) $a = 2$ (B) $a = 1$ (C) $L = \frac{1}{64}$ (D) $L = \frac{1}{32}$
84. If $\lim_{x \rightarrow 0} \left[1 + x \ln(1+b^2) \right]^{1/x} = 2b \sin^2 \theta$, $b > 0$ and $\theta \in (-\pi, \pi]$, then the value of θ is :
- (A) $\pm \frac{\pi}{4}$ (B) $\pm \frac{\pi}{3}$ (C) $\pm \frac{\pi}{6}$ (D) $\pm \frac{\pi}{2}$
85. If $\lim_{x \rightarrow \infty} \left(\frac{x^2 + x + 1}{x + 1} - ax - b \right) = 4$, then :
- (A) $a = 1, b = 4$ (B) $a = 1, b = -4$ (C) $a = 2, b = -3$ (D) $a = 2, b = 3$
86. Let $\alpha(a)$ and $\beta(a)$ be the roots of the equation $(\sqrt[3]{1+a} - 1)x^2 + (\sqrt{1+a} - 1)x + (\sqrt[6]{1+a} - 1) = 0$ where $a \geq -1$.
 Then $\lim_{a \rightarrow 0^+} \alpha(a)$ and $\lim_{a \rightarrow 0^+} \beta(a)$ are :
- (A) $-\frac{5}{2}$ and 1 (B) $-\frac{1}{2}$ and -1 (C) $-\frac{7}{2}$ and 2 (D) $-\frac{9}{2}$ and 3
87. Let $f(a) = g(a) = k$ and their n th derivatives $f^n(a), g^n(a)$ exist and are not equal for some n . Further if
 $\lim_{x \rightarrow a} \frac{f(a)g(x) - f(a) - g(a)f(x) + g(a)}{g(x) - f(x)} = 4$, then the value of k is equal to :
- (A) 4 (B) 2 (C) 1 (D) 0
88. If $f(x) = \begin{cases} xe^{-\left(\frac{1}{|x|} + \frac{1}{x}\right)}, & x \neq 0 \\ 0, & x = 0 \end{cases}$, then $f(x)$ is :
- (A) Continuous as well as differentiable for all x
 (B) Continuous for all x but not differentiable at $x = 0$
 (C) Neither differentiable nor continuous at $x = 0$
 (D) Discontinuous everywhere
89. The function $f: \mathbb{R} - \{0\} \rightarrow \mathbb{R}$ given by $f(x) = \frac{1}{x} - \frac{2}{e^{2x} - 1}$ can be made continuous at $x = 0$ by defining $f(0)$ as :
- (A) 2 (B) -1 (C) 0 (D) 1
90. Let $f(x) = \begin{cases} (x-1)\sin \frac{1}{x-1}, & x \neq 1 \\ 0, & x = 1 \end{cases}$. Then which one of the following is true ?
- (A) f is differentiable at $x = 0$ and at $x = 1$ (B) f is differentiable at $x = 0$ but not at $x = 1$
 (C) f is differentiable at $x = 1$ but not at $x = 0$ (D) f is neither differentiable at $x = 0$ nor at $x = 1$

91. Let $f: R \rightarrow R$ be a positive increasing function with $\lim_{x \rightarrow \infty} \frac{f(3x)}{f(x)} = 1$. Then $\lim_{x \rightarrow \infty} \frac{f(2x)}{f(x)}$.
- (A) $2/3$ (B) $3/2$ (C) 3 (D) 1
92. Let $f: R \rightarrow [0, \infty)$ be such that $\lim_{x \rightarrow 5} f(x)$ exists and $\lim_{x \rightarrow 5} \frac{(f(x))^2 - 9}{\sqrt{|x-5|}} = 0$. Then $\lim_{x \rightarrow 5} f(x)$ equals :
- (A) 0 (B) 1 (C) 2 (D) 3
93. The value of p and q for which the function $f(x) = \begin{cases} \frac{\sin(p+1)x + \sin x}{x}, & x < 0 \\ q, & x = 0 \\ \frac{\sqrt{x+x^2} - \sqrt{x}}{x^{3/2}}, & x > 0 \end{cases}$ is continuous for all $x \in R$ are :
- (A) $p = \frac{1}{2}, q = -\frac{3}{2}$ (B) $p = \frac{5}{2}, q = \frac{1}{2}$ (C) $p = -\frac{3}{2}, q = \frac{1}{2}$ (D) $p = \frac{1}{2}, q = \frac{3}{2}$
94. $\lim_{x \rightarrow \frac{\pi}{2}} \frac{\left(1 - \tan \frac{x}{2}\right)(1 - \sin x)}{\left(1 + \tan \frac{x}{2}\right)(\pi - 2x)^3}$ is equal to :
- (A) $\frac{1}{16}$ (B) $-\frac{1}{16}$ (C) $\frac{1}{32}$ (D) $-\frac{1}{32}$
95. If $\lim_{x \rightarrow 0} \frac{\ln(3+x) - \ln(3-x)}{x} = k$, then the value of k is :
- (A) 0 (B) $-\frac{1}{3}$ (C) $\frac{2}{3}$ (D) $-\frac{2}{3}$
96. If $\lim_{x \rightarrow \infty} \left(1 + \frac{a}{x} + \frac{b}{x^2}\right)^{2x} = e^2$, then the value of a and b are :
- (A) $a \in R, b \in R$ (B) $a = 1, b \in R$ (C) $a \in R, b = 2$ (D) $a = 1, b = 2$
97. Let α and β be the distinct roots of $ax^2 + bx + c = 0$, then $\lim_{x \rightarrow \alpha} \frac{1 - \cos(ax^2 + bx + c)}{(x - \alpha)^2}$ is equal to :
- (A) $\frac{1}{2}(\alpha - \beta)^2$ (B) $-\frac{a^2}{2}(\alpha - \beta)^2$ (C) 0 (D) $\frac{a^2}{2}(\alpha - \beta)^2$
98. The set of points, where $f(x) = \frac{x}{1+|x|}$ is differentiable, is :
- (A) $(-\infty, -1) \cup (-1, \infty)$ (B) $(-\infty, \infty)$
(C) $(0, \infty)$ (D) $(-\infty, 0) \cup (0, \infty)$
99. If function $f(x)$ is differentiable at $x = a$, then $\lim_{x \rightarrow a} \frac{x^2 f(a) - a^2 f(x)}{x - a}$ is :
- (A) $-a^2 f'(a)$ (B) $af(a) - a^2 f'(a)$
(C) $2af(a) - a^2 f'(a)$ (D) $2af(a) + a^2 f'(a)$

100. If $f: \mathbb{R} \rightarrow \mathbb{R}$ is a function defined by $f(x) = [x] \cos\left(\frac{2x-1}{2}\right)\pi$, where $[x]$ denotes the greatest integer function then f is:

- (A) Continuous for every real x (B) Discontinuous only at $x = 0$
(C) Discontinuous only at non-zero integral values of x (D) Continuous only at $x = 0$

101. Consider the function, $f(x) = |x-2| + |x-5|$, $x \in \mathbb{R}$.

Statement 1 : $f'(4) = 0$

Statement 2 : f is continuous in $[2, 5]$, differentiable in $(2, 5)$ and $f(2) = f(5)$

- (A) Statement-1 is True, Statement-2 is True and Statement-2 is a correct explanation for Statement-1
(B) Statement-1 is True, Statement-2 is True and Statement-2 is NOT a correct explanation for Statement-1
(C) Statement-1 is True, Statement-2 is False
(D) Statement-1 is False, Statement-2 is True

102. $\lim_{x \rightarrow 0} \frac{(1 - \cos 2x)(3 + \cos x)}{x \tan 4x}$ is equal to :

- (A) $-\frac{1}{4}$ (B) $\frac{1}{2}$ (C) 1 (D) 2

103. If $f'(x) = \sqrt{2x^2 - 1}$ and $y = f(x^2)$, then $\frac{dy}{dx}$ at $x = 1$ is equal to :

- (A) 2 (B) 1 (C) -2 (D) -1

104. If $f(x) = \log_x(\ln x)$, then $f'(x)$ at $x = e$ is equal to :

- (A) $\frac{1}{e}$ (B) e (C) 1 (D) Zero

105. If $x = \frac{1+t}{t^3}$, $y = \frac{3}{2t^2} + \frac{2}{t}$, then $x\left(\frac{dy}{dx}\right)^3 - \frac{dy}{dx}$ is equal to :

- (A) 0 (B) -1 (C) 1 (D) 2

106. If $f(x) = |x|^{\sin x}$, then $f'\left(\frac{\pi}{4}\right)$ is equal to :

- (A) $\left(\frac{\pi}{4}\right)^{1/\sqrt{2}} \left(\frac{\sqrt{2}}{2} \ln \frac{4}{\pi} - \frac{2\sqrt{2}}{\pi}\right)$ (B) $\left(\frac{\pi}{4}\right)^{1/\sqrt{2}} \left(\frac{\sqrt{2}}{2} \ln \frac{4}{\pi} + \frac{2\sqrt{2}}{\pi}\right)$
(C) $\left(\frac{\pi}{4}\right)^{1/\sqrt{2}} \left(\frac{\sqrt{2}}{2} \ln \frac{\pi}{4} - \frac{2\sqrt{2}}{\pi}\right)$ (D) $\left(\frac{\pi}{4}\right)^{1/\sqrt{2}} \left(\frac{\sqrt{2}}{2} \ln \frac{\pi}{4} + \frac{2\sqrt{2}}{\pi}\right)$

107. Let $f(x)$ be a polynomial in x . Then the second derivative of $f(e^x)$ w.r.t. x is :

- (A) $f''(e^x) \cdot e^x + f'(e^x)$ (B) $f''(e^x) \cdot e^{2x} + f'(e^x) \cdot e^{2x}$
(C) $f''(e^x) e^x$ (D) $f''(e^x) \cdot e^{2x} + f'(e^x) \cdot e^x$

108. If $x\sqrt{1+y} + y\sqrt{1+x} = 0$, then $\frac{dy}{dx}$ is equal to :

- (A) $\frac{1}{(1+x)^2}$ (B) $-\frac{1}{(1+x)^2}$ (C) $\frac{1}{(1+x^2)}$ (D) $\frac{1}{(1+x)}$

109. Let g is the inverse function of f and $f'(x) = \frac{x^{10}}{(1+x^2)}$. If $g(2) = a$, then $g'(2)$ is equal to :
- (A) $\frac{a}{2^{10}}$ (B) $\frac{1+a^2}{a^{10}}$ (C) $\frac{a^{10}}{1+a^2}$ (D) $\frac{1+a^{10}}{a^2}$
110. If $y = \sin^{-1}\left(x\sqrt{1-x} + \sqrt{x}\sqrt{1-x^2}\right)$ and $\frac{dy}{dx} = \frac{1}{2\sqrt{x}\sqrt{1-x}} + p$, then p is equal to :
- (A) 0 (B) $\frac{1}{\sqrt{1-x}}$ (C) $\sin^{-1}\sqrt{x}$ (D) $\frac{1}{\sqrt{1-x^2}}$
111. If $\frac{d}{dx}\left(\frac{1+x^2+x^4}{1+x+x^2}\right) = ax+b$, then the value of 'a' and 'b' are respectively :
- (A) 2 and 1 (B) -2 and 1 (C) 2 and -1 (D) None of these
112. If $y = f\left(\frac{2x-1}{x^2+1}\right)$ and $f'(x) = \sin x$, then $\frac{dy}{dx}$ is equal to :
- (A) $\frac{1+x-x^2}{(1+x^2)^2} \sin\left(\frac{2x-1}{x^2+1}\right)$ (B) $\frac{2(1+x-x^2)}{(1+x^2)^2} \sin\left(\frac{2x-1}{x^2+1}\right)$
- (C) $\frac{1-x+x^2}{(1+x^2)^2} \sin\left(\frac{2x-1}{x^2+1}\right)$ (D) None of these
- *113. If $\sqrt{x^2+y^2} = e^t$ where $t = \sin^{-1}\left(\frac{y}{\sqrt{x^2+y^2}}\right)$, then $\frac{dy}{dx}$ is equal to :
- (A) $\frac{x-y}{x+y}$ (B) $\frac{x+y}{x-y}$ (C) $\frac{y-x}{y+x}$ (D) $\frac{2-y}{2x+y}$
- *114. If f is twice differentiable such that $f''(x) = -f(x)$ and $f'(x) = g(x)$. If $h(x)$ is a twice differentiable function such that $h'(x) = (f(x))^2 + (g(x))^2$. If $h(0) = 2, h(1) = 4$, then the equation $y = h(x)$ represents :
- (A) a curve of degree 2 (B) a curve passing through the origin
- (C) a straight line with slope 2 (D) a straight line with y intercept equal to 2
- *115. $y = \cos^{-1}\sqrt{\frac{\sqrt{1+x^2}+1}{2\sqrt{1+x^2}}}$, then $\frac{dy}{dx}$ is equal to :
- (A) $\frac{1}{2(1+x^2)}, x \in R$ (B) $\frac{1}{2(1+x^2)}, x > 0$ (C) $\frac{-1}{2(1+x^2)}, x < 0$ (D) $\frac{1}{2(1+x^2)}, x < 0$
116. Let $f(x)$ be a polynomial function of second degree. If $f(1) = f(-1)$ and a, b, c are in AP, then $f'(a), f'(b)$ and $f'(c)$ are in :
- (A) AP (B) GP
- (C) HP (D) Arithmetico-Geometric Progression

Integer Answer Type Questions

The Answer to the following questions are positive integers of 1/2/3 digits and zero

117. Total number of integral points where the function $f(x) = [x]^2 - [x^2]$ is continuous is ____ . [Where $[x]$ represents greatest integer less than or equal to x .]
118. If $\lim_{x \rightarrow \frac{\pi}{4}} \frac{\int_{\frac{\pi}{4}}^{\sec^2 x} f(t) dt}{x^2 - \frac{\pi^2}{16}}$ equals $\frac{8}{\pi} f(r)$, then r equal ____ .
119. $\lim_{x \rightarrow 0^+} \left[(\sin x)^{1/x} + \left(\frac{1}{x} \right)^{\sin x} \right]$ is equal to ____ .
120. Total number of points of discontinuity of the function $f(x) = \min\{1, x^2, x^3\}$ is ____ .
121. The set of all points where the function $f(x) = \frac{\sin x}{1 + |x|}$ is non-differentiable is ____ .
122. Let $\mathbb{R}$ be the set of real numbers and $f: \mathbb{R} \rightarrow \mathbb{R}$ be such that for all x and y in $\mathbb{R}$, $|f(x) - f(y)|^2 \leq (x - y)^2$. If $f(0) = 1$, then find $f(10)$ ____ .
123. The total number of points where the function $f(x) = [x] \cos\left(\frac{2x-1}{2}\right)\pi$ is discontinuous is ____ . [Where $[x]$ represents greatest integer less than or equal to x]
124. $\lim_{x \rightarrow 1} (1-x) \tan\left(\frac{\pi x}{2}\right) \cdot \pi$ is equal to ____ .
125. $\lim_{x \rightarrow \infty} \sqrt{\frac{x - \sin^2 x}{x + \cos^2 x}}$ is equal to ____ .
126. $\lim_{x \rightarrow 0} \frac{\int_0^{x^2} \cos^2 t dt}{x \sin x}$ is equal to ____ .
127. The total number of points where the function $f(x) = (x^2 - 1) \left| x^2 - 3x + 2 \right| + \cos(|x|)$ is not differentiable is ____
128. The total number of points of non-differentiability of $f(x) = ||x| - 1|$ is ____ .
129. Let f be a twice differentiable function $f''(x) = -f(x)$ and $g(x) = f'(x)$. Given $F(x) = (f(x))^2 + (g(x))^2$ and $F(5) = 5$, find $F(10)$.
130. If $x^2 + y^2 = 1$ and $xy' + ax = 0$, then the value of a is ____ .
131. The total number of points of discontinuity of $f(x) = \frac{\tan \pi \left[\frac{(x-\pi)}{1+[x]^2} \right]}{1+[x]^2}$ is (where $[x]$ represents the greatest integer less than or equal to x)

Differential Calculus-2

CHOOSE THE CORRECT ALTERNATIVE. ONLY ONE CHOICE IS CORRECT. HOWEVER, QUESTIONS MARKED ‘*’ MAY HAVE MORE THAN ONE CORRECT OPTION.

- The minimum distance of a point on the curve $y = x^2 - 4$ from the origin is :
 (A) $\frac{\sqrt{19}}{2}$ (B) $\sqrt{\frac{15}{2}}$ (C) $\frac{\sqrt{15}}{2}$ (D) $\sqrt{\frac{19}{2}}$
- Let C be a curve given by $y(x) = 1 + \sqrt{4x-3}$, $x > \frac{3}{4}$. If P is a point on C , such that the tangent at P has slope $\frac{2}{3}$, then a point through which the normal at P passes, is :
 (A) $(2, 3)$ (B) $(4, -3)$ (C) $(1, 7)$ (D) $(3, -4)$
- The distance, from the origin, of the normal to the curve, $x = 2\cos t + 2t\sin t$, $y = 2\sin t - 2t\cos t$ at $t = \frac{\pi}{4}$, is :
 (A) 4 (B) $2\sqrt{2}$ (C) 2 (D) $\sqrt{2}$
- If Rolle's theorem holds for the function $f(x) = 2x^3 + bx^2 + cx$, $x \in [-1, 1]$, at the point $x = \frac{1}{2}$, then $2b + c$ equals
 (A) 1 (B) -1 (C) 2 (D) -3
- Let the tangents drawn to the circle, $x^2 + y^2 = 16$ from the point $P(0, h)$ meet the x -axis at points A and B . If the area of $\triangle APB$ is minimum, then h is equal to :
 (A) $4\sqrt{3}$ (B) $3\sqrt{3}$ (C) $3\sqrt{2}$ (D) $4\sqrt{2}$
- The equation of a normal to the curve, $\sin y = x \sin\left(\frac{\pi}{3} + y\right)$ at $x = 0$, is :
 (A) $2x + \sqrt{3}y = 0$ (B) $2y - \sqrt{3}x = 0$ (C) $2y + \sqrt{3}x = 0$ (D) $2x - \sqrt{3}y = 0$
- Let k and K be the minimum and the maximum values of the function $f(x) = \frac{(1+x)^{0.6}}{1+x^{0.6}}$ in $[0, 1]$ respectively, then the ordered pair (k, K) is equal to :
 (A) $(1, 2^{0.6})$ (B) $(2^{-0.4}, 2^{0.6})$ (C) $(2^{-0.6}, 1)$ (D) $(2^{-0.4}, 1)$
- If the Rolle's theorem holds for the function $f(x) = 2x^3 + ax^2 + bx$ in the interval $[-1, 1]$ for the point $c = \frac{1}{2}$, then the value of $2a + b$ is :
 (A) 1 (B) -1 (C) 2 (D) -2
- For the curve $y = 3\sin\theta\cos\theta$, $x = e^\theta\sin\theta$, $0 \leq \theta \leq \pi$, the tangent is parallel to x -axis when θ is :
 (A) $3\pi/4$ (B) $\pi/2$ (C) $\pi/4$ (D) $\pi/6$
- Two ships A and B are sailing away from a fixed point O along routes such that $\angle AOB$ is always 120° . At a certain instance, $OA = 8\text{ km}$, $OB = 6\text{ km}$ and the ship A is sailing at the rate of 20 km/hr while the ship B sailing at the rate of 30 km/hr . Then the distance between A and B is changing at the rate in (km/hr) :
 (A) $260/\sqrt{37}$ (B) $260/37$ (C) $80/\sqrt{37}$ (D) $80/37$

11. The volume of the largest possible right circular cylinder that can be inscribed in a sphere of radius $= \sqrt{3}$ is :
- (A) $\frac{4}{3}\sqrt{3}\pi$ (B) $\frac{8}{3}\sqrt{3}\pi$ (C) 4π (D) 2π
12. Let f and g be two differentiable functions on R such that $f'(x) > 0$ and $g'(x) < 0$, for all $x \in R$. Then for all x :
- (A) $f(g(x)) > f(g(x-1))$ (B) $f(g(x)) > f(g(x+1))$
 (C) $g(f(x)) > g(f(x+1))$ (D) $g(f(x)) > g(f(x-1))$
13. If the volume of a spherical ball is increasing at the rate of $4\pi cc/sec$, then the rate of increase of its radius (in cm/sec), when the volume is $288\pi cc$, is :
- (A) $\frac{1}{6}$ (B) $\frac{1}{9}$ (C) $\frac{1}{36}$ (D) $\frac{1}{24}$
14. **Statement I :** The equation $x \log x = 2 - x$ is satisfied by at least one value of x lying between 1 and 2.
Statement II : The function $f(x) = x \log x$ is an increasing function in $[1, 2]$ and $g(x) = 2 - x$ is a decreasing function in $[1, 2]$ and the graphs represented by these function intersect at a point in $[1, 2]$.
- (A) Statement I is true; Statement II is true; Statement II is a correct explanation for Statement I.
 (B) Statement I is true; Statement II is true; Statement II is not a correct explanation for Statement I.
 (C) Statement I is false; Statement II is true.
 (D) Statement I is true; Statement II is false.
15. If the surface area of a sphere of radius r is increasing uniformly at the rate $8cm^2/s$, then the rate of change of its volume is :
- (A) constant (B) proportional to $\sqrt{r}$
 (C) proportional to r^2 (D) proportional to r
16. **Statement I :** The function $x^2(e^x + e^{-x})$ is increasing for all $x > 0$.
Statement II : The function x^2e^x and x^2e^{-x} are increasing for all $x > 0$ and the sum of two increasing function in any interval (a, b) is an increasing function in (a, b) .
- (A) Statement I is false; Statement II is true.
 (B) Statement I is true; Statement II is true; Statement II is not a correct explanation for Statement I.
 (C) Statement I is true; Statement II is false.
 (D) Statement I is true; Statement II is true; Statement II is a correct explanation for Statement I.
17. The maximum area of a right angled triangle with hypotenuse h is :
- (A) $h^2/2\sqrt{2}$ (B) $h^2/2$ (C) $h^2/\sqrt{2}$ (D) $h^2/4$
18. The cost of running a bus from A to B , is Rs. $\left(av + \frac{b}{v}\right)$, where $v km/h$ is the average speed of the bus. When the bus travels at $30 km/h$, the cost comes out to be Rs. 75 while at $40 km/h$, it is Rs. 65. Then the most economical speed (in km/h) of the bus is :
- (A) 45 (B) 50 (C) 60 (D) 40
19. Water is poured into an inverted conical vessel of which the radius of the base is 2m and height 4m, at the rate of 77 litre/minute. The rate at which the water level is rising at the instant when the depth is 70 cm is (Use $\pi = 22/7$)
- (A) 10 cm/min (B) 20 cm/min (C) 40 cm/min (D) None

20. A kite is 300 m high and there are 500 m of cord out. If the wind moves the kite horizontally at the rate of 5 km/hr directly away from the person who is flying it, find the rate at which the cord is being paid ?
 (A) 4 (B) 8 (C) 3 (D) Cannot be determined
21. Number of tangents drawn from the point $\left(-\frac{1}{2}, 0\right)$ to the curve $y = e^{\{x\}}$. (Here $\{ \}$ denotes fractional part function).
 (A) 2 (B) 1 (C) 3 (D) 4
22. If tangent to curve $2y^3 = ax^2 + x^3$ at point (a, a) cuts off intercepts α, β on co-ordinate axes, where $\alpha^2 + \beta^2 = 61$, then the value of 'a' is equal to :
 (A) 20 (B) 25 (C) 30 (D) None of these
- *23. The co-ordinates of point(s) on the graph of the function, $f(x) = \frac{x^3}{3} - \frac{5x^2}{2} + 7x - 4$ where the tangent drawn cut off intercepts from the co-ordinate axes which are equal in magnitude but opposite in sign, is :
 (A) $\left(2, \frac{8}{3}\right)$ (B) $\left(3, \frac{7}{2}\right)$ (C) $\left(1, \frac{5}{6}\right)$ (D) None of these
- *24. The co-ordinates of a point on the parabola $2y = x^2$ which is nearest to the point $(0, 3)$ is :
 (A) $(2, 2)$ (B) $(-\sqrt{2}, 1)$ (C) $(\sqrt{2}, 1)$ (D) $(-2, 2)$
25. The values of p for which the function $f(x) = \left(\frac{\sqrt{p+4}}{1-p} - 1\right)x^5 - 3x + \ln 5$ decreases for all real x is :
 (A) $(-\infty, \infty)$ (B) $\left[-4, \frac{3-\sqrt{21}}{2}\right] \cup (1, \infty)$
 (C) $\left[-3, \frac{5-\sqrt{27}}{2}\right] \cup (2, \infty)$ (D) $[1, \infty)$
- *26. If $f(x) = 2x + \cot^{-1} x + \ln(\sqrt{1+x^2} - x)$, then $f(x)$:
 (A) Increases in $[0, \infty)$ (B) Decreases in $[0, \infty)$
 (C) Neither increases nor decreases in $[0, \infty)$ (D) Increases in $(-\infty, \infty)$
- *27. Let $g(x) = 2f\left(\frac{x}{2}\right) + f(1-x)$ and $f''(x) < 0$ in $0 \leq x \leq 1$ then $g(x)$
 (A) Decreases in $\left[0, \frac{2}{3}\right]$ (B) Decreases $\left[\frac{2}{3}, 1\right]$
 (C) Increases in $\left[0, \frac{2}{3}\right]$ (D) Increases in $\left[\frac{2}{3}, 1\right]$
- *28. If $f(x) = \begin{cases} -\sqrt{1-x^2} & , 0 \leq x \leq 1 \\ -x & , x > 1 \end{cases}$, then :
 (A) Maximum of $f(x)$ exist at $x = 1$ (B) Maximum of $f(x)$ doesn't exist
 (C) Minimum of $f^{-1}(x)$ exist at $x = -1$ (D) Minimum of $f^{-1}(x)$ exist at $x = 1$

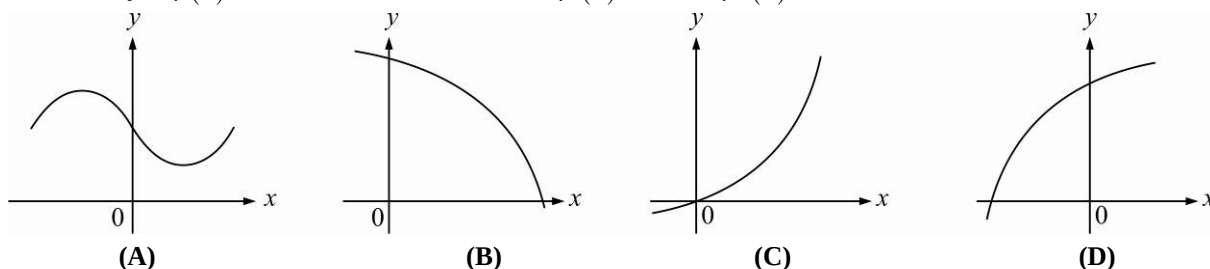
*29. If $f(x) = \tan^{-1} x - \left(\frac{1}{2}\right) \ln x$. Then :

- (A) The greatest value of $f(x)$ on $\left[\frac{1}{\sqrt{3}}, \sqrt{3}\right]$ is $\frac{\pi}{6} + \left(\frac{1}{4}\right) \ln 3$
 (B) The least value of $f(x)$ on $\left[\frac{1}{\sqrt{3}}, \sqrt{3}\right]$ is $\frac{\pi}{3} - \left(\frac{1}{4}\right) \ln 3$
 (C) $f(x)$ decreases on $(0, \infty)$ (D) $f(x)$ increases on $(-\infty, 0)$

30. The radius of a right circular cylinder of greatest curved surface which can be inscribed in a given right circular cone is :

- (A) One third of the radius of the cone (B) $\frac{1}{\sqrt{2}}$ times of the radius of the cone
 (C) $\frac{2}{3}$ of the radius of the cone (D) $\frac{1}{2}$ of the radius of the cone

31. The curve $y = f(x)$ which satisfies the condition $f'(x) > 0$ and $f''(x) < 0$ for all real x , is :



32. If the point $(1, 3)$ serves as the point of inflection of the curve $y = ax^3 + bx^2$ then the value of 'a' and 'b' are :

- (A) $a = \frac{3}{2}$ and $b = -\frac{9}{2}$ (B) $a = \frac{3}{2}$ and $b = \frac{9}{2}$
 (C) $a = -\frac{3}{2}$ and $b = -\frac{9}{2}$ (D) $a = -\frac{3}{2}$ and $b = \frac{9}{2}$

*33. If $f(x) = \ln(x-2) - \frac{1}{x}$, then : (M.I. : Monotonically increasing)

- (A) $f(x)$ is M.I. for $x \in (2, \infty)$ (B) $f(x)$ is M.I. for $x \in [-1, 2]$
 (C) $f(x)$ is always concave downwards (D) $f^{-1}(x)$ is M.I. wherever defined

34. The function $f(x) = x(x+3)e^{-x/2}$ satisfies all the conditions of Rolle's theorem on $[-3, 0]$. The value of c which verifies Rolle's theorem, is :

- (A) 0 (B) -1 (C) -2 (D) 3

For Questions 35 - 36

- (A) Statement-1 is True, Statement-2 is True and Statement-2 is a correct explanation for Statement-1
 (B) Statement-1 is True, Statement-2 is True and Statement-2 is NOT a correct explanation for Statement-1
 (C) Statement-1 is True, Statement-2 is False (D) Statement-1 is False, Statement-2 is True

35. **Statement 1 :** e^x is bigger than π^e .

Statement 2 : $f(x) = x^{1/x}$ is a increasing function when $x \in [e, \infty)$.

36. Let $f(x) = x^{50} - x^{20}$.

Statement 1 : Global maximum of $f(x)$ in $[0, 1]$ is 0

Statement 2 : $x = 0$ is a stationary point of $f(x)$.

37. Let $f(x) = \begin{cases} -x^2 & , x < 0 \\ x^2 + 8 & , x \geq 0 \end{cases}$ equation of tangent line touching both branches of $y = f(x)$ is :
 (A) $y = 4x + 1$ (B) $y = 4x + 4$ (C) $y = x + 4$ (D) $y = x + 1$
38. Equation of normal drawn to the graph of the function defined as $f(x) = \frac{\sin x^2}{4}$, $x \neq 0$ and $f(0) = 0$ at the origin is :
 (A) $x + y = 0$ (B) $x - y = 0$ (C) $y = 0$ (D) $x = 0$
39. All points on curve $y^2 = 4a\left(x + a \sin \frac{x}{a}\right)$ at which tangents are parallel to the axis of x , lie on a :
 (A) Circle (B) Parabola (C) Line (D) None of these
40. Let $f(x) = x^3 + ax^2 + bx + 5 \sin^2 x$ be an increasing function in the set of real numbers R . Then a and b satisfy the condition :
 (A) $a^2 - 3b - 15 > 0$ (B) $a^2 - 3b + 15 \leq 0$ (C) $a^2 + 3b - 15 < 0$ (D) $a > 0$ and $b > 0$
41. The line $\frac{x}{a} + \frac{y}{b} = 1$ touches the curve $y = be^{-x/a}$ at the points :
 (A) $(-a, 2b)$ (B) $\left(\frac{a}{2}, \frac{b}{2}\right)$ (C) $\left(a, \frac{b}{e}\right)$ (D) $(0, b)$
42. If $f(x) = a^{\{a^{|x|} \operatorname{sgn} x\}}$; $g(x) = a^{\left[a^{\lfloor x \rfloor} \operatorname{sgn} x\right]}$ for $a > 1, a \neq 1$ and $x \in R$, where $\{ \}$ and $\lfloor \rfloor$ denote the fractional part and integral part functions respectively, then which of the following statements holds good for the function $h(x)$, where $(\ell n a)h(x) = (\ell n f(x)) + (\ell n g(x))$.
 (A) 'h' is even and increasing (B) 'h' is odd and decreasing
 (C) 'h' is even and decreasing (D) 'h' is odd and increasing
43. Let $f(x) = \begin{cases} x^3 - x^2 + 10x - 5 & , x \leq 1 \\ -2x + \log_2(b^2 - 2) & , x > 1 \end{cases}$ the set of values of b for which $f(x)$ has greatest value at $x = 1$ is given by:
 (A) $1 \leq b \leq 2$ (B) $b = \{1, 2\}$
 (C) $b \in (-\infty, -1)$ (D) $\left[-\sqrt{130}, \sqrt{2}\right) \cup \left(\sqrt{2}, \sqrt{130}\right]$
44. In a regular triangular prism the distance from the centre of one base to one of the vertices of the other base is ℓ . The altitude of the prism for which the volume is greatest, is :
 (A) $\frac{\ell}{2}$ (B) $\frac{\ell}{\sqrt{3}}$ (C) $\frac{\ell}{3}$ (D) $\frac{\ell}{4}$
45. Consider the following statements :
 S_1 : The function $y = \frac{2x^2 - 1}{x^4}$ is neither increasing nor decreasing.
 S_2 : If $f(x)$ is strictly increasing real function defined on R and c is a real constant, then number of solutions of $f(x) = c$ is always equal to one.
 S_3 : Let $f(x) = x$; $x \in (0, 1)$. $f(x)$ does not has any point of local maxima/minima.
 S_4 : $f(x) = \{x\}$ has maximum at $x = 6$ (here $\{.\}$ denotes fractional part function).
 State, in order, whether S_1, S_2, S_3, S_4 are true or false :
 (A) TTFT (B) FTFT (C) TFTF (D) TTFT

- *46. Let $f(x) = x^{m/n}$ for $x \in \mathbb{R}$ where m and n are integers, m even and n odd and $0 < m < n$. Then :
- (A) $f(x)$ decreases on $(-\infty, 0]$ (B) $f(x)$ increases on $[0, \infty)$
 (C) $f(x)$ increases on $(-\infty, 0]$ (D) $f(x)$ decreases on $[0, \infty)$
- *47. Let $\phi(x) = (f(x))^3 - 3(f(x))^2 + 4f(x) + 5x + 3\sin x + 4\cos x \forall x \in \mathbb{R}$, then :
- (A) ϕ is increasing whenever f is increasing (B) ϕ is increasing whenever f is decreasing
 (C) ϕ is decreasing whenever f is decreasing (D) ϕ is decreasing whenever if $f'(x) = -11$
- *48. For the function $f(x) = x^4(12 \ln x - 7)$:
- (A) The point $(1, -7)$ is the point of inflection (B) $x = e^{1/3}$ is the point of minima
 (C) The graph is concave downwards in $(0, 1)$ (D) The graph is concave upwards in $(1, \infty)$
- *49. The curve $y = \frac{x+1}{x^2+1}$ has :
- (A) $x = 1$, as point of inflection (B) $x = -2 + \sqrt{3}$, as point of inflection
 (C) $x = -1$, as point of minimum (D) $x = -2 - \sqrt{3}$, as point of inflection
50. How many real solutions does the equation $x^7 + 14x^5 + 16x^3 + 30x - 560 = 0$ have ?
- (A) 5 (B) 7 (C) 1 (D) 3
- *51. Let $f(x) = \frac{40}{(3x^4 + 8x^3 - 18x^2 + 60)}$. Which of the following statement(s) about $f(x)$ is(are) correct ?
- (A) $f(x)$ has local minima at $x = 0$ (B) $f(x)$ has local maxima at $x = 0$
 (C) Absolute maximum value of $f(x)$ is not defined (D) $f(x)$ is local maxima at $x = -3, x = 1$

PARAGRAPH FOR QUESTIONS 52 - 54

Let $a(t)$ be a function of t such that $\frac{da}{dt} = 2$ for all values of t and $a = 0$ when $t = 0$. Further $y = m(t)x + c(t)$ is tangent to the curve $y = x^2 - 2ax + a^2 + a$ at the point whose abscissa is 0. Then :

52. If the rate of change of distance of vertex of $y = x^2 - 2ax + a^2 + a$ from the origin with respect to t is k , then $k =$
- (A) 2 (B) $2\sqrt{2}$ (C) $\sqrt{2}$ (D) $4\sqrt{2}$
53. If the rate of change of $c(t)$ with respect to t , when $t = k$, is ℓ , then $\ell =$
- (A) $16\sqrt{2} - 2$ (B) $8\sqrt{2} + 2$ (C) $10\sqrt{2} + 2$ (D) $16\sqrt{2} + 2$
54. The rate of change of $m(t)$, with respect to t , at $t = \ell$ is :
- (A) -2 (B) 2 (C) -4 (D) 4

PARAGRAPH FOR QUESTIONS 55 - 57

Consider a function f defined by $f(x) = \sin^{-1} \sin\left(\frac{x + \sin x}{2}\right), \forall x \in [0, \pi]$,

which satisfies $f(x) + f(2\pi - x) = \pi, \forall x \in [\pi, 2\pi]$ and $f(x) = f(4\pi - x)$ for all $x \in [2\pi, 4\pi]$, then :

55. If α is the length of the largest interval on which $f(x)$ is increasing, then $\alpha =$
- (A) $\frac{\pi}{2}$ (B) π (C) 2π (D) 4π

56. If $f(x)$ is symmetric about $x = \beta$, then $\beta =$
 (A) $\frac{\alpha}{2}$ (B) α (C) $\frac{\alpha}{4}$ (D) 2α
57. Maximum value of $f(x)$ on $[0, 4\pi]$ is :
 (A) $\frac{\beta}{2}$ (B) β (C) $\frac{\beta}{4}$ (D) 2β
58. Let f, g and h be real-valued functions defined on the interval $[0, 1]$ by $f(x) = e^{x^2} + e^{-x^2}$, $g(x) = xe^{x^2} + e^{-x^2}$ and $h(x) = x^2e^{x^2} + e^{-x^2}$. If a, b and c denote, respectively, the absolute maximum of f, g and h on $[0, 1]$, then :
 (A) $a = b$ and $c \neq b$ (B) $a = c$ and $a \neq b$ (C) $a \neq b$ and $c \neq b$ (D) $a = b = c$
59. The number of points in $(-\infty, \infty)$, for which $x^2 - x \sin x - \cos x = 0$, is :
 (A) 6 (B) 4 (C) 2 (D) 0
- *60. A rectangular sheet of fixed perimeter with sides having their lengths in the ratio 8 : 15 is converted into an open rectangular box by folding after removing squares of equal area from all four corners. If the total area of removed squares is 100, the resulting box has maximum volume. The lengths of the sides of the rectangular sheet are :
 (A) 24 (B) 32 (C) 45 (D) 60
61. The function $f(x) = 2|x| + |x + 2| - ||x + 2| - 2|x||$ has a local minimum or a local maximum at $x =$
 (A) -2 (B) $-\frac{2}{3}$ (C) 2 (D) $\frac{2}{3}$
- *62. If the function $f(x) = 2x^3 - 9ax^2 + 12a^2x + 1$, where $a > 0$, attains its maximum and minimum at p and q respectively such that $p^2 = q$, then a equals :
 (A) 3 (B) 1 (C) 2 (D) $1/2$
63. Angle between the tangents to the curve $y = x^2 - 5x + 6$ at the points (2, 0) and (3, 0) is :
 (A) $\frac{\pi}{2}$ (B) $\frac{\pi}{6}$ (C) $\frac{\pi}{4}$ (D) $\frac{\pi}{3}$
64. A function is matched below against an interval where it is supposed to be increasing. Which of the following pairs is incorrectly matched ?
- | Interval | Function | Interval | Function |
|---------------------|-------------------------|---|-----------------------|
| (A) $(-\infty, -4]$ | $x^3 + 6x^2 + 6$ | (B) $\left(-\infty, \frac{1}{3}\right]$ | $3x^2 - 2x + 1$ |
| (C) $[2, \infty)$ | $2x^3 - 3x^2 - 12x + 6$ | (D) $(-\infty, \infty)$ | $x^3 - 3x^2 + 3x + 3$ |
65. Area of the greatest rectangle that can be inscribed in the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is :
 (A) $2ab$ (B) ab (C) $\sqrt{ab}$ (D) $\frac{a}{b}$
66. Let $f(x)$ be differentiable for all x . If $f(1) = -2$ and $f'(x) \geq 2$ for $x \in (1, 6)$ then :
 (A) $f(6) = 5$ (B) $f(6) < 5$ (C) $f(6) \geq 8$ (D) $f(6) < 8$

- *67. The normal to the curve $x = a(\cos \theta + \theta \sin \theta)$, $y = a(\sin \theta - \theta \cos \theta)$ at any point ' θ ' is such that :
- (A) It is at a constant distance from the origin (B) It passes through $\left(\frac{a\pi}{2}, -a\right)$
- (C) It makes angle $\frac{\pi}{2} + \theta$ with the x-axis (D) It passes through the origin
68. A value of c for which the conclusion of Mean Value Theorem holds for the function $f(x) = \log_e x$ on the interval $[1, 3]$ is :
- (A) $2\log_3 e$ (B) $\frac{1}{2}\log_e 3$ (C) $\log_3 e$ (D) $\log_e 3$
69. The function $f(x) = \tan^{-1}(\sin x + \cos x)$ is an increasing function in :
- (A) $\left(\frac{\pi}{4}, \frac{\pi}{2}\right)$ (B) $\left(-\frac{\pi}{2}, \frac{\pi}{4}\right)$ (C) $\left(0, \frac{\pi}{2}\right)$ (D) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
70. Given $P(x) = x^4 + ax^3 + bx^2 + cx + d$ such that $x = 0$ is the only real root of $P'(x) = 0$. If $P(-1) < P(1)$, then in the interval $[-1, 1]$:
- (A) $P(-1)$ is the minimum and $P(1)$ is the maximum of P
- (B) $P(-1)$ is not minimum but $P(1)$ is the maximum of P
- (C) $P(-1)$ is the minimum and $P(1)$ is not the maximum of P
- (D) Neither $P(-1)$ is the minimum nor $P(1)$ is the maximum of P
71. Let $f: R \rightarrow R$ be defined by : $f(x) = \begin{cases} k - 2x & \text{if } x \leq -1 \\ 2x + 3 & \text{if } x > -1 \end{cases}$
- If f has a local minimum at $x = -1$, then a possible value of k is :
- (A) 0 (B) $-\frac{1}{2}$ (C) -1 (D) 1
72. A spherical balloon is filled with 4500π cubic meters of helium gas. If a leak in the balloon causes the gas escape at the rate of 72π cubic meters per minute, then the rate (in meters per minute) at which the radius of the balloon decreases 49 minutes after the leakage began is :
- (A) $\frac{9}{7}$ (B) $\frac{7}{9}$ (C) $\frac{2}{9}$ (D) $\frac{9}{2}$
73. The real number k for which the equation, $2x^3 + 3x + k = 0$ has two distinct real roots in $[0, 1]$:
- (A) Lies between 1 and 2 (B) Lies between 2 and 3
- (C) Lies between -1 and 0 (D) Does not exist
74. A spherical iron ball 10 cm in radius is coated with a layer of ice of uniform thickness that melts at a rate of $50 \text{ cm}^3/\text{min}$. When the thickness of ice is 5 cm, then the rate at which the thickness of ice decreases, is :
- (A) $\frac{5}{6\pi} \text{ cm/min}$ (B) $\frac{1}{54\pi} \text{ cm/min}$ (C) $\frac{1}{18\pi} \text{ cm/min}$ (D) $\frac{1}{36\pi} \text{ cm/min}$

75. A point on the parabola $y^2 = 18x$ at which the ordinate increases at twice the rate of the abscissa, is :
- (A) $(2, 4)$ (B) $(2, -4)$ (C) $\left(-\frac{9}{8}, \frac{9}{2}\right)$ (D) $\left(\frac{9}{8}, \frac{9}{2}\right)$
76. Let $f: R \rightarrow R$ be a continuous function defined by $f(x) = \frac{1}{e^x + 2e^{-x}}$
- Statement 1 :** $f(c) = \frac{1}{3}$, for some $c \in R$
- Statement 2 :** $0 < f(x) \leq \frac{1}{2\sqrt{2}}$ for all $x \in R$
- (A) Statement-1 is True, Statement-2 is True and Statement-2 is a correct explanation for Statement-1
 (B) Statement-1 is True, Statement-2 is True and Statement-2 is NOT a correct explanation for Statement-1
 (C) Statement-1 is True, Statement-2 is False
 (D) Statement-1 is False, Statement-2 is True
77. If f and g are differentiable function in $(0, 1)$ satisfying $f(0) = 2 = g(1)$, $g(0) = 0$ and $f(1) = 6$, then for some $c \in (0, 1)$
- (A) $2f''(c) = g''(c)$ (B) $2f'(c) = 3g'(c)$ (C) $f'(c) = g'(c)$ (D) $f'(c) = 2g'(c)$
78. If $x = -1$ and $x = 2$ are extreme points of $f(x) = \alpha \log|x| + \beta x^2 + x$, then :
- (A) $\alpha = -6, \beta = \frac{1}{2}$ (B) $\alpha = -6, \beta = -\frac{1}{2}$ (C) $\alpha = 2, \beta = -\frac{1}{2}$ (D) $\alpha = 2, \beta = \frac{1}{2}$
79. Let $a, b \in R$ be such that the function f given by $f(x) = \log|x| + bx^2 + ax$, $x \neq 0$ has extreme values at $x = -1$ and $x = 2$.
- Statement 1 : f has focal maximum at $x = -1$ and at $x = 2$.
- Statement 2 : $a = \frac{1}{2}$ and $b = \frac{-1}{2}$.
- (A) Statement-1 is True, Statement-2 is True and Statement-2 is a correct explanation for Statement-1
 (B) Statement-1 is True, Statement-2 is True and Statement-2 is NOT a correct explanation for Statement-1
 (C) Statement-1 is True, Statement-2 is False
 (D) Statement-1 is False, Statement-2 is True

Integer Answer Type Questions

The Answer to the following questions are positive integers of 1/2/3 digits and zero

80. Let $f(x)$ be a polynomial of degree four having extreme values at $x=1$ and $x=2$. If $\lim_{x \rightarrow 0} \left[1 + \frac{f(x)}{x^2} \right] = 3$, then $f(2)$ is equal to:
81. The number of points in $(-\infty, \infty)$ for which $x^2 - x \sin x - \cos x = 0$, is:
82. The number of values of x , where the function $f(x) = \cos x + \cos(\sqrt{2}x)$ attains its maximum, is :
83. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined as $f(x) = |x| + |x^2 - 1|$. The total number of points at which f attains either a local maximum or a local minimum is.....
84. Let $p(x)$ be a real polynomial of least degree which has a local maximum at $x=1$ and a local minimum at $x=3$. If $p(1) = 6$ and $p(3) = 2$, then $p'(0)$ is equal to.....
85. The number of distinct real roots of $x^4 - 4x^3 + 12x^2 + x - 1 = 0$ is
86. Let f be a function defined on $\mathbb{R}$ (the set of all real numbers) such that $f'(x) = 2010(x-2009)(x-2010)^2(x-2011)^3(x-2012)^4$, $\forall x \in \mathbb{R}$. If g is a function defined on $\mathbb{R}$ with values in the interval $(0, \infty)$ such that $f(x) = \ln(g(x))$, $\forall x \in \mathbb{R}$, then the number of points in $\mathbb{R}$ at which g has a local maximum is:
87. The maximum value of the expression $\frac{1}{\sin^2 \theta + 3 \sin \theta \cos \theta + 5 \cos^2 \theta}$ is
88. The maximum value of the function $f(x) = 2x^3 - 15x^2 + 36x - 48$ on the set $A = \{x \mid x^2 + 20 \leq 9x\}$ is:
89. Number of integers in the range of 'c' so that the equation $x^3 - 3x + c = 0$ has all its roots real and distinct, is:
90. The total number of point(s) of minima of the function, $f(x) = 4x^3 - x|x-2|$, $x \in [0, 3]$ is:
91. Number of solution of the equation $3 \tan x + x^3 = 2$ in $\left(0, \frac{\pi}{4}\right)$ is:
92. At the point $P(a, a^n)$ on the graph of $y = x^n$ ($n \in \mathbb{N}$) in the first quadrant a normal is drawn. The normal intersects the y-axis at the point $(0, b)$. If $\lim_{a \rightarrow 0} b = \frac{1}{2}$, then n equals:
93. P is a point on positive x-axis, Q is a point on the positive y-axis and 'O' is the origin. If the line passing through P and Q is tangent to the curve $y = 3 - x^2$ then the minimum area of the triangle OPQ, is:
94. Number of solutions of the equation $4x^2 e^{-x} - 1 = 0$, is:

Matrices & Determinants

CHOOSE THE CORRECT ALTERNATIVE. ONLY ONE CHOICE IS CORRECT.

- If $x \neq 0$, $y \neq 0$, $z \neq 0$ and $\begin{vmatrix} 1+x & 1 & 1 \\ 1+y & 1+2y & 1 \\ 1+z & 1+z & 1+3z \end{vmatrix} = 0$, then $x^{-1} + y^{-1} + z^{-1}$ is equal to :

(A) -1 (B) -2 (C) -3 (D) $1/3$
- If $f = \begin{vmatrix} x & y & z \\ x^2 & y^2 & z^2 \\ yz & zx & xy \end{vmatrix}$ and $g = (x-y)(y-z)(z-x)$, then $\frac{f}{g}$ is :

(A) $xy + yz + zx$ (B) $x^2 + y^2 + z^2$ (C) $x^2 + y^2 + z^2 - xy - yz - zx$ (D) None of these
- If $g(x) = \begin{vmatrix} a^{-x} & e^{x \log_e a} & x^2 \\ a^{-3x} & e^{3x \log_e a} & x^4 \\ a^{-5x} & e^{5x \log_e a} & 1 \end{vmatrix}$, then :

(A) $g(x) + g(-x) = 0$ (B) $g(x) - g(-x) = 0$ (C) $g(x) \times g(-x) = 0$ (D) None of these
- If $[p]$ stands for the greatest integer less than or equal to p , then in order that the set of equations : $2x - 2y = 4$, $7x - 3y = 2$, $[3\pi]x - [2e]y = [4a]$ may be consistent is that a should lie in :

(A) $\left[\frac{1}{2}, 1\right]$ (B) $[2, 3]$ (C) $\left[\frac{3}{2}, \frac{7}{4}\right]$ (D) $\left[1, \frac{3}{2}\right]$
- If $f(x)$, $g(x)$ and $h(x)$ are three polynomials of degree 3, then $\phi(x) = \begin{vmatrix} f'(x) & g'(x) & h'(x) \\ f''(x) & g''(x) & h''(x) \\ f'''(x) & g'''(x) & h'''(x) \end{vmatrix}$ is a polynomial of degree :

(A) 3 (B) 4 (C) 5 (D) None of these
- $\begin{vmatrix} x^2 + x & x+1 & x-2 \\ 2x^2 + 3x - 1 & 3x & 3x - 3 \\ x^2 + 2x + 3 & 2x - 1 & 2x - 1 \end{vmatrix} = Ax + B$ where A and B are determinants of order 3. Then $A + 2B$ is equal to :

(A) 0 (B) 4 (C) 1 (D) 2
- The value of a, b, c when $\begin{bmatrix} 0 & 2b & c \\ a & b & -c \\ a & -b & c \end{bmatrix}$ is orthogonal, are :

(A) $\pm \frac{1}{\sqrt{2}}, \pm \frac{1}{\sqrt{6}}, \pm \frac{1}{\sqrt{2}}$ (B) $\pm \frac{1}{\sqrt{3}}, \pm \frac{1}{\sqrt{2}}, \pm \frac{1}{\sqrt{6}}$
 (C) $\pm \frac{1}{\sqrt{2}}, \pm \frac{1}{\sqrt{6}}, \pm \frac{1}{\sqrt{3}}$ (D) $\pm \frac{1}{\sqrt{2}}, \pm \frac{1}{\sqrt{2}}, \pm \frac{1}{\sqrt{2}}$

8. If $A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ p & q & r \end{bmatrix}$ then $A^3 - rA^2 - qA$ equals :
 (A) pI (B) qI (C) rI (D) None of these
 where I is third order unit matrix.
9. If $A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$ and $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, then which one of the following holds for all $n \geq 1$?
 (A) $A^n = 2^{n-1}A + (n-1)I$ (B) $A^n = nA + (n-1)I$
 (C) $A^n = 2^{n-1}A - (n-1)I$ (D) $A^n = nA - (n-1)I$
10. A square matrix P satisfies $P^2 = I - P$, where I is the identity matrix. If $P^n = 5I - 8P$, then n is equal to
 (A) 4 (B) 5 (C) 6 (D) 7
11. If $3A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ x & 2 & y \end{bmatrix}$ and $AA^T = I$, then $x + y$ is equal to :
 (A) -3 (B) -2 (C) -1 (D) 0
12. If $P = \begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}$, $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $Q = PAP^T$, then $P^T(Q^{2005})P$ is equal to :
 (A) $\begin{bmatrix} 1 & 2005 \\ 0 & 1 \end{bmatrix}$ (B) $\begin{bmatrix} \frac{\sqrt{3}}{2} & 2005 \\ 1 & 0 \end{bmatrix}$ (C) $\begin{bmatrix} 1 & 2005 \\ \frac{\sqrt{3}}{2} & 1 \end{bmatrix}$ (D) $\begin{bmatrix} 1 & \frac{\sqrt{3}}{2} \\ 0 & 2005 \end{bmatrix}$
13. If $2x - y = \begin{bmatrix} 3 & -3 & 0 \\ 3 & 3 & 2 \end{bmatrix}$ and $2y + x = \begin{bmatrix} 4 & 1 & 5 \\ -1 & 4 & -4 \end{bmatrix}$, then :
 (A) $x + y = \begin{bmatrix} 3 & 0 & -1 \\ 0 & 3 & -2 \end{bmatrix}$ (B) $x = \begin{bmatrix} 2 & -1 & 1 \\ 1 & 2 & 0 \end{bmatrix}$
 (C) $x - y = \begin{bmatrix} 1 & -2 & 3 \\ 2 & 1 & 2 \end{bmatrix}$ (D) $y = \begin{bmatrix} 1 & 1 & -2 \\ -1 & 1 & -2 \end{bmatrix}$
14. The equation $2x + y = 5$, $x + 3y = 5$, $x - 2y = 0$ have :
 (A) No solution (B) One solution (C) Two solutions (D) Infinitely many solutions
15. If $f(x) = \begin{bmatrix} \cos x & -\sin x & 0 \\ \sin x & \cos x & 0 \\ 0 & 0 & 1 \end{bmatrix}$, then $f(x+y)$ is equal to :
 (A) $f(x) + f(y)$ (B) $f(x) - f(y)$ (C) $f(x) \cdot f(y)$ (D) None of these
16. If $AB = 0$ where $A = \begin{bmatrix} \cos^2 \theta & \cos \theta \sin \theta \\ \cos \theta \sin \theta & \sin^2 \theta \end{bmatrix}$ and $B = \begin{bmatrix} \cos^2 \phi & \cos \phi \sin \phi \\ \cos \phi \sin \phi & \sin^2 \phi \end{bmatrix}$, then $|\theta - \phi|$ can be equal to :
 (A) 0 (B) $\frac{\pi}{2}$ (C) $\frac{\pi}{4}$ (D) π

17. If $A = \begin{bmatrix} 1 & \omega & \omega^2 \\ \omega & \omega^2 & 1 \\ \omega^2 & 1 & \omega \end{bmatrix}$, $B = \begin{bmatrix} \omega & \omega^2 & 1 \\ \omega^2 & 1 & \omega \\ \omega & \omega^2 & 1 \end{bmatrix}$ and $C = \begin{bmatrix} 1 \\ \omega \\ \omega^2 \end{bmatrix}$ where ω is the non-real cube root of 1, then $(A+B)C$ is

equal to :

(A) $\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

(B) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

(C) $\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$

(D) $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

18. If $A = \begin{bmatrix} 0 & c & -b \\ -c & 0 & a \\ b & -a & 0 \end{bmatrix}$ and $B = \begin{bmatrix} a^2 & ab & ac \\ ba & b^2 & bc \\ ca & cb & c^2 \end{bmatrix}$ then AB is equal to :

(A) O

(B) I

(C) $2I$

(D) None of these

Paragraph for Questions 19 - 21

Consider the determinant, $\Delta = \begin{vmatrix} p & q & r \\ x & y & z \\ l & m & n \end{vmatrix}$. M_{ij} denotes the minor of an element in i^{th} row and j^{th} column and C_{ij} denotes the

cofactor of an element in i^{th} row and j^{th} column.

19. The value of $p \cdot C_{21} + q \cdot C_{22} + r \cdot C_{23}$ is equal to :

(A) 0

(B) $-\Delta$

(C) Δ

(D) Δ^2

20. The value of $x \cdot C_{21} + y \cdot C_{22} + z \cdot C_{23}$ is equal to :

(A) 0

(B) $-\Delta$

(C) Δ

(D) Δ^2

21. The value of $q \cdot M_{12} - y \cdot M_{22} + m \cdot M_{32}$ is equal to :

(A) 0

(B) $-\Delta$

(C) Δ

(D) Δ^2

22. If a, b, c are in A.P. and $f(x) = \begin{vmatrix} x+a & x^2+1 & 1 \\ x+b & 2x^2-1 & 1 \\ x+c & 3x^2-2 & 1 \end{vmatrix}$, then $f'(x)$ is :

(A) 0

(B) 1

(C) $a+bc$

(D) $\frac{abc}{a+b+c}$

23. If $\Delta = \begin{vmatrix} f(x) & f\left(\frac{1}{x}\right) + f(x) \\ 1 & f\left(\frac{1}{x}\right) \end{vmatrix} = 0$ where, $f(x) = a + bx^n$ and $f(2) = 17$, then $f(5)$ is :

(A) 126

(B) 326

(C) 428

(D) 626

24. The set of equations $\lambda x - y + (\cos \theta)z = 0$; $3x + y + 2z = 0$; $(\cos \theta)x + y + 2z = 0$, where $0 \leq \theta < 2\pi$, has non-trivial solutions.

(A) For no value of λ and θ

(B) For all value of λ and θ

(C) For all values of λ and only two value of θ

(D) For only one value of λ and all values of θ

25. The value of x , so that $\begin{bmatrix} 1 & x & 1 \end{bmatrix} \begin{bmatrix} 1 & 3 & 2 \\ 0 & 5 & 1 \\ 0 & 3 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ x \end{bmatrix} = 0$, is :
- (A) $\frac{-7 \pm \sqrt{35}}{2}$ (B) $\frac{-9 \pm \sqrt{53}}{2}$ (C) ± 2 (D) 0
26. The value of $\begin{vmatrix} 1 & 1 & 1 \\ (2^x + 2^{-x})^2 & (3^x + 3^{-x})^2 & (5^x + 5^{-x})^2 \\ (2^x - 2^{-x})^2 & (3^x - 3^{-x})^2 & (5^x - 5^{-x})^2 \end{vmatrix}$ is :
- (A) 0 (B) 30^x (C) 30^{-x} (D) None of these
27. If $A = \begin{bmatrix} 1 & -1 & 1 \\ 0 & 2 & -3 \\ 2 & 1 & 0 \end{bmatrix}$, $B = (\text{adj } A)$ and $C = 5A$, then $\frac{|\text{adj } B|}{|C|}$ is equal to :
- (A) 5 (B) 25 (C) -1 (D) 1
28. If $D = \text{diagonal } \{d_1, d_2, d_3, \dots, d_n\}$ where $d_i \neq 0 \forall i = 1, 2, 3, \dots, n$ then D^{-1} is equal to :
- (A) O (B) I_n
(C) $\text{Diagonal } (d_1^{-1}, d_2^{-1}, \dots, d_n^{-1})$ (D) None of these
29. If $a \neq p, b \neq q, c \neq r$ and the system of equations $px + by + cz = 0$, $ax + qy + cz = 0$, $ax + by + rz = 0$ has a non-zero solution, then value of $\frac{p}{p-a} + \frac{q}{q-b} + \frac{r}{r-c}$ is :
- (A) -1 (B) -2 (C) 1 (D) 2
30. A and B are square matrices and A is non-singular matrix, $(A^{-1}BA)^n$, $n \in I^+$, is equal to :
- (A) $A^{-n}B^nA^n$ (B) $A^nB^nA^{-n}$ (C) $A^{-1}B^nA$ (D) $A^{-n}BA^n$
31. Total number of possible matrices of order 3×3 with each entry 2 or 0 is :
- (A) 9 (B) 27 (C) 81 (D) 512
32. If A is matrix of order $m \times n$ and B is a matrix such that AB' and $B'A$ are both defined, then order of matrix B is :
- (A) $m \times m$ (B) $n \times n$ (C) $n \times m$ (D) $m \times n$
33. If A and B are matrices of same order, then $(AB' - BA')$ is a :
- (A) Skew symmetric matrix (B) Null matrix
(C) Symmetric matrix (D) Unit matrix
34. The value of determinant $\begin{vmatrix} a-b & b+c & a \\ b-a & c+a & b \\ c-a & a+b & c \end{vmatrix}$:
- (A) $a^3 + b^3 + c^3$ (B) $3bc$
(C) $a^3 + b^3 + c^3 - 3abc$ (D) None of these

35. The determinant $\begin{vmatrix} b^2 - ab & b - c & bc - ac \\ ab - a^2 & a - b & b^2 - ab \\ bc - ac & c - a & ab - a^2 \end{vmatrix}$ equals :
- (A) $abc(b-c)(c-a)(a-b)$ (B) $(b-c)(c-a)(a-b)$
 (C) $(a+b+c)(b-c)(c-a)(a-b)$ (D) None of these
36. Let $f(t) = \begin{vmatrix} \cos t & t & 1 \\ 2 \sin t & t & 2t \\ \sin t & t & t \end{vmatrix}$, then $\lim_{t \rightarrow 0} \frac{f(t)}{t^2}$ is equal to :
- (A) 0 (B) -1 (C) 2 (D) 3
37. The value of the determinant $\begin{vmatrix} x & x+y & x+2y \\ x+2y & x & x+y \\ x+y & x+2y & x \end{vmatrix}$ is :
- (A) $9x^2(x+y)$ (B) $9y^2(x+y)$ (C) $3y^2(x+y)$ (D) $7x^2(x+y)$
39. If A and B are two square matrices such that $B = -A^{-1}BA$, then $(A+B)^2$ is equal to :
- (A) $A^2 + B^2$ (B) 0 (C) $A^2 + 2AB + B^2$ (D) $A+B$
38. The matrix $A^2 + 4A - 5I$, where I is identity matrix and $A = \begin{bmatrix} 1 & 2 \\ 4 & -3 \end{bmatrix}$, equals :
- (A) $4 \begin{bmatrix} 2 & 1 \\ 2 & 0 \end{bmatrix}$ (B) $4 \begin{bmatrix} 0 & -1 \\ 2 & 2 \end{bmatrix}$ (C) $32 \begin{bmatrix} 2 & 1 \\ 2 & 0 \end{bmatrix}$ (D) $32 \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$
40. If A and B are square matrices of the same order such that $A^2 = A$, $B^2 = B$, $AB = BA = 0$, then :
- (A) $(A-B)^2 = B-A$ (B) $(A-B)^2 = A-B$
 (C) $(A+B)^2 = A+B$ (D) None of these
41. The system of homogeneous equations $tx + (t+1)y + (t-1)z = 0$, $(t+1)x + ty + (t+2)z = 0$ and $(t-1)x + (t+2)y + tz = 0$ has non-trivial solutions for :
- (A) Exactly three real values of t (B) Exactly two real values of t
 (C) Exactly one real value of t (D) Infinite number of values of t
42. If in a square matrix $A = [a_{ij}]$, we find that $a_{ij} = a_{ji} \forall i, j$ then A is a :
- (A) Symmetric matrix (B) Diagonal matrix
 (C) Orthogonal matrix (D) Skew-symmetric matrix
43. If a matrix $\begin{bmatrix} (x-a)^2 & (x-b)^2 & (x-c)^2 \\ (y-a)^2 & (y-b)^2 & (y-c)^2 \\ (z-a)^2 & (z-b)^2 & (z-c)^2 \end{bmatrix}$ is a zero matrix, then a, b, c, x, y, z are connected by :
- (A) $a+b+c=0, x+y+z=0$ (B) $a+b+c=0, x=y=z$
 (C) $a=b=c, x+y+z=0$ (D) None of these

44. If $A = \begin{bmatrix} 3 & 4 \\ 4 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} -2 & -2 \\ 0 & -2 \end{bmatrix}$, then $(A+B)^{-1}$ is :
- (A) A skew-symmetric matrix (B) $A^{-1} + B^{-1}$
 (C) Does not exist (D) None of these
45. If $|A|$ denotes the determinant of matrix $A = \begin{bmatrix} 1 & 1 & 1 \\ -1 & 1 & 1 \\ -1 & -1 & 1 \end{bmatrix}$, then $|5A|$ equals :
- (A) 4 (B) 0 (C) 20 (D) 500
46. If A and B are any 2×2 matrices, then $|A+B| = 0$ implies :
- (A) $|A| + |B| = 0$ (B) $|A| = 0$ or $|B| = 0$
 (C) $|A| = 0$ and $|B| = 0$ (D) None of these
47. If A is a 3×3 non-singular matrix, then $\det[adj A]$ is equal to :
- (A) $(\det A)^2$ (B) $(\det A)^3$ (C) $(\det A)$ (D) $(\det A)^{-1}$
48. The number of values of k for which the linear equations $4x + ky + 2z = 0$; $kx + 4y + z = 0$; $2x + 2y + z = 0$ possess a non-zero solution is :
- (A) 2 (B) 1 (C) zero (D) 3
49. Let $A = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{pmatrix}$. If u_1 and u_2 are column matrices such that $Au_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ and $Au_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$, then $u_1 + u_2$ is equal to :
- (A) $\begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}$ (B) $\begin{pmatrix} -1 \\ 1 \\ -1 \end{pmatrix}$ (C) $\begin{pmatrix} -1 \\ -1 \\ 0 \end{pmatrix}$ (D) $\begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix}$
50. Let P and Q be 3×3 matrices $P \neq Q$. If $P^3 = Q^3$ and $P^2Q = Q^2P$, then determinant of $(P^2 + Q^2)$ is equal to :
- (A) -2 (B) 1 (C) 0 (D) -1
51. If A is 3×3 non-singular matrix such that $AA' = A'A$ and $B = A^{-1}A'$, then BB' equals :
- (A) B^{-1} (B) $(B^{-1})'$ (C) $I + B$ (D) I
52. If $P = \begin{bmatrix} 1 & \alpha & 3 \\ 1 & 3 & 3 \\ 2 & 4 & 4 \end{bmatrix}$ is the adjoint of 3×3 matrix A and $|A| = 4$, then α is equal to :
- (A) 4 (B) 11 (C) 5 (D) 0
53. If the system of linear equations $x_1 + 2x_2 + 3x_3 = 6$, $x_1 + 3x_2 + 5x_3 = 9$, $2x_1 + 5x_2 + ax_3 = b$ is consistent and has infinite number of solutions, then :
- (A) $a = 8$, b can be any real number (B) $b = 15$, a can be any real number
 (C) $a \in \mathbb{R} - \{8\}$ and $b \in \mathbb{R} - \{15\}$ (D) $a = 8$, $b = 15$

54. The number of distinct real roots of the equation, $\begin{vmatrix} \cos x & \sin x & \sin x \\ \sin x & \cos x & \sin x \\ \sin x & \sin x & \cos x \end{vmatrix} = 0$ in the interval $\left[-\frac{\pi}{4}, \frac{\pi}{4}\right]$ is :
- (A) 4 (B) 3 (C) 2 (D) 1
55. Let A be 3×3 matrix such that $A^2 - 5A + 7I = O$.
- Statement-I :** $A^{-1} = \frac{1}{7}(5I - A)$.
- Statement-II :** The polynomial $A^3 - 2A^2 - 3A + I$ can be reduced to $5(A - 4I)$.
- Then :
- (A) Statement-I is true, but Statement-II is false (B) Statement-I is false, but Statement-II is true
(C) Both the statements are true (D) Both the statements are false
56. If $A = \begin{bmatrix} -4 & -1 \\ 3 & 1 \end{bmatrix}$, then the determinant of the matrix $(A^{2016} - 2A^{2015} - A^{2014})$ is :
- (A) 2014 (B) -174 (C) 2016 (D) -25
57. If p, q, r are 3 real numbers satisfying the matrix equation, $\begin{bmatrix} p & q & r \end{bmatrix} \begin{bmatrix} 3 & 4 & 1 \\ 3 & 2 & 3 \\ 2 & 0 & 2 \end{bmatrix} = \begin{bmatrix} 3 & 0 & 1 \end{bmatrix}$, then $2p + q - r$ equals:
- (A) -3 (B) -1 (C) 4 (D) 2
58. If $A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$, then which one of the following statements is not correct ?
- (A) $A^4 - I = A^2 + I$ (B) $A^3 - I = A(A - I)$
(C) $A^2 + I = A(A^2 - I)$ (D) $A^3 + I = A(A^3 - I)$
59. If A is a 3×3 matrix such that $|5 \cdot \text{adj } A| = 5$, then $|A|$ is equal to :
- (A) $\pm \frac{1}{5}$ (B) ± 5 (C) ± 1 (D) $\pm \frac{1}{25}$
60. If $\begin{vmatrix} x^2 + x & x + 1 & x - 2 \\ 2x^2 + 3x - 1 & 3x & 3x - 3 \\ x^2 + 2x + 3 & 2x - 1 & 2x - 1 \end{vmatrix} = ax - 12$, then 'a' is equal to :
- (A) 12 (B) 24 (C) -12 (D) -24
61. If a, b, c are non-zero real numbers and if the system of equations
 $(a-1)x = y + z, (b-1)y = z + x, (c-1)z = x + y$
 has a non-trivial solution, then $ab + bc + ca$ equals :
- (A) $a + b + c$ (B) abc (C) 1 (D) -1

62. If B is a 3×3 matrix such that $B^2 = 0$, then $\det[(I+B)^{50} - 50B]$ is equal to :
 (A) 1 (B) 2 (C) 3 (D) 50
63. Let A be a 3×3 matrix such that $A \begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & 3 \\ 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$. Then A^{-1} is :
 (A) $\begin{bmatrix} 3 & 1 & 2 \\ 3 & 0 & 2 \\ 1 & 0 & 1 \end{bmatrix}$ (B) $\begin{bmatrix} 3 & 2 & 1 \\ 3 & 2 & 0 \\ 1 & 1 & 0 \end{bmatrix}$ (C) $\begin{bmatrix} 0 & 1 & 3 \\ 0 & 2 & 3 \\ 1 & 1 & 1 \end{bmatrix}$ (D) $\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \\ 0 & 2 & 3 \end{bmatrix}$
64. Let for $i = 1, 2, 3$, $p_i(x)$ be a polynomial of degree 2 in x , $p_i'(x)$ and $p_i''(x)$ be the first and second order derivatives of $p_i(x)$ respectively. Let, $A(x) = \begin{bmatrix} p_1(x) & p_1'(x) & p_1''(x) \\ p_2(x) & p_2'(x) & p_2''(x) \\ p_3(x) & p_3'(x) & p_3''(x) \end{bmatrix}$ and $B(x) = [A(x)]^T A(x)$.
 Then determinant of $B(x)$:
 (A) is a polynomial of degree 6 in x (B) is a polynomial of degree 3 in x
 (C) is a polynomial of degree 2 in x (D) does not depend on x
65. If $A = \begin{bmatrix} 1 & 2 & x \\ 3 & -1 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} y \\ x \\ 1 \end{bmatrix}$ be such that $AB = \begin{bmatrix} 6 \\ 8 \end{bmatrix}$, then :
 (A) $y = -x$ (B) $y = x$ (C) $y = -2x$ (D) $y = 2x$
66. If $\begin{vmatrix} a^2 & b^2 & c^2 \\ (a+\lambda)^2 & (b+\lambda)^2 & (c+\lambda)^2 \\ (a-\lambda)^2 & (b-\lambda)^2 & (c-\lambda)^2 \end{vmatrix} = k\lambda \begin{vmatrix} a^2 & b^2 & c^2 \\ a & b & c \\ 1 & 1 & 1 \end{vmatrix}$, $\lambda \neq 0$, then k is equal to :
 (A) $4\lambda^2$ (B) $-4\lambda^2$ (C) $4\lambda abc$ (D) $-4\lambda abc$
67. Let A and B be any two 3×3 matrices. If A is symmetric and B is skew symmetric, then the matrix $AB - BA$ is :
 (A) skew symmetric (B) symmetric
 (C) neither symmetric nor skew symmetric (D) I or $-I$, where I is an identity matrix
68. If a, b, c are sides of a scalene triangle, then the value of $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$ is :
 (A) non-negative (B) negative (C) positive (D) non-positive

FOR QUESTIONS 69 - 70

- (A) Statement I is true; Statement II is false.
 (B) Statement I is true; Statement II is true; Statement II is not a correct explanation for Statement I.
 (C) Statement I is true; Statement II is true; Statement II is a correct explanation for Statement I.
 (D) Statement I is false; Statement II is true.
69. **Statement I :** The system of linear equations $x + (\sin \alpha)y + (\cos \alpha)z = 0$; $x - (\sin \alpha)y - (\cos \alpha)z = 0$ and $x + (\cos \alpha)y + (\sin \alpha)z = 0$ has a non-trivial solution for only one value of α lying in the interval $(0, \pi/2)$.

Statement II : The equation in α is $\begin{vmatrix} \cos \alpha & \sin \alpha & \cos \alpha \\ \sin \alpha & \cos \alpha & \sin \alpha \\ \cos \alpha & -\sin \alpha & -\cos \alpha \end{vmatrix} = 0$ has only one solution lying in the interval $(0, \pi/2)$.

70. Let A , other than I or $-I$, be a 2×2 real matrix such that $A^2 = I$, I being the unit matrix. Let $Tr(A)$ be the sum of diagonal elements of A .

Statement I : $Tr(A) = 0$

Statement II : $\det(A) = -1$

Integer Answer Type Questions

The Answer to the following questions are positive integers of 1/2/3 digits and zero

71. $A_{3 \times 3}$ is a matrix such that $|A| = a$, $B = adj(A)$ such that $|B| = b$ find the value of $(ab^2 + a^2b + 1)S$ where $\frac{S}{2} = \frac{a}{b} + \frac{a^2}{b^3} + \frac{a^3}{b^5} + \dots$ upto ∞ and $a = 3$. ($|A|$ denotes determinant of A)

72. If A and B are square matrix of order 3, where $|A| = 2$ and $|B| = 1$ then find $|A^{-1}adj(B^{-1})adj(2A^{-1})|$.

73. Let A be a 3×3 matrix such that $a_{11} = a_{33} = 2$ and all other $a_{ij} = 1$. Let $A^{-1} = xA^2 + yA + 2I$ then find the value of $x + y + z$ where I is unit matrix of order '3'.

74. Let $A = \begin{bmatrix} 2 & 0 & 7 \\ 0 & 1 & 0 \\ 1 & -2 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} -x & 14x & 7x \\ 0 & 1 & 0 \\ x & -4x & -2x \end{bmatrix}$ are two matrices such that $AB = (AB)^{-1}$ and $AB \neq I$ where I is an identify matrix of order 3×3 .

Find value of $Tr(AB + (AB)^2 + (AB)^3 + \dots + (AB)^{100})$ where $Tr(A)$ denotes the trace of matrix A .

75. If $A^{-1} = \begin{bmatrix} \sin^2 \alpha & 0 & 0 \\ 0 & \sin^2 \beta & 0 \\ 0 & 0 & \sin^2 \gamma \end{bmatrix}$ and $B^{-1} = \begin{bmatrix} \cos^2 \alpha & 0 & 0 \\ 0 & \cos^2 \beta & 0 \\ 0 & 0 & \cos^2 \gamma \end{bmatrix}$ where α, β, γ are real number and

$C = (A^{-3} + B^{-3}) + 3A^{-1}B^{-1}(B^{-1} + A^{-1})$ then value of determinant of matrix C .

76. If A is an idempotent matrix satisfying, $(I - 0.4A)^{-1} = I - \alpha A$, where I is the unit matrix of the same order as that of A , $\alpha \in \mathbb{R}$ then the value of $|9\alpha|$ is equal to _____.
77. Let $P = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 4 & 2 & 1 \end{bmatrix}$ and I be the identity matrix of order 3. If $Q = [q_{ij}]$ is matrix such that $P^{50} - Q = I$, then $\frac{q_{31} + q_{32}}{q_{21}}$ equals :
78. The total number of distinct $x \in \mathbb{R}$ for which $\begin{vmatrix} x & x^2 & 1+x^3 \\ 2x & 4x^2 & 1+8x^3 \\ 3x & 9x^2 & 1+27x^3 \end{vmatrix} = 10$ is :
79. Let M be a 3×3 matrix satisfying $M \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \\ 3 \end{bmatrix}$, $M \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$ and $M \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 12 \end{bmatrix}$. Then the sum of the diagonal entries of M is :
80. Given $x = cy + bz$, $y = az + cx$, $z = bx + ay$ where x, y, z are not all zero, then the value of $a^2 + b^2 + c^2 + 2abc$ is :
81. Absolute value of sum of roots of the equation $\begin{vmatrix} x+2 & 2x+3 & 3x+4 \\ 2x+3 & 3x+4 & 4x+5 \\ 3x+5 & 5x+8 & 10x+17 \end{vmatrix} = 0$ is _____.
82. If $\begin{vmatrix} x & x+y & x+y+z \\ 2x & 3x+2y & 4x+3y+2z \\ 3x & 6x+3y & 10x+6y+3z \end{vmatrix} = 64$, then the real value of x is _____.
83. $A = \begin{bmatrix} 0 & 1 \\ 3 & 0 \end{bmatrix}$ and $A^8 + A^6 + A^4 + A^2 + I) V = \begin{bmatrix} 0 \\ 11 \end{bmatrix}$ (where I is the 2×2 identity matrix), then the product of all elements of matrix V is _____.
84. If $\begin{bmatrix} a & b \\ c & 1-a \end{bmatrix}$ is an idempotent matrix and $f(x) = x - x^2$ and $bc = \frac{1}{4}$, then the value of $\frac{1}{f(a)}$ is _____.
85. Suppose A is any 3×3 non-singular matrix and $(A - 3I)(A - 5I) = O$, where $I = I_3$ and $O = O_3$. If $\alpha A + \beta A^{-1} = 4I$, then $\alpha + \beta$ is equal to :

Integral Calculus-1

CHOOSE THE CORRECT ALTERNATIVE. ONLY ONE CHOICE IS CORRECT. HOWEVER, QUESTIONS MARKED ‘*’ MAY HAVE MORE THAN ONE CORRECT OPTION.

1. If $\int \frac{dx}{\cos^3 x \sqrt{2 \sin 2x}} = (\tan x)^A + C (\tan x)^B + k$, where k is a constant of integration, then $A + B + C$ equals :

(A) $\frac{21}{5}$ (B) $\frac{16}{5}$ (C) $\frac{7}{10}$ (D) $\frac{27}{10}$

2. The integral $\int \frac{dx}{(1+\sqrt{x})\sqrt{x-x^2}}$ is equal to : (where C is a constant of integration)

(A) $-2\sqrt{\frac{1+\sqrt{x}}{1-\sqrt{x}}} + C$ (B) $-2\sqrt{\frac{1-\sqrt{x}}{1+\sqrt{x}}} + C$ (C) $-\sqrt{\frac{1-\sqrt{x}}{1+\sqrt{x}}} + C$ (D) $2\sqrt{\frac{1+\sqrt{x}}{1-\sqrt{x}}} + C$

3. The integral $\int \frac{dx}{(x+1)^{3/4}(x-2)^{5/4}}$ is equal to :

(A) $4\left(\frac{x+1}{x-2}\right)^{1/4} + C$ (B) $4\left(\frac{x-2}{x+1}\right)^{1/4} + C$ (C) $-\frac{4}{3}\left(\frac{x+1}{x-2}\right)^{1/4} + C$ (D) $-\frac{4}{3}\left(\frac{x-2}{x+1}\right)^{1/4} + C$

4. If $\int \frac{\log(t + \sqrt{1+t^2})}{\sqrt{1+t^2}} dt = \frac{1}{2}(g(t))^2 + C$, where C is a constant, the $g(2)$ is equal to :

(A) $2\log(2 + \sqrt{5})$ (B) $\log(2 + \sqrt{5})$ (C) $\frac{1}{\sqrt{5}}\log(2 + \sqrt{5})$ (D) $\frac{1}{2}\log(2 + \sqrt{5})$

5. $\int \frac{\sin^8 x - \cos^8 x}{(1 - 2\sin^2 x \cos^2 x)} dx$ is equal to :

(A) $\frac{1}{2}\sin 2x + c$ (B) $-\frac{1}{2}\sin 2x + c$ (C) $-\frac{1}{2}\sin x + c$ (D) $-\sin^2 x + c$

6. The integral $\int x \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right) dx (x > 0)$ is equal to :

(A) $-x + (1+x^2)\tan^{-1} x + c$ (B) $x - (1+x^2)\cot^{-1} x + c$
(C) $-x + (1+x^2)\cot^{-1} x + c$ (D) $x - (1+x^2)\tan^{-1} x + c$

7. The integral $\int \frac{\sin^2 x \cos^2 x}{(\sin^3 x + \cos^3 x)^2} dx$ is equal to :

(A) $\frac{1}{(1+\cot^3 x)} + c$ (B) $-\frac{\cos^3 x}{3(1+\sin^3 x)} + c$ (C) $-\frac{1}{3(1+\tan^3 x)} + c$ (D) $\frac{\sin^3 x}{(1+\cos^3 x)} + c$

8. If m is a non-zero number and $\int \frac{x^{5m-1} + 2x^{4m-1}}{(x^{2m} + x^m + 1)^3} dx = f(x) + c$, then $f(x)$ is :
- (A) $\frac{x^{5x}}{2m(x^{2m} + x^m + 1)^2}$ (B) $\frac{x^{4m}}{2m(x^{2m} + x^m + 1)^2}$
- (C) $\frac{2m(x^{5m} + x^{4m})}{(x^{2m} + x^m + 1)^2}$ (D) $\frac{(x^{5m} - x^{4m})}{2m(x^{2m} + x^m + 1)^2}$
9. If $\int \frac{dx}{x+x^7} = p(x)$ then, $\int \frac{x^6}{x+x^7} dx$ is equal to :
- (A) $\ln|x| - p(x) + c$ (B) $\ln|x| + p(x) + c$ (C) $x - p(x) + c$ (D) $x + p(x) + c$
10. If $\int \frac{x^2 - x + 1}{x^2 + 1} e^{\cot^{-1} x} dx = A(x) e^{\cot^{-1} x} + C$, then $A(x)$ is equal to :
- (A) $-x$ (B) x (C) $\sqrt{1-x}$ (D) $\sqrt{1+x}$
11. The integral $\int \frac{x dx}{2-x^2 + \sqrt{2-x^2}}$ equals :
- (A) $\log|1 + \sqrt{2+x^2}| + c$ (B) $-\log|1 + \sqrt{2-x^2}| + c$
- (C) $-x \log|1 - \sqrt{2-x^2}| + c$ (D) $x \log|1 - \sqrt{2+x^2}| + c$
12. The value of $\int \frac{dx}{\sqrt{x}(1+\sqrt{x})}$ is :
- (A) $2 \log|\sqrt{x}| + C$ (B) $2 \log|1 + \sqrt{x}| + C$
- (C) $\log|1 + \sqrt{x}| + C$ (D) None of these
13. The value of $\int \frac{1 + \cot x}{(x + \log \sin x)} dx$ is :
- (A) $\log|x + \log \sin x| + C$ (B) $\log|x - \log \sin x| + C$
- (C) $\log|x^2 + \log \sin x| + C$ (D) None of these
14. The value of $\int \frac{\log(\sin x)}{\tan x} dx$ is :
- (A) $\frac{[\log(\sin x)]^2}{2} + C$ (B) $\frac{[\log(\sin x)]}{2} + C$
- (C) $\frac{[(\sin x)]^2}{2} + C$ (D) None of these

15. The value of $\int x\sqrt{x+x^2} dx$ is :
- (A) $\frac{1}{3}(x^2+x)^{3/2} - \frac{1}{4}\left[\left(x+\frac{1}{2}\right)\sqrt{x^2+x} - \frac{1}{4}\log\left(x+\frac{1}{2}+\sqrt{x^2+x}\right)\right] + c$
- (B) $\frac{1}{3}(x^2+x)^{3/2} + \frac{1}{4}\left[\left(x+\frac{1}{2}\right)\sqrt{x^2+x} + \frac{1}{4}\log\left(x+\frac{1}{2}+\sqrt{x^2+x}\right)\right] + c$
- (C) $\frac{1}{3}(x^2+x)^{3/2} + \frac{1}{4}\left[\left(x^2+\frac{1}{2}\right)\sqrt{x^2+x} - \frac{1}{4}\log\left(x+\frac{1}{2}+\sqrt{x^2+x}\right)\right] + c$
- (D) None of these
16. The value of $\int \frac{dx}{\sin x \cdot \sin(x+\alpha)}$ is equal to :
- (A) $\operatorname{cosec} \alpha \ln \left| \frac{\sin x}{\sin(x+\alpha)} \right| + C$
- (B) $\operatorname{cosec} \alpha \ln \left| \frac{\sin(x+\alpha)}{\sin x} \right| + C$
- (C) $\operatorname{cosec} \alpha \ln \left| \frac{\sec(x+\alpha)}{\sec x} \right| + C$
- (D) $\operatorname{cosec} \alpha \ln \left| \frac{\sec x}{\sec(x+\alpha)} \right| + C$
17. If $\int (\sin 2x - \cos 2x) dx = \frac{1}{\sqrt{2}} \sin(2x-a) + b$, then :
- (A) $a = \frac{5\pi}{4}, b \in R$
- (B) $a = -\frac{5\pi}{4}, b \in R$
- (C) $a = \frac{\pi}{4}, b \in R$
- (D) None of these
18. The value of $\int \frac{\cos 2x}{\cos x} dx$ is equal to :
- (A) $2 \sin x - \ln |\sec x + \tan x| + C$
- (B) $2 \sin x - \ln |\sec x - \tan x| + C$
- (C) $2 \sin x + \ln |\sec x + \tan x| + C$
- (D) None of these
19. The value of $\int 5^{5^{5^x}} \cdot 5^{5^x} \cdot 5^x dx$ is equal to :
- (A) $\frac{5^{5^x}}{(\ln 5)^3} + C$
- (B) $5^{5^{5^x}} (\ln 5)^3 + C$
- (C) $\frac{5^{5^{5^x}}}{(\ln 5)^3} + C$
- (D) None of these
20. The value of $\int \frac{\sqrt{\tan x}}{\sin x \cos x} dx$ is equal to :
- (A) $2\sqrt{\tan x} + C$
- (B) $2\sqrt{\cot x} + C$
- (C) $\frac{\sqrt{\tan x}}{2} + C$
- (D) None of these
21. The value of $\int \tan^3 2x \sec 2x dx$ is equal to :
- (A) $\frac{1}{3} \sec^3 2x - \frac{1}{2} \sec 2x + C$
- (B) $-\frac{1}{6} \sec^3 2x - \frac{1}{2} \sec 2x + C$
- (C) $\frac{1}{6} \sec^3 2x - \frac{1}{2} \sec 2x + C$
- (D) $\frac{1}{3} \sec^3 2x + \frac{1}{2} \sec 2x + C$
22. The value of $\int \frac{\cos 2x}{(\sin x + \cos x)^2} dx$ is equal to :
- (A) $\frac{-1}{\sin x + \cos x} + C$
- (B) $\ln |(\sin x + \cos x)| + C$
- (C) $\ln |(\sin x - \cos x)| + C$
- (D) $\ln (\sin x + \cos x)^2 + C$

23. If $\int x^{13/2} \cdot (1+x^{5/2})^{1/2} dx = A(1+x^{5/2})^{7/2} + B(1+x^{5/2})^{5/2} + C(1+x^{5/2})^{3/2}$, then A, B and C are :
- (A) $A = \frac{4}{35}, B = -\frac{8}{25}, C = \frac{4}{15}$ (B) $A = \frac{4}{35}, B = \frac{8}{25}, C = \frac{4}{15}$
 (C) $A = -\frac{4}{35}, B = -\frac{8}{25}, C = \frac{4}{15}$ (D) $A = \frac{4}{35}, B = -\frac{8}{25}, C = -\frac{4}{15}$
- *24. If $\int e^{3x} \cos 4x dx = e^{3x} (A \sin 4x + B \cos 4x) + C$ then :
- (A) $4A = 3B$ (B) $2A = 3B$ (C) $3A = 4B$ (D) $4A + 3B = 1$
25. The value of $\int \frac{1}{x^2(x^4+1)^{3/4}} dx$ is equal to :
- (A) $\left(1 + \frac{1}{x^4}\right)^{1/4} + C$ (B) $(x^4+1)^{1/4} + C$ (C) $\left(1 - \frac{1}{x^4}\right)^{1/4} + C$ (D) $-\left(1 + \frac{1}{x^4}\right)^{1/4} + C$
26. If $\int \frac{dx}{x^4+x^3} = \frac{A}{x^2} + \frac{B}{x} + \ell n \left| \frac{x}{x+1} \right| + C$, then :
- (A) $A = \frac{1}{2}, B = 1$ (B) $A = 1, B = -\frac{1}{2}$ (C) $A = -\frac{1}{2}, B = 1$ (D) None of these
27. If $\int \sqrt{\frac{\cos^3 x}{\sin^{11} x}} dx = -2(A \tan^{-9/2} x + B \tan^{-5/2} x) + C$, then :
- (A) $A = \frac{1}{9}, B = \frac{-1}{5}$ (B) $A = \frac{1}{9}, B = \frac{1}{5}$ (C) $A = -\frac{1}{9}, B = \frac{1}{5}$ (D) None of these
28. The value of $\int \sqrt{\sec x - 1} dx$ is equal to :
- (A) $2\ell n \left(\cos \frac{x}{2} + \sqrt{\cos^2 \frac{x}{2} - \frac{1}{2}} \right) + C$ (B) $\ell n \left(\cos \frac{x}{2} + \sqrt{\cos^2 \frac{x}{2} - \frac{1}{2}} \right) + C$
 (C) $-2\ell n \left(\cos \frac{x}{2} + \sqrt{\cos^2 \frac{x}{2} - \frac{1}{2}} \right) + C$ (D) None of these
- *29. Let $f'(x) = 3x^2 \sin \frac{1}{x} - x \cos \frac{1}{x}$, if $x \neq 0$, $f(0) = 0$ and $f\left(\frac{1}{\pi}\right) = 0$ then :
- (A) $f(x)$ is continuous at $x = 0$ (B) $f(x)$ is non-derivable at $x = 0$
 (C) $f'(x)$ is continuous at $x = 0$ (D) $f'(x)$ is non-derivable at $x = 0$
30. **Statement 1 :** If $x > 0, x \neq 1$ then $\int (\log_x e - (\log_x e)^2) dx = x \log_x e + C$
Statement 2 : $\int e^x (f(x) + f'(x)) dx = e^x f(x) + C$ and $e^t = x$ iff $t = \ell n x$
- (A) Statement-1 is True, Statement-2 is True and Statement-2 is a correct explanation for Statement-1
 (B) Statement-1 is True, Statement-2 is True and Statement-2 is NOT a correct explanation for Statement-1
 (C) Statement-1 is True, Statement-2 is False
 (D) Statement-1 is False, Statement-2 is True

31. The value of $\int \tan x \cdot \tan 2x \cdot \tan 3x dx$ is :
- (A) $-\ln|\sec x| + \frac{1}{3}\ln|\sec 3x| - \frac{1}{2}\ln|\sec 2x| + C$ (B) $\ln|\sec x| + \frac{1}{3}\ln|\sec 3x| - \frac{1}{2}\ln|\sec 2x| + C$
- (C) $\ln|\sec x| - \frac{1}{3}\ln|\sec 3x| + \frac{1}{2}\ln|\sec 2x| + C$ (D) None of these
32. The value of $\int e^{\sin x} \frac{x \cos^3 x - \sin x}{\cos^2 x} dx$ is :
- (A) $e^{\sin x}(x + \sec x) + C$ (B) $e^{\sin x}(x + \cos x) + C$
- (C) $e^{\sin x}(x - \sec x) + C$ (D) None of these
33. The value of $\int \frac{\sqrt{4+x^2}}{x^6} dx$ is $\frac{1}{120} \left(\frac{(x^2+4)^{b/a} (x^2-6)}{x^5} \right) + C$ then the value of $a + b$ is :
- (A) 5 (B) 2 (C) 3 (D) None of these
34. The value of $\int \frac{1+x \cos x}{x(1-x^2 e^{2 \sin x})} \cdot dx$ is :
- (A) $\ln|xe^{\sin x}| + \frac{1}{2} \log|1+x^2 e^{2 \sin x}| + K$ (B) $\ln|xe^{\sin x}| - \frac{1}{2} \log|1-x^2 e^{2 \sin x}| + K$
- (C) $\ln|xe^{\sin x}| - \frac{1}{2} \log|1+x e^{2 \sin x}| + K$ (D) None of these
35. If $f(x) = \int \frac{2 \sin x - \sin 2x}{x^3} dx$, where $x \neq 0$, then $\lim_{x \rightarrow 0} f'(x)$ has the value :
- (A) 0 (B) 1 (C) 2 (D) Not defined
36. The value of $\int \frac{e^{\sqrt{x}}}{\sqrt{x}} (x + \sqrt{x}) dx$ is equal to :
- (A) $2e^{\sqrt{x}} [\sqrt{x} - x + 1] + C$ (B) $2e^{\sqrt{x}} [x - 2\sqrt{x} + 1] + C$
- (C) $2e^{\sqrt{x}} [x - \sqrt{x} + 1] + C$ (D) $2e^{\sqrt{x}} (x + \sqrt{x} + 1) + C$
37. The value of $\int e^{\tan \theta} (\sec \theta - \sin \theta) d\theta$ is equal to :
- (A) $-e^{\tan \theta} \sin \theta + C$ (B) $e^{\tan \theta} \sin \theta + C$ (C) $e^{\tan \theta} \sec \theta + C$ (D) $e^{\tan \theta} \cos \theta + C$
38. The value of $\int \frac{1-x^7}{x(1+x^7)} dx$ is equal to :
- (A) $\ln|x| + \frac{2}{7} \ln|1+x^7| + C$ (B) $\ln|x| - \frac{2}{7} \ln|1-x^7| + C$
- (C) $\ln|x| - \frac{2}{7} \ln|1+x^7| + C$ (D) $\ln|x| + \frac{2}{7} \ln|1-x^7| + C$
39. The value of $\int (x e^{\ln \sin x} - \cos x) dx$ is equal to :
- (A) $x \cos x + C$ (B) $\sin x - x \cos x + C$
- (C) $-e^{\ln x} \cos x + C$ (D) $\sin x + x \cos x + C$

40. The value of $\int \left\{ \ln(1 + \sin x) + x \tan\left(\frac{\pi}{4} - \frac{x}{2}\right) \right\} dx$ is equal to :
- (A) $x \ln(1 + \sin x) + C$ (B) $\ln(1 + \sin x) + C$ (C) $-x \ln(1 + \sin x) + C$ (D) $\ln(1 - \sin x) + C$

41. The value of $\int \sqrt{\frac{x-1}{x+1}} \cdot \frac{1}{x^2} dx$ is equal to :

- (A) $\sin^{-1} \frac{1}{x} + \frac{\sqrt{x^2-1}}{x} + C$ (B) $\frac{\sqrt{x^2-1}}{x} + \cos^{-1} \frac{1}{x} + C$
- (C) $\sec^{-1} x - \frac{\sqrt{x^2-1}}{x} + C$ (D) $\tan^{-1} \sqrt{x^2+1} - \frac{\sqrt{x^2-1}}{x} + C$

42. Consider the following statements :

S₁ : The anti-derivative of every even function is an odd function.

S₂ : Primitive of $\frac{3x^4-1}{(x^4+x+1)^2}$ w.r.t. x is $\frac{x}{x^4+x+1} + C$ S₃ : $\int \frac{1}{\sqrt{\sin^3 x \cos x}} dx = \frac{-2}{\sqrt{\tan x}} + C$

S₄ : The value of $\int \left(\sqrt{\frac{a+x}{a-x}} - \sqrt{\frac{a-x}{a+x}} \right) dx$ is equal to $-2\sqrt{a^2-x^2} + C$

State, in order, whether S₁, S₂, S₃, S₄ are true or false :

- (A) FTTT (B) TTTT (C) FFFF (D) TFTF

- *43. The value of $\int \frac{\sin 2x}{\sin^4 x + \cos^4 x} dx$ is equal to :

- (A) $\cot^{-1}(\cot^2 x) + C$ (B) $-\cot^{-1}(\tan^2 x) + C$
- (C) $\tan^{-1}(\tan^2 x) + C$ (D) $-\tan^{-1}(\cos 2x) + C$

- *44. The value of $\int \frac{dx}{\sqrt{x-x^2}}$ is equal to :

- (A) $2 \sin^{-1} \sqrt{x} + C$ (B) $\sin^{-1}(2x-1) + C$
- (C) $C - 2 \cos^{-1}(2x-1)$ (D) $\cos^{-1} 2\sqrt{x-x^2} + C$

- *45. If $I_n = \int \cot^n x dx$ and $I_0 + I_1 + 2(I_2 + \dots + I_8) + I_9 + I_{10} = A \left(u + \frac{u^2}{2} + \dots + \frac{u^9}{9} \right) + C$, where $u = \cot x$ and C is an arbitrary constant, then :

- (A) A is constant (B) $A = -1$ (C) $A = 1$ (D) A is dependent on x

PARAGRAPH FOR QUESTIONS 46 - 48

It is known that $\sqrt{\tan x} + \sqrt{\cot x} = \begin{cases} \frac{\sqrt{\sin x}}{\sqrt{\cos x}} + \frac{\sqrt{\cos x}}{\sqrt{\sin x}} & \text{if } 0 < x < \frac{\pi}{2} \\ \frac{\sqrt{-\sin x}}{\sqrt{-\cos x}} + \frac{\sqrt{-\cos x}}{\sqrt{-\sin x}} & \text{if } \pi < x < \frac{3\pi}{2} \end{cases}$

$$\frac{d}{dx}(\sqrt{\tan x} - \sqrt{\cot x}) = \frac{1}{2}(\sqrt{\tan x} + \sqrt{\cot x})(\tan x + \cot x), \forall x \in \left(0, \frac{\pi}{2}\right) \cup \left(\pi, \frac{3\pi}{2}\right)$$

and $\frac{d}{dx}(\sqrt{\tan x} + \sqrt{\cot x}) = \frac{1}{2}(\sqrt{\tan x} - \sqrt{\cot x})(\tan x + \cot x), \forall x \in \left(0, \frac{\pi}{2}\right) \cup \left(\pi, \frac{3\pi}{2}\right)$

46. Value of integral $I = \int (\sqrt{\tan x} + \sqrt{\cot x}) dx$, where $x \in \left(0, \frac{\pi}{2}\right) \cup \left(\pi, \frac{3\pi}{2}\right)$ is :
- (A) $\sqrt{2} \tan^{-1} \left(\frac{\sqrt{\tan x} - \sqrt{\cot x}}{\sqrt{2}} \right) + C$ (B) $\sqrt{2} \tan^{-1} \left(\frac{\sqrt{\tan x} + \sqrt{\cot x}}{\sqrt{2}} \right) + C$
 (C) $-\sqrt{2} \tan^{-1} \left(\frac{\sqrt{\tan x} - \sqrt{\cot x}}{\sqrt{2}} \right) + C$ (D) $-\sqrt{2} \tan^{-1} \left(\frac{\sqrt{\tan x} + \sqrt{\cot x}}{\sqrt{2}} \right) + C$
47. Value of the integral $I = \int (\sqrt{\tan x} + \sqrt{\cot x}) dx$, where $x \in \left(0, \frac{\pi}{2}\right)$ is :
- (A) $\sqrt{2} \sin^{-1} (\cos x - \sin x) + C$ (B) $\sqrt{2} \sin^{-1} (\sin x - \cos x) + C$
 (C) $\sqrt{2} \sin^{-1} (\sin x + \cos x) + C$ (D) $-\sqrt{2} \sin^{-1} (\sin x + \cos x) + C$
48. Value of the integral $I = \int (\sqrt{\tan x} + \sqrt{\cot x}) dx$, where $x \in \left(\pi, \frac{3\pi}{2}\right)$, is :
- (A) $\sqrt{2} \sin^{-1} (\cos x - \sin x) + C$ (B) $\sqrt{2} \sin^{-1} (\sin x - \cos x) + C$
 (C) $\sqrt{2} \sin^{-1} (\sin x + \cos x) + C$ (D) $-\sqrt{2} \sin^{-1} (\sin x + \cos x) + C$
49. For any natural number m , the value of $\int (x^{3m} + x^{2m} + x^m) (2x^{2m} + 3x^m + 6)^{1/m} dx$, $x > 0$ is :
- (A) $\frac{1}{6(m+1)} \cdot (2x^{3m} + 3x^{2m} + 6x^m)^{(m+1)/m} + C$ (B) $\frac{1}{6(m+1)} \cdot (2x^{3m} + 3x^{2m} + 6x^m)^{(m+1)} + C$
 (C) $\frac{1}{6(m+1)} \cdot (2x^{3m} + 3x^{2m} + 6x^m)^{m/m+1} + C$ (D) None of these
50. Let $f(x) = \frac{x}{(1+x^n)^{1/n}}$ for $n \geq 2$ and $g(x) = \underbrace{(f \circ f \circ \dots \circ f)}_{f \text{ occurs } n \text{ times}}(x)$. Then $\int x^{n-2} g(x) dx$ equals :
- (A) $\frac{1}{n(n-1)} (1+nx^n)^{1-\frac{1}{n}} + K$ (B) $\frac{1}{(n-1)} (1+nx^n)^{1-\frac{1}{n}} + K$
 (C) $\frac{1}{n(n+1)} (1+nx^n)^{1+\frac{1}{n}} + K$ (D) $\frac{1}{(n+1)} (1+nx^n)^{1+\frac{1}{n}} + K$
51. The integral $\int \frac{\sec^2 x}{(\sec x + \tan x)^{9/2}} dx$ equals (for some arbitrary constant K)
- (A) $\frac{-1}{(\sec x + \tan x)^{11/2}} \left\{ \frac{1}{11} - \frac{1}{7} (\sec x + \tan x)^2 \right\} + K$ (B) $\frac{1}{(\sec x + \tan x)^{11/2}} \left\{ \frac{1}{11} - \frac{1}{7} (\sec x + \tan x)^2 \right\} + K$
 (C) $\frac{-1}{(\sec x + \tan x)^{11/2}} \left\{ \frac{1}{11} + \frac{1}{7} (\sec x + \tan x)^2 \right\} + K$ (D) $\frac{1}{(\sec x + \tan x)^{11/2}} \left\{ \frac{1}{11} + \frac{1}{7} (\sec x + \tan x)^2 \right\} + K$
52. $\int \frac{dx}{x(x^n+1)}$ is equal to :
- (A) $\frac{1}{n} \log \left| \frac{x^n}{x^n+1} \right| + c$ (B) $\frac{1}{n} \log \left| \frac{x^n+1}{x^n} \right| + c$ (C) $\log \left| \frac{x^n}{x^n+1} \right| + c$ (D) None of these

53. If $\int \frac{\sin x}{\sin(x-\alpha)} dx = Ax + B \log \sin(x-\alpha) + c$, then value of (A, B) is :
 (A) $(\sin \alpha, \cos \alpha)$ (B) $(\cos \alpha, \sin \alpha)$ (C) $(-\sin \alpha, \cos \alpha)$ (D) $(-\cos \alpha, \sin \alpha)$
54. $\int \frac{dx}{\cos x - \sin x}$, is equal to :
 (A) $\frac{1}{\sqrt{2}} \log \left| \tan \left(\frac{x}{2} - \frac{\pi}{8} \right) \right| + c$ (B) $\frac{1}{\sqrt{2}} \log \left| \cot \left(\frac{x}{2} \right) \right| + c$
 (C) $\frac{1}{\sqrt{2}} \log \left| \tan \left(\frac{x}{2} - \frac{3\pi}{8} \right) \right| + c$ (D) $\frac{1}{\sqrt{2}} \log \left| \tan \left(\frac{x}{2} + \frac{3\pi}{8} \right) \right| + c$
55. $\int \left\{ \frac{(\log x - 1)}{1 + (\log x)^2} \right\}^2 dx$ is equal to :
 (A) $\frac{x}{(\log x)^2 + 1} + c$ (B) $\frac{xe^x}{1 + x^2} + c$ (C) $\frac{x}{x^2 + 1} + c$ (D) $\frac{\log x}{(\log x)^2 + 1}$
56. $\int \frac{dx}{\cos x + \sqrt{3} \sin x}$, equals :
 (A) $\frac{1}{2} \log \tan \left(\frac{x}{2} + \frac{\pi}{12} \right) + c$ (B) $\frac{1}{2} \log \tan \left(\frac{x}{2} - \frac{\pi}{12} \right) + c$
 (C) $\log \tan \left(\frac{x}{2} + \frac{\pi}{12} \right) + c$ (D) $\log \tan \left(\frac{x}{2} - \frac{\pi}{12} \right) + c$
57. The value of $\sqrt{2} \int \frac{\sin x dx}{\sin \left(x - \frac{\pi}{4} \right)}$ is :
 (A) $x + \log \left| \cos \left(x - \frac{\pi}{4} \right) \right| + c$ (B) $x - \log \left| \sin \left(x - \frac{\pi}{4} \right) \right| + x$
 (C) $x + \log \left| \sin \left(x - \frac{\pi}{4} \right) \right| + c$ (D) $x - \log \left| \cos \left(x - \frac{\pi}{4} \right) \right| + c$
58. If the integral $\int \frac{5 \tan x}{\tan x - 2} dx = x + a \ln |\sin x - 2 \cos x| + k$ then a is equal to :
 (A) -1 (B) -2 (C) 1 (D) 2
59. If $\int f(x) dx = \psi(x)$, then $\int x^5 f(x^3) dx$ is equals to :
 (A) $\frac{1}{3} \left[x^3 \psi(x^3) - \int x^2 \psi(x^3) dx \right] + C$ (B) $\frac{1}{3} x^3 \psi(x^3) - 3 \int x^3 \psi(x^3) dx + C$
 (C) $\frac{1}{3} x^3 \psi(x^3) - \int x^2 \psi(x^3) dx + C$ (D) $\frac{1}{3} \left[x^3 \psi(x^3) - \int x^3 \psi(x^3) dx \right] + C$
60. The integral $\int \left(1 + x - \frac{1}{x} \right) e^{x + \frac{1}{x}}$ is equal to :
 (A) $(x-1)e^{x + \frac{1}{x}} + C$ (B) $xe^{x + \frac{1}{x}} + C$ (C) $(x+1)e^{x + \frac{1}{x}} + C$ (D) $-xe^{x + \frac{1}{x}} + C$

Integer Answer Type Questions

The Answer to the following questions are positive integers of 1/2/3 digits and zero

61. Let $f(x)$ be a polynomial of degree three satisfying $f(0) = -1$ and $f(1) = 0$. Also, 0 is a stationary point of $f(x)$. If $f(x)$ does not have an extremum at $x = 0$, then $\int \frac{f(x)}{x^3 - 1} dx = kh + C$ then k is.
62. If it is known that at the point $x = 1$ two anti-derivatives of $f(x) = e^x$ differ by 2, the difference of these anti-derivatives at $x = 100$ is:
63. If $\int \frac{1}{x} \log_{ex} e \cdot \log_{e^2 x} e dx = \log \left(\frac{1 + f(x)}{2 + f(x)} \right) + C$ then $\lim_{x \rightarrow 0} \frac{f(1+x)}{x}$ equals:
64. If $f(x) = \frac{1}{\cos^2 x \sqrt{1 + \tan x}}$, then its anti-derivative $F(x)$ satisfying $F(0) = 4$ is $A(\sqrt{1 + \tan x} + B)$ then $A + B$ is:
65. If $\int \frac{x dx}{x^2 - 4x + 8} = k \log(x^2 - 4x + 8) + \tan^{-1} \left(\frac{x-2}{2} \right) + C$, then the value of $2k$ is :
66. $\int \sqrt{1 + \sec x} dx$ is equal to $\frac{k}{101} \tan^{-1} \sqrt{\frac{1 - \cos x}{\cos x}} + C$ then k is :
67. If $f(x) = \int \frac{2 \sin x - \sin 2x}{x^3} dx$, $x \neq 0$ then $\lim_{x \rightarrow 0} f'(x)$ is equal to:
68. $\int \sqrt[3]{x} \sqrt[7]{1 + \sqrt[3]{x^4}} dx$ is equal to $\frac{a}{32} (1 + \sqrt[3]{x^6})^{8/c} + C$ then $a + b + c$ equal to :
69. If $f(x) = \int \frac{(5x^4 + 4x^5)}{(x^5 + x + 1)^2} dx$ and $f(0) = 0$, then the value of $30 f(1)$ is :
70. If $\int \frac{\sqrt{\sin x}}{\cos^{9/2} x} dx = \frac{2}{3} \tan^{a/4} x + \frac{1}{7} \tan^{b/4} x + C$ then exponent of 2 in $(a + b)!$ is :
71. If $\int \sin x \cdot \cos x \cdot \cos 2x \cdot \cos 4x \cdot \cos 8x \cdot \cos 16x dx$ is $-\frac{\cos 32x}{k} + C$, then k is :
72. If $\int \frac{x + (\cos^{-1} 3x)^2}{\sqrt{1 - 9x^2}} dx = A\sqrt{1 - 9x^2} + B(\cos^{-1} 3x)^3 + C$, then $A - B$ is :
73. If $\int x^{26} \cdot (x-1)^{17} \cdot (5x-3) dx = \frac{x^{27} \cdot (x-1)^{18}}{k} + C$ where C is a constant of integration, then the value of k is equal to:
74. Let $f(x)$ is a quadratic function such that $f(0) = 1$ and $\int \frac{f(x) dx}{x^2(x+1)^3}$ is a rational function, find the value of $f'(0)$
75. If $\int \frac{(x - x^3)^{1/3}}{x^4} dx = a \left(\frac{1}{x^2} - 1 \right)^b + c$, then the value of $\frac{1}{|ab|}$ is :

Integral Calculus-2

CHOOSE THE CORRECT ALTERNATIVE. ONLY ONE CHOICE IS CORRECT. HOWEVER, QUESTIONS MARKED ‘*’ MAY HAVE MORE THAN ONE CORRECT OPTION.

- If $2 \int_0^1 \tan^{-1} x \, dx = \int_0^1 \cot^{-1}(1-x+x^2) \, dx$, then $\int_0^1 \tan^{-1}(1-x+x^2) \, dx$ is equal to :

(A) $\log 4$ (B) $\frac{\pi}{2} + \log 2$ (C) $\log 2$ (D) $\frac{\pi}{2} - \log 4$
- The area (in sq. units) of the region described by $A = \{(x, y) \mid y \geq x^2 - 5x + 4, x + y \geq 1, y \leq 0\}$ is :

(A) $\frac{7}{2}$ (B) $\frac{19}{6}$ (C) $\frac{13}{6}$ (D) $\frac{17}{6}$
- The value of the integral $\int_4^{10} \frac{[x^2] \, dx}{[x^2 - 28x + 196] + [x^2]}$, where $[x]$ denotes the greatest integer less than or equal to x , is

(A) 6 (B) 3 (C) 7 (D) $\frac{1}{3}$
- For $x \in \mathbb{R}, x \neq 0$, if $y(x)$ is a differential function such that $x \int_1^x y(t) \, dt = (x+1) \int_1^x t y(t) \, dt$, then $y(x)$ equals : (where C is a constant.)

(A) $\frac{C}{x} e^{-\frac{1}{x}}$ (B) $\frac{C}{x^2} e^{-\frac{1}{x}}$ (C) $\frac{C}{x^3} e^{-\frac{1}{x}}$ (D) $Cx^3 e^{\frac{1}{x}}$
- For $x > 0$, let $f(x) = \int_1^x \frac{\log t}{1+t} \, dt$. Then $f(x) + f\left(\frac{1}{x}\right)$ is equal to :

(A) $\frac{1}{4}(\log x)^2$ (B) $\frac{1}{2}(\log x)^2$ (C) $\log x$ (D) $\frac{1}{4} \log x^2$
- The area (in square units) of the region bounded by the curves $y + 2x^2 = 0$ and $y + 3x^2 = 1$, is equal to :

(A) $\frac{3}{5}$ (B) $\frac{3}{4}$ (C) $\frac{1}{3}$ (D) $\frac{4}{3}$
- Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function such that $f(2-x) = f(2+x)$ and $f(4-x) = f(4+x)$, for all $x \in \mathbb{R}$ and $\int_0^2 f(x) \, dx = 5$. Then the value of $\int_{10}^{50} f(x) \, dx$ is :

(A) 80 (B) 100 (C) 125 (D) 200
- Let $f: (-1, 1) \rightarrow \mathbb{R}$ be a continuous function. If $\int_0^{\sin x} f(t) \, dt = \frac{\sqrt{3}}{2}x$, then $f\left(\frac{\sqrt{3}}{2}\right)$ is equal to :

(A) $\frac{\sqrt{3}}{2}$ (B) $\sqrt{3}$ (C) $\sqrt{\frac{3}{2}}$ (D) $\frac{1}{2}$
- The integral $\int_0^{\frac{1}{2}} \frac{\ln(1+2x)}{1+4x^2} \, dx$, equals :

(A) $\frac{\pi}{4} \ln 2$ (B) $\frac{\pi}{8} \ln 2$ (C) $\frac{\pi}{16} \ln 2$ (D) $\frac{\pi}{32} \ln 2$

10. If for $n \geq 1$, $P_n = \int_1^e (\log x)^n dx$, then $P_{10} - 90P_8$ is equal to :
 (A) -9 (B) $10e$ (C) $-9e$ (D) 10
11. If $[]$ denotes the greatest integer function, then the integral $\int_0^\pi [\cos x] dx$ is equal to :
 (A) $-\frac{\pi}{2}$ (B) 0 (C) $\frac{\pi}{2}$ (D) -1
12. If for a continuous function $f(x)$, $\int_{-\pi}^t (f(x) + x) dx = \pi^2 - t^2$, for all $t \geq -\pi$, then $f\left(-\frac{\pi}{3}\right)$ is equal to :
 (A) π (B) $\pi/3$ (C) $\pi/2$ (D) $\pi/6$
13. Let function F be defined as $F(x) = \int_1^x \frac{e^t}{t} dt$, $x > 0$ then the value of the integral $\int_1^x \frac{e^t}{t+a} dt$, where $a > 0$, is :
 (A) $e^a [F(x) - F(1+a)]$ (B) $e^{-a} [F(x+a) - F(a)]$
 (C) $e^a [F(x+a) - F(1+a)]$ (D) $e^{-a} [F(x+a) - F(1+a)]$
14. If $x = \int_0^y \frac{dt}{\sqrt{1+t^2}}$, then $\frac{d^2y}{dx^2}$ is equal to :
 (A) y (B) $\sqrt{1+y^2}$ (C) $y/\sqrt{1+y^2}$ (D) y^2
15. The area bounded by the curve $y = \ln(x)$ and the lines $y = 0$, $y = \ln(3)$ and $x = 0$ is equal to :
 (A) 3 (B) $3\ln(3) - 2$ (C) $3\ln(3) + 2$ (D) 2
16. The area of the origin region (in sq. units), in the quadrant, bounded by the parabola $y = 9x^2$ and the lines $x = 0$, $y = 1$ and $y = 4$, is :
 (A) $7/9$ (B) $14/3$ (C) $7/3$ (D) $14/9$
17. The integral $\int_{7\pi/4}^{7\pi/3} \sqrt{\tan^2 x} dx$ is equal to :
 (A) $\log 2\sqrt{2}$ (B) $\log 2$ (C) $2\log 2$ (D) $\log \sqrt{2}$
18. The value of $\int_{-\pi/2}^{\pi/2} \frac{\sin^2 x}{1+2^x} dx$ is :
 (A) π (B) $\pi/2$ (C) 4π (D) $\pi/4$
19. The area under the curve $y = |\cos x - \sin x|$, $0 \leq x \leq \frac{\pi}{2}$, and above x-axis is :
 (A) $2\sqrt{2}$ (B) $2\sqrt{2} - 2$ (C) $2\sqrt{2} + 2$ (D) 0
20. The value of $\int_{-1}^2 \{2x\} dx$ (where function $\{.\}$ denotes fractional part function) is :
 (A) $1/2$ (B) 1 (C) $3/2$ (D) 0

21. The value of $\int_0^{10\pi} (|\sin x| + |\cos x|) dx$ is :
 (A) 20 (B) 40 (C) 30 (D) None of these
22. If $a \leq \int_1^3 \sqrt{3+x^3} \leq b$ then value of a and b are :
 (A) $a=3, b=\sqrt{30}$ (B) $a=3, b=2\sqrt{30}$ (C) $a=4, b=2\sqrt{30}$ (D) None of these
23. The value of $\lim_{n \rightarrow \infty} \frac{3}{n} \left[1 + \sqrt{\frac{n}{n+3}} + \sqrt{\frac{n}{n+6}} + \sqrt{\frac{n}{n+9}} + \dots + \sqrt{\frac{n}{n+3(n-1)}} \right]$ is :
 (A) π (B) $-\frac{\pi}{2}$ (C) $\frac{\pi}{2}$ (D) None of these
24. Let $f(x) = \text{Maximum} \{x^2, (1-x)^2, 2x(1-x)\}$ where $0 \leq x \leq 1$. The area of the region bounded by the curves $y = f(x)$, x -axis, $x=0$ and $x=1$ is :
 (A) 15/27 square units (B) 17/27 square units
 (C) 15/14 square units (D) None of these
25. The area bounded by the y -axis and the curve $x = e^y \sin \pi y$, $y=0$, and $y=1$ is :
 (A) $\frac{(e+1)\pi}{1+\pi^2}$ (B) $\frac{(e+1)}{1+\pi^2}$ (C) $\frac{(e+1)\pi}{1+\pi}$ (D) None of these
26. If $f(x) = \min\{x+1, \sqrt{1-x}\}$, then the value of $\int_{-1}^1 \frac{12}{7} f(x) dx$.
 (A) 2 (B) -2 (C) 1 (D) None of these
27. The area of the region bounded by $y = \{x\}$ and $2x-1=0, y=0$, ($\{ \}$ stands for fraction part of x)
 (A) 1/2 (B) 1/4 (C) 1/8 (D) None of these
28. If $f(x) = \begin{cases} x & x < 1 \\ x-1 & x \geq 1 \end{cases}$, then $\int_0^2 x^2 f(x) dx$ is equal to :
 (A) 1 (B) 4/3 (C) 5/3 (D) 5/2
29. If $f(0)=1, f(2)=3, f'(2)=5$ and $f'(0)$ is finite, then $\int_0^1 x \cdot f''(2x) dx$ is equal to :
 (A) Zero (B) 1 (C) 2 (D) None of these
30. The value of $\int_{-1}^3 (|x-2| + [x]) dx$ is ($[x]$ stands for greatest integer less than or equal to x)
 (A) 7 (B) 5 (C) 4 (D) 3
31. The value of $\int_{\ln \pi - \ln 2}^{\ln \pi} \frac{e^x}{1 - \cos\left(\frac{2}{3}e^x\right)} dx$ is equal to :
 (A) $\sqrt{3}$ (B) $-\sqrt{3}$ (C) $\frac{1}{\sqrt{3}}$ (D) $-\frac{1}{\sqrt{3}}$

32. If $I_1 = \int_e^{e^2} \frac{dx}{\ln x}$ and $I_2 = \int_1^2 \frac{e^x}{x} dx$, then :
- (A) $I_1 = I_2$ (B) $2I_1 = I_2$ (C) $I_1 = 2I_2$ (D) None of these
33. Let $I_1 = \int_0^{3\pi} f(\cos^2 x) dx$, $I_2 = \int_0^{2\pi} f(\cos^2 x) dx$ and $I_3 = \int_0^{\pi} f(\cos^2 x) dx$, then :
- (A) $I_1 + 2I_3 + 3I_2 = 0$ (B) $I_1 = 2I_2 + I_3$
 (C) $I_2 + I_3 = I_1$ (D) $I_1 = 2I_3$
- *34. The value of $\int_0^{\infty} \frac{x}{(1+x)(1+x^2)} dx$ equal to :
- (A) $\pi/4$ (B) $\pi/2$
 (C) is same as $\int_0^{\infty} \frac{dx}{(1+x)(1+x^2)}$ (D) Cannot be evaluated
- *35. The value of integral $\int_a^b \frac{|x|}{x} dx$, $a < b$ is :
- (A) $b-a$ if $0 < a < b$ (B) $a-b$ if $a < b < 0$ (C) $b+a$ if $a < 0 < b$ (D) $|b| - |a|$
36. Let $f: R \rightarrow R$, $g: R \rightarrow R$ be continuous functions. Then the value of integral $\int_{\ell n \lambda}^{\ell n l / \lambda} \frac{f\left(\frac{x^2}{4}\right) [f(x) - f(-x)]}{g\left(\frac{x^2}{4}\right) [g(x) + g(-x)]} dx$ is :
- (A) Depends on λ (B) A non-zero constant
 (C) Zero (D) None of these
37. $\int_{2-\ell n 3}^{3+\ell n 3} \frac{\ln(4+x)}{\ell n(4+x) + \ell n(9-x)} dx$ is equal to :
- (A) Cannot be evaluate (B) is equal to $5/2$
 (C) is equal to $1 + 2\ell n 3$ (D) is equal to $1/2 + \ell n 3$
38. The value of integral $\int_0^{\pi} x f(\sin x) dx$ is :
- (A) $\frac{\pi}{2} \int_0^{\pi} f(\sin x) dx$ (B) $\pi \int_0^{\pi/2} f(\sin x) dx$ (C) 0 (D) None of these
39. If $f(x) = \int_0^x (\cos^4 t + \sin^4 t) dt$, then $f(x + \pi)$ is equal to :
- (A) $f(x) + f(\pi)$ (B) $f(x) + 2f(\pi)$ (C) $f(x) + f\left(\frac{\pi}{2}\right)$ (D) $f(x) + 2f\left(\frac{\pi}{2}\right)$

40. The value of $\lim_{h \rightarrow 0} \frac{\int_a^{x+h} \ell n^2 t dt - \int_a^x \ell n^2 t dt}{h}$ equals to :
- (A) 0 (B) $\ell n^2 x$ (C) $\frac{2\ell n x}{x}$ (D) Does not exist
41. The value of x , $f(x) = 1 + x + \int_1^x (\ell n^2 t + 2\ell n t) dt$, where $f'(x)$ vanishes is :
- (A) e^{-1} (B) 0 (C) $2e^{-1}$ (D) $1 + 2e^{-1}$
42. If $\int_0^y \cos t^2 dt = \int_0^{x^2} \frac{\sin t}{t} dt$, then the value of $\frac{dy}{dx}$ is :
- (A) $\frac{2\sin^2 x}{x\cos^2 y}$ (B) $\frac{2\sin x^2}{x\cos y^2}$ (C) $\frac{2\sin x^2}{x\left(1 + 2\sin \frac{y^2}{2}\right)}$ (D) None of these
43. The value of $\lim_{n \rightarrow \infty} \frac{\pi}{n} \left[\sin \frac{\pi}{n} + \sin \frac{2\pi}{n} + \dots + \sin \frac{(n-1)\pi}{n} \right]$ is :
- (A) 0 (B) π (C) 2 (D) None of these
44. $\lim_{n \rightarrow \infty} \left[\left(1 + \frac{1}{n^2}\right) \left(1 + \frac{2^2}{n^2}\right) \dots \left(1 + \frac{n^2}{n^2}\right) \right]^{1/n}$ is equal to :
- (A) $\frac{e^{\pi/2}}{2e^2}$ (B) $2e^2 e^{\pi/2}$ (C) $\frac{2}{e^2} e^{\pi/2}$ (D) None of these
45. The value of $\lim_{n \rightarrow \infty} \sum_{r=1}^n \left(\frac{r^3}{r^4 + n^4} \right)$ is :
- (A) $\log 2$ (B) $\frac{1}{2} \log 2$ (C) $\frac{1}{3} \log 2$ (D) $\frac{1}{4} \log 2$
46. The area bounded by curve $y^3 - 9y + x = 0$ and y -axis is :
- (A) $\frac{9}{2}$ (B) 9 (C) $\frac{81}{2}$ (D) 81
47. The area bounded by $y = 2 - |2 - x|$ and $y = \frac{3}{|x|}$ is :
- (A) $\frac{4 + 3\ell n 3}{2}$ (B) $\frac{4 - 3\ell n 3}{2}$ (C) $\frac{3}{2} + \ell n 3$ (D) $\frac{1}{2} + \ell n 3$
- *48. For which of the following values of m , is the area of the region bounded by the curve $y = x - x^2$ and the line $y = mx$ equals $\frac{9}{2}$?
- (A) -4 (B) -2 (C) 2 (D) 4
49. The area of the region bounded by $x = 0$, $y = 0$, $x = 2$, $y = 2$, $y \leq e^x$ and $y \geq \ell n x$, is :
- (A) $6 - 4\ell n 2$ (B) $4\ell n 2 - 2$ (C) $2\ell n 2 - 4$ (D) $6 - 2\ell n 2$

50. The area between two arms of the curve $|y| = x^3$ from $x = 0$ to $x = 2$ is :
 (A) 2 (B) 4 (C) 8 (D) 16
51. **Statement 1 :** If $\{.\}$ represents fractional part function, then $\int_0^{5.5} \{x\} dx = \frac{21}{8}$.
Statement 2 : If $[.]$ and $\{.\}$ represents greatest integer and fractional part functions respectively, then

$$\int_0^t \{x\} dx = \frac{[t]}{2} + \frac{\{t\}^2}{2}$$

 (A) Statement-1 is True, Statement-2 is True and Statement-2 is a correct explanation for Statement-1
 (B) Statement-1 is True, Statement-2 is True and Statement-2 is NOT a correct explanation for Statement-1
 (C) Statement-1 is True, Statement-2 is False (D) Statement-1 is False, Statement-2 is True
52. $\int_0^{2n\pi} \left(|\sin x| - \left\lfloor \frac{\sin x}{2} \right\rfloor \right) dx$ (where $[]$ denotes the greatest integer function and $n \in I$) is equal to :
 (A) 0 (B) $2n$ (C) $2n\pi$ (D) $4n$
53. If $f(\pi) = 2$ and $\int_0^\pi (f(x) + f''(x)) \sin x dx = 5$, $f(x)$ is continuous in $[0, \pi]$. then $f(0)$ is equal to :
 (A) 7 (B) 3 (C) 5 (D) 1
54. The value of $\int_{1/e}^{\tan x} \frac{t}{1+t^2} dt + \int_{1/e}^{\cot x} \frac{1}{t(1+t^2)} dt$, where $x \in (\pi/6, \pi/3)$, is equal to :
 (A) 0 (B) 2 (C) 1 (D) None of these
- *55. If $I_n = \int_0^1 \frac{dx}{(1+x^2)^n}$; $n \in N$, then which of the following statements hold good?
 (A) $2n I_{n+1} = 2^{-n} + (2n-1) I_n$ (B) $I_2 = \frac{\pi}{8} + \frac{1}{4}$
 (C) $I_2 = \frac{\pi}{8} - \frac{1}{4}$ (D) $I_3 = \frac{\pi}{16} - \frac{5}{48}$
56. The area bounded by the curve $x = a \cos^3 t$, $y = a \sin^3 t$ is :
 (A) $\frac{3\pi a^2}{8}$ (B) $\frac{3\pi a^2}{16}$ (C) $\frac{3\pi a^2}{32}$ (D) $3\pi a^2$
57. The area bounded by the curve $y = f(x)$, x -axis and the ordinates $x = 1$ and $x = b$ is $(b-1) \sin(3b+4) \forall b \in R$, then $f(x) =$
 (A) $(x-1) \cos(3x+4)$ (B) $\sin(3x+4)$
 (C) $\sin(3x+4) + 3(x-1) \cos(3x+4)$ (D) None of these

58. Consider the following statements :

S_1 : The value $\int_0^{2\pi} \cos^{-1}(\cos x) dx$ is π^2

S_2 : Area enclosed by the curve $|x-2| + |y+1| = 1$ is equal to 3 square unit

S_3 : If $\frac{d}{dx} f(x) = g(x)$ for $a \leq x \leq b$, then $\int_a^b f(x)g(x)dx$ equals to $\frac{[f(b)]^2 - [f(a)]^2}{2}$

S_4 : Area of the region $R \equiv \{(x, y); x^2 \leq y \leq x\}$ is $1/6$

State, in order, where S_1, S_2, S_3, S_4 are true or false

(A) TFFT (B) TTTT (C) FFFF (D) TFTF

59. If $I_{(m, n)} = \int_0^1 t^m (1+t)^n dt$, then the expression of $I_{(m, n)}$ in terms of $I_{(m+1, n-1)}$ is :

(A) $\frac{2^n}{m+1} - \frac{n}{m+1} I_{(m+1, n-1)}$ (B) $\frac{n}{m+1} I_{(m+1, n-1)}$

(C) $\frac{2^n}{m-1} - \frac{n}{m+1} I_{(m+1, n-1)}$ (B) $\frac{n}{m+1} I_{(m+1, n-1)}$

60. If $f(x) = \int_{x^2}^{x^2+1} e^{-t^2} dt$, then the function $f(x)$ decreases in :

(A) $(-2, 2)$ (B) $(0, \infty)$ (C) No value of x (D) $(-\infty, 0)$

61. $y(x) = \int_{\pi^2/16}^{x^2} \frac{\cos x \cos \sqrt{\theta}}{1 + \sin^2 \sqrt{\theta}} \cdot d\theta$. Find $y'(x)$ at $x = \pi$.

(A) π (B) 2π (C) 3π (D) 4π

62. The value of $\int_{-\pi/3}^{\pi/3} \frac{\pi + 4x^3}{2 - \cos\left(|x| + \frac{\pi}{3}\right)} dx$ is :

(A) $\frac{4\pi}{\sqrt{3}} \tan^{-1} \frac{1}{2}$ (B) $\frac{4}{\sqrt{3}} \tan^{-1} \frac{1}{2}$ (C) $\frac{4\pi}{\sqrt{3}} \tan^{-1} 2$ (D) None of these

63. $\int_{\sin x}^1 t^2 (f(t)) dt = (1 - \sin x)$, then $f\left(\frac{1}{\sqrt{3}}\right)$ is :

(A) $1/3$ (B) $1/\sqrt{3}$ (C) 3 (D) $\sqrt{3}$

*64. $f(x) = \begin{cases} 0 & , 0 \leq x \leq 1 \\ 2 - e^{x-1} & , 1 < x \leq 2 \text{ and } g(x) = \int_0^x f(t) dt, x \in [1, 3] \text{ then :} \\ x - e & , 2 < x \leq 3 \end{cases}$

(A) $g(x)$ has no local maxima (B) $g(x)$ has no local minima
(C) $g(x)$ has a local maxima at $x = 1 + \log 2$ (D) $g(x)$ has a local minima at $x = e$

65. The value of $\int_{-2}^0 \{x^3 + 3x^2 + 3x + 3 + (x+1)\cos(x+1)\}$ is equal to :
- (A) -4 (B) 0 (C) 4 (D) 6
- *66. If $I_n = \int_{-\pi}^{\pi} \frac{\sin nx}{(1+\pi^x)\sin x} dt$, $n = 0, 1, 2, \dots$, then :
- (A) $I_n = I_{n+2}$ (B) $\sum_{m=1}^{10} I_{2m+1} = 10\pi$ (C) $\sum_{m=1}^{10} I_{2m} = 0$ (D) $I_n = I_{n+1}$
67. The value $\int_{\sqrt{\ell n 2}}^{\sqrt{\ell n 3}} \frac{x \sin x^2}{\sin x^2 + \sin(\ell n 6 - x^2)} dx$ is :
- (A) $\frac{1}{4} \ell n \frac{3}{2}$ (B) $\frac{1}{2} \ell n \frac{3}{2}$ (C) $\ell n \frac{3}{2}$ (D) $\frac{1}{6} \ell n \frac{3}{2}$
68. The value of integral $\int_{-\pi/2}^{\pi/2} \left(x^2 + \ell n \frac{\pi+x}{\pi-x} \right) \cos x dx$ is :
- (A) 0 (B) $\frac{\pi^2}{2} - 4$ (C) $\frac{\pi^2}{2} + 4$ (D) $\frac{\pi^2}{2}$
69. The area (in square units) bounded by the curves $y = \sqrt{x}$, $2y - x + 3 = 0$, x -axis and lying in the first quadrant is :
- (A) 9 (B) 36 (C) 18 (D) 27/4
70. **Statement 1** The value of the integral $\int_{\pi/6}^{\pi/3} \frac{dx}{1+\sqrt{\tan x}}$ is equal to $\pi/6$.
- Statement 2 :** $\int_a^b f(x) dx = \int_a^b f(a+b-x) dx$
- (A) Statement-1 is True, Statement-2 is True and Statement-2 is a correct explanation for Statement-1
 (B) Statement-1 is True, Statement-2 is True and Statement-2 is NOT a correct explanation for Statement-1
 (C) Statement-1 is True, Statement-2 is False
 (D) Statement-1 is False, Statement-2 is True
- *71. If $g(x) = \int_0^x \cos 4t dt$, then $g(x+\pi)$ equals :
- (A) $\frac{g(x)}{g(\pi)}$ (B) $g(x) + g(\pi)$ (C) $g(x) - g(\pi)$ (D) $g(x) \cdot g(\pi)$
72. The area bounded between that parabolas $x^2 = \frac{y}{4}$, $x^2 = 9y$ and the straight line $y = 2$ is :
- (A) $20\sqrt{2}$ (B) $\frac{10\sqrt{2}}{3}$ (C) $\frac{20\sqrt{2}}{3}$ (D) $10\sqrt{2}$
73. The area bounded by the curves $y^2 = 4x$ and $x^2 = 4y$ is:
- (A) 32/3 (B) 16/3 (C) 8/3 (D) 0
74. The area of the region enclosed by the curves $y = x$, $x = e$, $y = 1/x$ and the positive x -axis is :
- (A) $1/2$ square units (B) 1 square units (C) $3/2$ square units (D) $5/2$ square units

75. Let $[.]$ denote the greatest integer function then the value of $\int_0^{1.5} x[x^2]dx$ is :
 (A) 0 (B) $3/2$ (C) $3/4$ (D) $5/4$
76. For $x \in \left(0, \frac{5\pi}{2}\right)$, define $f(x) = \int_0^x \sqrt{t} \sin t dt$. Then f has :
 (A) Local maximum at π and 2π (B) Local minimum at π and 2π
 (C) Local minimum at π and local maximum at 2π
 (D) Local maximum at π and local minimum at 2π
77. The area of the region bounded by the parabola $(y-2)^2 = x-1$, the tangent to the parabola at the point $(2, 3)$ and the x -axis is :
 (A) 6 square unit (B) 9 square unit (C) 12 square unit (D) 3 square unit
78. Let $p(x)$ be a function defined on R such that $p'(x) = p'(1-x)$, for all $x \in [0, 1]$, $p(0) = 1$ and $p(1) = 41$.
 Then $\int_0^1 p(x) dx$
 (A) 21 (B) 41 (C) 42 (D) $\sqrt{41}$
79. $\int_0^\pi [cot x] dx$, where $[.]$ denotes the greatest integer function, is equal to :
 (A) 1 (B) -1 (C) $-\pi/2$ (D) $\pi/2$
80. The area of the plane region bounded by the curves $x+2y^2=0$ and $x+3y^2=1$ is equal to :
 (A) $5/3$ square unit (B) $1/3$ square unit (C) $2/3$ square unit (D) $4/3$ square unit
81. The integral $\int_0^\pi \sqrt{1+4\sin^2 \frac{x}{2} - 4\sin \frac{x}{2}} dx$ is :
 (A) $\pi - 4$ (B) $\frac{2\pi}{3} - 4 - 4\sqrt{3}$ (C) $4\sqrt{3} - 4$ (D) $4\sqrt{3} - 4 - \frac{\pi}{3}$
82. The area of the region described by $A = \{(x, y) : x^2 + y^2 \leq 1 \text{ and } y^2 \leq 1-x\}$ is :
 (A) $\frac{\pi}{2} + \frac{4}{3}$ (B) $\frac{\pi}{2} - \frac{4}{3}$ (C) $\frac{\pi}{2} - \frac{2}{3}$ (D) $\frac{\pi}{2} + \frac{2}{3}$
83. The value of $\int_0^1 \left(\frac{8 \log(1+x)}{1+x^2} \right) dx$ is :
 (A) $\frac{\pi}{8} \log 2$ (B) $\frac{\pi}{2} \log 2$ (C) $\log 2$ (D) $\pi \log 2$
84. The value of $\int_1^a [x] f'(x) dx$, $a > 1$ where $[x]$ denotes the greatest integer not exceeding x , is :
 (A) $[a] f(a) - \{f(1) + f(2) + \dots + f([a])\}$ (B) $|a| - f([a]) - \{f(1) + f(2) + \dots + f(a)\}$
 (C) $af([a]) - \{f(1) + f(2) + \dots + f(a)\}$ (D) $af(a) - \{f(1) + f(2) + \dots + f([a])\}$

Integer Answer Type Questions

The Answer to the following questions are positive integers of 1/2/3 digits and zero

85. Suppose for every integer n , $\int_n^{n+1} f(x)dx = n^2$. The value of $\int_{-2}^4 f(x)dx$ is :
86. If $f(x) = Ax^2 + Bx + C$ satisfies the conditions $f'(1) = 8$, $f(2) + f''(2) = 33$ and $\int_0^1 f(x)dx = 7/3$, then value of $A + B + C$ is :
87. Let $f: R \rightarrow R$ be a differentiable function and $f(1) = 1$, $f'(1) = 3$. Then the value of $\lim_{x \rightarrow 1} \frac{\int_1^{x^2} (f(t) - t)dt}{(x-1)^2}$ is:
88. If $\lim_{n \rightarrow \infty} \sum_{r=1}^{4n} \frac{\sqrt{n}}{\sqrt{r}(3\sqrt{r} + 4\sqrt{n})^2}$ is $\frac{1}{k}$, then k is:
89. $\int_{1/e}^{\tan t} \frac{x dx}{1+x^2} + \int_{1/e}^{\cot t} \frac{dx}{x(1+x^2)}$ is:
90. The value of $\int_{-10}^{20} [\cot^{-1} x] dx$ is : (where $[.]$ is the greatest integer function) $= A + \cot B + \cot C + \cot D$ then $A + B + C + D$ is :
91. If $I_{10} = \int_0^{\pi/2} x^{10} \sin x dx$ then the value of $I_{10} + 90 I_8 = a \left(\frac{\pi}{6} \right)^c$ then $a + b + c$ is:
92. If $I_1 = \int_{-100}^{101} \frac{dx}{(5+2x-2x^2)(1+e^{2-4x})}$ and $I_2 = \int_{-100}^{101} \frac{dx}{5+2x-2x^2}$, then $\frac{I_2}{I_1}$ is :
93. The value of $\int_{-2}^0 [x^3 + 3x^2 + 3x + 3 + (x+1)\cos(x+1)] dx$ is :
94. If $\int_{\sin x}^1 t^2 f(t) dt = 1 - \sin x, \forall x \in (0, \pi/2)$, then $f\left(\frac{1}{\sqrt{3}}\right)$ is:
95. If $\int_0^x \phi(x) dx = x + \int_x^2 x \phi(x) dx$, the value of $10 \phi(4)$
96. If $\lim_{n \rightarrow \infty} \frac{1^a + 2^a + 3^a + \dots + n^a}{n^{a+1}} = \frac{1}{5}$
(where $a > -1$) then the value of 'a' is:
97. Let $I_n = \int_0^1 (x \ln x)^n dx$, if $I_4 = \int_0^1 x^4 \ln x^3 dx$, then 'k' is equal to:
98. The area bounded by the curve $y = \ln x$ and the lines $y = 0, y = \ln 3$ and $x = 0$ is equal to :
99. If function $y = f(x)$ satisfy condition of Rolle's theorem in $[1, 3]$ then the value of $\int_1^3 f'(x) dx$ is equal to:

Differential Equations

CHOOSE THE CORRECT ALTERNATIVE. ONLY ONE CHOICE IS CORRECT. HOWEVER, QUESTIONS MARKED ‘*’ MAY HAVE MORE THAN ONE CORRECT OPTION.

- The solution of the differential equation $\frac{dy}{dx} + \frac{y}{2} \sec x = \frac{\tan x}{2}$, where $0 \leq x < \frac{\pi}{2}$, and $y(0) = 1$, is given by :

(A) $y = 1 - \frac{x}{\sec x + \tan x}$ (B) $y^2 = 1 + \frac{x}{\sec x + \tan x}$

(C) $y^2 = 1 - \frac{x}{\sec x + \tan x}$ (D) $y = 1 + \frac{x}{\sec x + \tan x}$
- If $y(x)$ is the solution of the differential equation $(x+2)\frac{dy}{dx} = x^2 + 4x - 9$, $x \neq 2$ and $y(0) = 0$, then $y(-4)$ is equal to :

(A) 0 (B) 1 (C) -1 (D) 2
- The solution of the differential equation $y dx - (x + 2y^2) dy = 0$ is $x = f(y)$. If $f(-1) = 1$, then $f(1)$ is equal to:

(A) 4 (B) 3 (C) 2 (D) 1
- The differential equation representing the family of all circles touching x-axis at the origin is $(x^2 - y^2)\frac{dy}{dx} = g(x)y$, then $g(x)$ equals :

(A) $\frac{1}{2}x$ (B) $2x^2$ (C) $2x$ (D) $\frac{1}{2}x^2$
- If the general solution of the differential equation $y' = \frac{y}{x} + \phi\left(\frac{x}{y}\right)$, for some function ϕ , is given by $y \ln|cx| = x$, where c is an arbitrary constant, then $\phi(2)$ is equal to :

(A) 4 (B) $\frac{1}{4}$ (C) -4 (D) $-\frac{1}{4}$
- The general solution of the differential equation, $\sin 2x + \left(\frac{dy}{dx} - \sqrt{\tan x}\right) - y = 0$, is :

(A) $y\sqrt{\tan x} = x + c$ (B) $y\sqrt{\cot x} = x + c$

(C) $y\sqrt{\tan x} = \cot x + c$ (D) $y\sqrt{\cot x} = \tan x + c$
- If $\frac{dy}{dx} + y \tan x = \sin 2x$ and $y(0) = 1$, then $y(\pi)$ is equal to :

(A) 1 (B) -1 (C) -5 (D) 5
- The order and degree of the given differential equation: $e^{\frac{d^3y}{dx^3}} - x \frac{d^2y}{dx^2} + y = 0$

(A) (3, 2) (B) (2, 3) (C) 3, degree is not defined (D) None of these

9. The order and degree of the given differential equation: $\frac{d^2y}{dx^2} = \sin\left(x + \frac{dy}{dx}\right)$
- (A) (2, 1) (B) 2, degree is not defined
(C) 1, degree is not defined (D) None of these
10. Differential equations to the curve : $ax^2 + by^2 = 1$, is : (where a and b are arbitrary constants)
 $\left(\text{where } y_1 = \frac{dy}{dx}, y_2 = \frac{d^2y}{dx^2}\right)$
- (A) $xyy_2 + (xy_1 + y)y_1 = 0$ (B) $xyy_2 + (xy_1 - y)y_1 = 0$
(C) $xyy_2 + (xy_1 + y)y = 0$ (D) None of these
11. Differential equations to the curve : $xy = ae^{-x} + be^x$ is : (where a and b are arbitrary constants)
- (A) $xy_2 - 2y_1 = xy$ (B) $xy_2 - 2y_1 = xy_1$ (C) $xy_2 + 2y_1 = xy$ (D) None of these
12. The solution of differential equation : $\frac{d^2y}{dx^2} = \frac{dy}{dx}$
- (A) $y = c_1e^x + c_2$ (B) $y = c_1e^{-x} + c_2$ (C) $y = c_1e^{x/2} + c_2$ (D) None of these
13. The solution of differential equation : $\frac{d^3y}{dx^3} = 8\frac{d^2y}{dx^2}$ satisfying $y(0) = \frac{1}{8}$, $y_1(0) = 0$ and $y_2(0) = 1$
- (A) $y = \frac{e^{8x}}{64} + \frac{x}{8} + \frac{7}{64}$ (B) $y = \frac{e^{8x}}{64} - \frac{x}{8} + \frac{7}{64}$
(C) $y = \frac{e^{8x}}{8} + \frac{x}{8}$ (D) None of these
14. The order and degree of the differential equation $r = \frac{\left[r + \left(\frac{dy}{dx}\right)^2\right]^{3/2}}{\frac{d^2y}{dx^2}}$ are respectively :
- (A) 2, 2 (B) 2, 3 (C) 2, 1 (D) None of these
15. If p and q are order and degree of differential equation $y^2\left(\frac{d^2y}{dx^2}\right)^2 + 3x\left(\frac{dy}{dx}\right)^{1/3} + x^2y^2 = \sin x$, then :
- (A) $p > q$ (B) $\frac{p}{q} = \frac{1}{2}$ (C) $p = q$ (D) $p < q$
16. Family $y = Ax + A^3$ of curve represented by the differential equation of degree : (where A is arbitrary constant)
- (A) Three (B) Two (C) One (D) None of these
17. The differential equation whose solution is $(x-h)^2 + (y-k)^2 = a^2$ is : (a is a constant)
- (A) $\left[1 + \left(\frac{dy}{dx}\right)^2\right]^3 = a^2 \frac{d^2y}{dx^2}$ (B) $\left[1 + \left(\frac{dy}{dx}\right)^2\right]^3 = a^2 \left(\frac{d^2y}{dx^2}\right)^2$
(C) $\left[1 + \left(\frac{dy}{dx}\right)^2\right]^3 = a^2 \left(\frac{d^2y}{dx^2}\right)^2$ (D) None of these

18. If $\phi(x) = \phi'(x)$ and $\phi(1) = 2$, then $\phi(3)$ equals :

- (A) e^2 (B) $2e^2$ (C) $3e^2$ (D) None

*19. The solution of $x^2 y_1^2 + xy_1 - 6y^2 = 0$ are : (where $y = \frac{dy}{dx}$)

- (A) $y = Cx^2$ (B) $x^2 y = C$ (C) $\frac{1}{2} \ln y = C + \log x$ (D) $x^3 y = C$

20. The solution of $\frac{dv}{dt} + \frac{k}{m}v = -g$ is :

- (A) $v = ce^{-\frac{k}{m}t} - \frac{mg}{k}$ (B) $v = c - \frac{mg}{k} e^{-\frac{k}{m}t}$ (C) $ve^{-\frac{k}{m}t} = c - \frac{mg}{k}$ (D) $ve^{\frac{k}{m}t} = c - \frac{mg}{k}$

21. The differential equation for all the straight lines which are at a unit distance from the origin is :

- (A) $\left(y - x \frac{dy}{dx}\right)^2 = 1 - \left(\frac{dy}{dx}\right)^2$ (B) $\left(y + x \frac{dy}{dx}\right)^2 = 1 + \left(\frac{dy}{dx}\right)^2$
(C) $\left(y - x \frac{dy}{dx}\right)^2 = 1 + \left(\frac{dy}{dx}\right)^2$ (D) $\left(y + x \frac{dy}{dx}\right)^2 = 1 - \left(\frac{dy}{dx}\right)^2$

For Question 22 - 23

- (A) Statement-1 is True, Statement-2 is True and Statement-2 is a correct explanation for Statement-1
(B) Statement-1 is True, Statement-2 is True and Statement-2 is NOT a correct explanation for Statement-1
(C) Statement-1 is True, Statement-2 is False
(D) Statement-1 is False, Statement-2 is True

22. **Statement 1 :** The solution of D.E. $\frac{\frac{xdy}{dx} - y}{\sqrt{x^2 - y^2}} = mx^2$ is given by $\tan^{-1} \frac{y}{x} = \frac{mx^2}{2} + c$.

Statement 2 : The solution of differential equation $\frac{dy}{dx} + \frac{y}{x} = \sin x$ is $x(y + \cos x) = \sin x + c$

23. **Statement 1 :** Solution of $\left(1 + x\sqrt{x^2 + y^2}\right)dx + y\left(-1 + \sqrt{x^2 + y^2}\right)dy = 0$ is $x - \frac{y^2}{2} + \frac{1}{3}(x^2 + y^2)^{3/2} + c = 0$

Statement 2 : Solution of $(1 + xy)y dx + (1 - xy)x dy = 0$ is $\ln \frac{x}{y} - \frac{1}{xy} = c$.

24. All the curves possessing the following property; the segment of the tangent between the point of tangency and the x-axis is bisected at the point of intersection with the y-axis are :

- (A) $y^2 = kx$ (B) $y = kx$ (C) $y^2 = kx^2$ (D) None of these

25. If $y_1(x)$ is a solution of the differential equation $\frac{dy}{dx} + f(x)y = 0$, then a solution of differential equation

$\frac{dy}{dx} + f(x)y = r(x)$ is :

- (A) $\frac{1}{y(x)} \int y_1(x) dx$ (B) $y_1(x) \int \frac{r(x)}{y_1(x)} dx$ (C) $\int r(x)y_1(x) dx$ (D) None of these

26. If $y_1(x)$ and $y_2(x)$ are two solutions of $\frac{dy}{dx} + f(x)y = r(x)$ then $y_1(x) + y_2(x)$ is solution of :
- (A) $\frac{dy}{dx} + f(x)y = 0$ (B) $\frac{dy}{dx} + 2f(x)y = r(x)$
 (C) $\frac{dy}{dx} + f(x)y = 2r(x)$ (D) $\frac{dy}{dx} + 2f(x)y = 2r(x)$
27. The value of $\lim_{x \rightarrow \infty} y(x)$ obtained from the differential equation $\frac{dy}{dx} = y - y^2$, where $y(0) = 2$ is :
- (A) Zero (B) 1 (C) ∞ (D) None of these
28. The solution of $y dx - x dy + 3x^2 y^2 e^{x^3} dx = 0$ is :
- (A) $\frac{x}{y} + e^{x^3} = C$ (B) $\frac{x}{y} - e^{x^3} = C$ (C) $-\frac{x}{y} + e^{x^3} = C$ (D) None of these
29. Consider the following statements :
- S₁ : $f(x, y) = e^{y/x} + \tan \frac{y}{x}$ is homogeneous of degree zero.
- S₂ : Solution of differential equation $(2x \ln y) dx + \left(\frac{x^2}{y} + 3y^2 \right) dy = 0$ is $x^2 \ln y + y^3 = c$
- S₃ : Solution of differential equation $x^2 = 1 + \left(\frac{x}{y} \right)^{-1} \frac{dy}{dx} + \frac{\left(\frac{x}{y} \right)^{-2} \left(\frac{dy}{dx} \right)^2}{2!} + \frac{\left(\frac{x}{y} \right)^{-3} \left(\frac{dy}{dx} \right)^3}{3!} + \dots$
 is $y^2 = x^2 (\ln x^2 - 1) + c$.
- S₄ : A curve which passes through (1, 2) and whose sub-normal at every point is 2, is $y^2 = 4x$.
- State, in order, whether S₁, S₂, S₃, S₄ are true or false :
- (A) FTFT (B) TTTT (C) FFFF (D) TFTF
30. Solution of the differential equation $\frac{dy}{dx} + \frac{1+y^2}{\sqrt{1-x^2}} = 0$ is :
- (A) $\tan^{-1} y + \sin^{-1} x = c$ (B) $\tan^{-1} x + \sin^{-1} y = c$
 (C) $\tan^{-1} y \cdot \sin^{-1} x = c$ (D) None
31. Solution of differential equation $f(x) \frac{dy}{dx} = f^2(x) + f(x)y + f'(x)y$ is :
- (A) $y = f(x) + ce^x$ (B) $y = -f(x) + ce^x$ (C) $y = -f(x) + ce^x f(x)$ (D) $y = cf(x) + e^x$
32. If $y = f(x)$ and $\left(\frac{2 + \sin x}{y+1} \right) \frac{dy}{dx} = -\cos x$; $y(0) = 1$, then $y\left(\frac{\pi}{2}\right)$ is equal to :
- (A) 1/3 (B) -2/3 (C) 2/3 (D) -1/3
- *33. A tangent drawn to the curve $y = f(x)$ at P(x, y) cuts the x-axis and y-axis at A and B respectively such that BP : AP = 3 : 1, given that $f(1) = 1$, then :
- (A) Differential equation of curve is $x \frac{dy}{dx} - 3y = 0$ (B) Normal at (1, 1) is $3y - x = 2$
 (C) Curve passes through $\left(2, \frac{1}{8}\right)$ (D) Differential equation of curve is $x \frac{dy}{dx} + 3y = 0$

34. Let $f(x)$ be differential on the interval $(0, \infty)$ such that $f(1)=1$ and $\lim_{t \rightarrow x} \frac{t^2 f(x) - x^2 f(t)}{t-x} = 1$ for each $x > 0$. Then $f(x)$ is :
- (A) $\frac{1}{3x} + \frac{2x^2}{3}$ (B) $\frac{-1}{3x} + \frac{4x^2}{3}$ (C) $\frac{-1}{x} + \frac{2}{x^2}$ (D) $\frac{1}{x}$
35. A curve passes through the point $\left(1, \frac{\pi}{6}\right)$. Let the slope of the curve at each point (x, y) be $\frac{y}{x} + \sec\left(\frac{y}{x}\right)$, $x > 0$. Then the equation of the curve is :
- (A) $\sin\left(\frac{y}{x}\right) = \log x + \frac{1}{2}$ (B) $\cos\left(\frac{y}{x}\right) = \log x + 2$
 (C) $\sec\left(\frac{2y}{x}\right) = \log x + 2$ (D) $\cos\left(\frac{2y}{x}\right) = \log x + \frac{1}{2}$
36. The solution of the differential equation $y dx + (x + x^2 y) dy = 0$ is :
- (A) $-\frac{1}{xy} = c$ (B) $-\frac{1}{xy} + \log y = c$ (C) $\frac{1}{xy} + \log y = c$ (D) $\log y = cx$
37. The differential equation representing the family of curves $y^2 = 2c(x + \sqrt{c})$, where $c > 0$, is a parameter, is of order and degree as follows :
- (A) Order 2, degree 2 (B) Order 1, degree 3 (C) Order 1, degree 1 (D) Order 1, degree 2
38. The differential equation which represents the family of curves $y = c_1 e^{c_2 x}$, where c_1 and c_2 are arbitrary constants is:
- (A) $y' = y^2$ (B) $y'' = y'y$ (C) $y \cdot y'' = y'$ (D) $y \cdot y'' = (y')^2$
39. Let the population of rabbits surviving at a time t be governed by the differential equation $\frac{dp(t)}{dt} = \frac{1}{2} p(t) - 200$. If $p(0) = 100$, then $p(t)$ is equal to :
- (A) $400 - 300e^{t/2}$ (B) $300 - 200e^{-t/2}$ (C) $600 - 500e^{t/2}$ (D) $400 - 300e^{-t/2}$
40. Let I be the purchase value of an equipment and $V(t)$ be the value after it has been used for t years. The value $V(t)$ depreciates at a rate given by differential equation $\frac{dV(t)}{dt} = -k(T-t)$, where $k > 0$ is a constant and T is the total life in years of the equipment. Then the scrap value $V(T)$ of the equipment is :
- (A) $T^2 - \frac{1}{k}$ (B) $I - \frac{kT^2}{2}$ (C) $I - \frac{k(T-t)^2}{2}$ (D) e^{-kT}
41. If $\frac{dy}{dx} = y + x$ and $y(0) = 2$, then $y(\ln 2)$ is equal to :
- (A) $5 - \ln 2$ (B) $6 - \ln 2$ (C) $7 - \ln 2$ (D) $8 - \ln 2$
42. The curve that passes through the point $(2, 3)$, and has the property that the segment of any tangent to it lying between the coordinate axes is bisected by the point of contact is given by :
- (A) $2y - 3x = 0$ (B) $y = \frac{6}{x}$ (C) $x^2 + y^2 = 13$ (D) $\left(\frac{x}{2}\right)^2 + \left(\frac{y}{3}\right)^2 = 2$
43. Consider the differential equation $y^2 dx + \left(x - \frac{1}{y}\right) dy = 0$. If $y(1) = 1$, then x is given by :
- (A) $4 - \frac{2}{y} - \frac{e^{1/y}}{e}$ (B) $3 - \frac{1}{y} - \frac{e^{1/y}}{e}$ (C) $1 + \frac{1}{y} - \frac{e^{1/y}}{e}$ (D) $1 - \frac{1}{y} + \frac{e^{1/y}}{e}$

Integer Answer Type Questions

The Answer to the following questions are positive integers of 1/2/3 digits and zero

44. The differential equation satisfied by all the circle in the x - y plane is $(1+y_1^2)y_3 = \lambda y_1 y_2^2$ then λ is _____.
45. The order of the differential equation whose general solution is given by $y = a \cos x + b \sin x + ce^{-x}$ is _____.
46. Sum of order and degree of the differential equation $\left[4 + \left(\frac{dy}{dx}\right)^2\right]^{2/3} = \frac{d^2y}{dx^2}$ is _____.
47. The solution of the differential equation $\frac{dy}{dx} = e^{x-y} + x^2 e^{-y}$ passes through origin is $y = f(x)$ then $[f(2)]$ is _____.
(where $[\cdot]$ denotes greatest integer function)
48. The solution of the differential equation $\frac{dy}{dx} = \frac{3y-7x-3}{3x-7y+7}$ is $(y-x-1)^m(y+x-1)^n = \text{constant}$ (m, n are lowest possible natural numbers) then $m \times n$ is _____.
49. The solution of the differential equation $(1+y^2)dx = (\tan^{-1}y - x)dy$ is $x = f(y)$, such that $f(0) = 1$ then $[f(-\tan 1)]$ is _____. ($[\cdot]$ denotes greater integer function)
50. The solution of the differential equation $(2x-10y^3)\frac{dy}{dx} + y = 0$ is $x = f(y)$ such that $f(1) = 3$, then $[f(2)]$ is _____.
($[\cdot]$ denotes greater integer function)
51. If m and n are the order & degree respectively of the equation $\left(\frac{d^2y}{dx^2}\right)^5 + 4 \cdot \frac{\left(\frac{d^2y}{dx^2}\right)^3}{\frac{d^3y}{dx^3}} + \frac{d^3y}{dx^3} = x^2 - 1$, then $m+n$ is ____.
52. The degree of differential equation of the family of all parabola whose axis is x -axis is _____.
53. The solution of the equation $\frac{dy}{dx} = e^{y+x} + e^{y-x}$ is passing through $(1, 1)$, then if it is passing through $(0, k)$ then value of $|k|$ is _____.
54. The solution of the differential equation $(x+y)dy - (x-y)dx = 0$ is $ax^2 + by^2 + 2xy = c$, then $a+b$ is _____.
55. The orthogonal trajectory of $x^{2/3} + y^{2/3} = a^{2/3}$ is $x^{b/3} - y^{b/3} = \text{constant}$, where $b \in \mathbb{N}$, then value of b is _____.
56. A curve passes through $(2, 0)$ and the slope of the tangent at $P(x, y)$ is equal to $\frac{(x+1)^2 + y - 3}{x+1}$ then the equation of the curve is $y = ax^2 + bx$, then value of $a-b$ is _____.
57. The differential equation $\frac{dy}{dx} = -\left(\frac{x-2y+5}{2x-y+4}\right)$ can be converted to homogeneous by substitution $x = x+h, y = y+k$ then $h^2 + k^2$ is _____.
58. If $\frac{xdy}{dx} + y = x \frac{f(xy)}{f'(xy)}$ then $f(xy) = ke^{ax^2}$, then value of $\frac{1}{a}$ is _____.
59. The differential equation corresponding to family of curves $y = \frac{x+c_1}{x+c_2}$ (for arbitrary c_1, c_2) is $y'' + \frac{a}{1-y}(y')^2 = 0$, then value of a is _____.

Vectors

CHOOSE THE CORRECT ALTERNATIVE. ONLY ONE CHOICE IS CORRECT.

- If the non-zero vectors $\vec{a}$ and $\vec{b}$ are perpendicular to each other, then the solution of the equation, $\vec{r} \times \vec{a} = \vec{b}$ is given by: (where x is any scalar)
 (A) $\vec{r} = x\vec{a} + \frac{\vec{a} \times \vec{b}}{|\vec{a}|^2}$ (B) $\vec{r} = x\vec{b} - \frac{\vec{a} \times \vec{b}}{|\vec{b}|^2}$ (C) $\vec{r} = x(\vec{a} \times \vec{b})$ (D) $\vec{r} = x(\vec{b} \times \vec{a})$
- The volume of the tetrahedron whose vertices have position vectors $\hat{i} - 6\hat{j} + 10\hat{k}$, $-\hat{i} - 3\hat{j} + 7\hat{k}$, $5\hat{i} - \hat{j} + \lambda\hat{k}$ and $7\hat{i} - 4\hat{j} + 7\hat{k}$ is 11 cubic units if λ equals :
 (A) -3 (B) 3 (C) 7 (D) -1
- If $\vec{a}, \vec{b}, \vec{c}$ are non-coplanar vectors and λ is a real number, then $[\lambda(\vec{a} + \vec{b}), \lambda^2\vec{b}, \lambda\vec{c}] = [\vec{a}, \vec{b} + \vec{c}, \vec{b}]$ for :
 (A) no value of λ (B) exactly one value of λ
 (C) exactly two values of λ (D) exactly three values of λ
- Let $\vec{b}$ and $\vec{c}$ be non-collinear vectors. If $\vec{a}$ is a vector such that $\vec{a} \cdot (\vec{b} + \vec{c}) = 4$ and $\vec{a} \times (\vec{b} \times \vec{c}) = (x^2 - 2x + 6)\vec{b} + \sin y \vec{c}$ then (x, y) lies on the line :
 (A) $x + y = 0$ (B) $x - y = 0$ (C) $x = 1$ (D) $y = \pi$
- If $\vec{\alpha} = x(\vec{a} \times \vec{b}) + y(\vec{b} \times \vec{c}) + z(\vec{c} \times \vec{a})$ and $[\vec{a} \ \vec{b} \ \vec{c}] = \frac{1}{8}$, then $x + y + z$ is equal to :
 (A) $8\vec{\alpha} \cdot (\vec{a} + \vec{b} + \vec{c})$ (B) $\vec{\alpha} \cdot (\vec{a} + \vec{b} + \vec{c})$ (C) $8(\vec{a} + \vec{b} + \vec{c})$ (D) None of these
- If $\vec{a} = (\hat{i} + \hat{j} + \hat{k})$, $\vec{a} \cdot \vec{b} = 1$ and $\vec{a} \times \vec{b} = \vec{j} - \vec{k}$ then $\vec{b}$ is :
 (A) $\hat{i} - \hat{j} + \hat{k}$ (B) $2\hat{j} - \hat{k}$ (C) $\hat{i}$ (D) $2\hat{i}$
- If $\vec{a}, \vec{b}$ and $\vec{c}$ are unit vectors, then $|\vec{a} - \vec{b}|^2 + |\vec{b} - \vec{c}|^2 + |\vec{c} - \vec{a}|^2$ does not exceed :
 (A) 4 (B) 9 (C) 8 (D) 6
- Let $\vec{a}, \vec{b}$ and $\vec{c}$ be non-zero vectors such that $(\vec{a} \times \vec{b}) \times \vec{c} = \frac{1}{3}|\vec{b}||\vec{c}|\vec{a}$. If θ is the angle between the vectors $\vec{b}$ and $\vec{c}$, then $\sin \theta$ equals :
 (A) $\frac{1}{3}$ (B) $\frac{\sqrt{2}}{3}$ (C) $\frac{2}{3}$ (D) $\frac{2\sqrt{2}}{3}$
- Let $\vec{u} = \hat{i} + \hat{j}$, $\vec{v} = \hat{i} - \hat{j}$ and $\vec{w} = \hat{i} + 2\hat{j} + 3\hat{k}$. If $\hat{n}$ is unit vector such that $\vec{u} \cdot \hat{n} = 0$ and $\vec{v} \cdot \hat{n} = 0$, then $|\vec{w} \cdot \hat{n}|$ is equal to:
 (A) 0 (B) 1 (C) 2 (D) 3

10. $\vec{a}, \vec{b}$ and $\vec{c}$ be three non coplanar vectors and $\vec{r}$ be any arbitrary vector, then
 $(\vec{a} \times \vec{b}) \times (\vec{r} \times \vec{c}) + (\vec{b} \times \vec{c}) \times (\vec{r} \times \vec{a}) + (\vec{c} \times \vec{a}) \times (\vec{r} \times \vec{b})$ is equal to :
 (A) $[\vec{a} \vec{b} \vec{c}] \vec{r}$ (B) $2[\vec{a} \vec{b} \vec{c}] \vec{r}$ (C) $3[\vec{a} \vec{b} \vec{c}] \vec{r}$ (D) None of these
11. If V is the volume of the parallelopiped having three coterminus edges as $\vec{a}, \vec{b}$ and $\vec{c}$, then the volume of the parallelopiped having three coterminus edges as
 $\vec{\alpha} = (\vec{a} \cdot \vec{a})\vec{a} + (\vec{a} \cdot \vec{b})\vec{b} + (\vec{a} \cdot \vec{c})\vec{c}, \vec{\beta} = (\vec{a} \cdot \vec{b})\vec{a} + (\vec{b} \cdot \vec{b})\vec{b} + (\vec{b} \cdot \vec{c})\vec{c}, \vec{\gamma} = (\vec{a} \cdot \vec{c})\vec{a} + (\vec{b} \cdot \vec{c})\vec{b} + (\vec{c} \cdot \vec{c})\vec{c}$ is :
 (A) V^3 (B) $3V$ (C) V^2 (D) $2V$
12. Let G_1, G_2, G_3 be the centroids of the triangular face OBC, OCA, OAB of a tetrahedron $OABC$. If V_1 denotes the volume of the tetrahedron $OABC$ and V_2 that of the parallelepiped with OG_1, OG_2, OG_3 as three concurrent edges, then :
 (A) $4V_1 = 9V_2$ (B) $9V_1 = 4V_2$ (C) $3V_1 = 2V_2$ (D) $3V_2 = 2V_1$
13. If the vectors $\vec{a} = \hat{i} + a\hat{j} + a^2\hat{k}, \vec{b} = \hat{i} + b\hat{j} + b^2\hat{k}, \vec{c} = \hat{i} + c\hat{j} + c^2\hat{k}$ are three non-coplanar vectors and
 $\begin{vmatrix} a & a^2 & 1+a^3 \\ b & b^2 & 1+b^3 \\ c & c^2 & 1+c^3 \end{vmatrix} = 0$, then the value of abc is :
 (A) 0 (B) 1 (C) 2 (D) -1
14. A vector $\vec{a}$ has components $2p$ and 1 with respect to a rectangular Cartesian system. This system is rotated through a certain angle about the origin in the counterclockwise sense. If with respect to new system, $\vec{a}$ has components $p+1$ and 1 , then :
 (A) $p = 0$ (B) $p = 1$ or $p = -1/3$
 (C) $p = -1$ or $p = \frac{1}{3}$ (D) $p = 1$ or $p = -1$
15. $\vec{a}, \vec{b}, \vec{c}$ are non-coplanar vectors and $\vec{a}_1, \vec{b}_1, \vec{c}_1$ constitute the corresponding reciprocal system of vectors, then we have
 $\vec{a}_1 \times \vec{b}_1 + \vec{b}_1 \times \vec{c}_1 + \vec{c}_1 \times \vec{a}_1 = [\vec{a} \vec{b} \vec{c}]^\lambda (\vec{a} + \vec{b} + \vec{c})$ where ' λ ' is equal to :
 (A) 1 (B) 0 (C) -1 (D) 2
16. If A, B, C, D are any four points in space, then $|\overrightarrow{AB} \times \overrightarrow{CD} + \overrightarrow{BC} \times \overrightarrow{AD} + \overrightarrow{CA} \times \overrightarrow{BD}|$ where $\Delta = \text{Ar}(ABC)$ is equal to :
 (A) 2Δ (B) 4Δ (C) 3Δ (D) 5Δ
17. The vertices of triangle are $A(1, -1, -3), B(2, 1, -2)$ and $C(-5, 2, -6)$. The length of bisector of its interior angle at the vertex A is :
 (A) $\frac{2}{3}\sqrt{10}$ (B) $\frac{3}{4}\sqrt{10}$ (C) $\frac{4}{3}\sqrt{10}$ (D) $\frac{3}{2}\sqrt{10}$

18. Let $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$, $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$ and $\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}$ be three non-zero vectors such that $|\vec{c}| = 1$; angle between $\vec{a}$ and $\vec{b}$ is $\frac{\pi}{4}$ and $\vec{c}$ is perpendicular to $\vec{a}$ and $\vec{b}$, then $\left| \begin{matrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{matrix} \right|^2 = \lambda (a_1^2 + a_2^2 + a_3^2)(b_1^2 + b_2^2 + b_3^2)$ where λ is equal to :
- (A) $1/2$ (B) $1/4$ (C) 1 (D) 2
19. $\vec{a}, \vec{b}, \vec{c}$ are three non-coplanar vectors such that $\vec{a} + \vec{b} + \vec{c} = \alpha \vec{d}$ and $\vec{b} + \vec{c} + \vec{d} = \beta \vec{a}$ then $\vec{a} + \vec{b} + \vec{c} + \vec{d}$ is equal to:
- (A) $\vec{0}$ (B) $\alpha \vec{a}$ (C) $\beta \vec{b}$ (D) $(\alpha + \beta)\vec{c}$
20. If $|\vec{a}| = 4, |\vec{b}| = 4$ and $|\vec{c}| = 5$ such that $\vec{a} \perp (\vec{b} + \vec{c}), \vec{b} \perp (\vec{c} + \vec{a})$ and $\vec{c} \perp (\vec{a} + \vec{b})$, then $|\vec{a} + \vec{b} + \vec{c}|$ is :
- (A) 7 (B) 5 (C) 13 (D) $\sqrt{57}$
21. If $\vec{u} = \vec{a} - \vec{b}, \vec{v} = \vec{a} + \vec{b}$ and $|\vec{a}| = |\vec{b}| = 2$, then $|\vec{u} \times \vec{v}|$ is :
- (A) $2\sqrt{16 - (\vec{a} \cdot \vec{b})^2}$ (B) $2\sqrt{4 - (\vec{a} \cdot \vec{b})^2}$ (C) $\sqrt{16 - (\vec{a} \cdot \vec{b})^2}$ (D) $\sqrt{4 - (\vec{a} \cdot \vec{b})^2}$
22. If $\vec{a}, \vec{b}, \vec{c}$ are three non-zero vectors such that $\vec{a} + \vec{b} + \vec{c} = \vec{0}$ and $m = \vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}$, then :
- (A) $m < 0$ (B) $m > 0$ (C) $m = 0$ (D) $m = 3$
23. Let α, β, γ be distinct real numbers. The points with position vectors $\alpha\hat{i} + \beta\hat{j} + \gamma\hat{k}, \beta\hat{i} + \gamma\hat{j} + \alpha\hat{k}, \gamma\hat{i} + \alpha\hat{j} + \beta\hat{k}$:
- (A) Are collinear (B) Form an equilateral triangle
(C) Form a scalene triangle (D) Form a right angled triangle
24. Given any vector $\vec{r}, |\vec{r} \times \hat{i}|^2 + |\vec{r} \times \hat{j}|^2 + |\vec{r} \times \hat{k}|^2$ equals :
- (A) $4|\vec{r}|^2$ (B) $|\vec{r}|^2$ (C) $2|\vec{r}|^2$ (D) None of these
25. If $\vec{a}$ and $\vec{b}$ are unit vectors and θ is the angle between them, then $\left| \frac{\vec{a} - \vec{b}}{2} \right|$ is :
- (A) $\sin \frac{\theta}{2}$ (B) $\sin \theta$ (C) $2 \sin \theta$ (D) $\sin 2\theta$
26. The work done by the force $\vec{F} = 2\hat{i} - \hat{j} - \hat{k}$ in moving an object along the vector $3\hat{i} + 2\hat{j} - 5\hat{k}$ is :
- (A) -9 units (B) 15 units (C) 9 units (D) None of these
27. The moment of the couple formed by the forces $5\hat{i} + \hat{k}$ and $-5\hat{i} - \hat{k}$ acting at the points $(9, -1, 2)$ and $(3, -2, 1)$ respectively, is :
- (A) $-\hat{i} + \hat{j} + 5\hat{k}$ (B) $\hat{i} - 11\hat{j} - 5\hat{k}$ (C) $-\hat{i} + 11\hat{j} - 5\hat{k}$ (D) $\hat{i} - \hat{j} - 5\hat{k}$
28. ABCD is a parallelogram with $\vec{AC} = \hat{i} - 2\hat{j} + \hat{k}$ and $\vec{BD} = -\hat{i} + 2\hat{j} - 5\hat{k}$. Area of this parallelogram is equal to :
- (A) $\sqrt{5} / 2 \text{ sq. units}$ (B) $2\sqrt{5} \text{ sq. units}$ (C) $4\sqrt{5} \text{ sq. units}$ (D) $\sqrt{5} \text{ sq. units}$

29. The values of x for which the angle between the vectors $\vec{a} = x\hat{i} - 3\hat{j} - \hat{k}$ and $\vec{b} = 2x\hat{i} + x\hat{j} - \hat{k}$ is acute and the angle between $\vec{b}$ and Y-axis lies between $\frac{\pi}{2}$ and π are :
- (A) -1 (B) $x > 0$ (C) 1 (D) $x < 0$
30. $\hat{x}$ and $\hat{y}$ are two mutually perpendicular unit vectors. If the vectors $a\hat{x} + a\hat{y} + c(\hat{x} \times \hat{y})$, $\hat{x} + (\hat{x} \times \hat{y})$ and $c\hat{x} + c\hat{y} + b(\hat{x} \times \hat{y})$, lie in a plane then c is :
- (A) A.M. of a and b (B) G.M. of a and b (C) H.M. of a and b (D) equal to zero
31. For any vector $\vec{a}$, the value of $(\vec{a} \times \hat{i})^2 + (\vec{a} \times \hat{j})^2 + (\vec{a} \times \hat{k})^2$ is equal to :
- (A) $|\vec{a}|^2$ (B) $3|\vec{a}|^2$ (C) $4|\vec{a}|^2$ (D) $2|\vec{a}|^2$
32. Let G be the centroid of $\triangle ABC$. If $\vec{AB} = \vec{a}$ and $\vec{AC} = \vec{b}$, then $\vec{AG}$, in terms of $\vec{a}$ and $\vec{b}$, is :
- (A) $\frac{2}{3}(\vec{a} + \vec{b})$ (B) $\frac{1}{6}(\vec{a} + \vec{b})$ (C) $\frac{1}{3}(\vec{a} + \vec{b})$ (D) $\frac{1}{2}(\vec{a} + \vec{b})$
33. If point A is $\vec{a} + 2\vec{b}$, P is $\vec{a}$ and P divides AB in the ratio $2 : 3$, then the position vector of B is :
- (A) $2\vec{a} - \vec{b}$ (B) $\vec{b} - 2\vec{a}$ (C) $\vec{a} - 3\vec{b}$ (D) $\vec{b}$
34. The line joining the points $6\vec{a} - 4\vec{b} - 5\vec{c}$, $-4\vec{c}$ and the line joining the points $-\vec{a} - 2\vec{b} - 3\vec{c}$, $\vec{a} + 2\vec{b} - 5\vec{c}$ intersect at :
- (A) $2\vec{c}$ (B) $-4\vec{c}$ (C) $8\vec{c}$ (D) None of these
35. If $\vec{r} \cdot \hat{i} = \vec{r} \cdot \hat{j} = \vec{r} \cdot \hat{k}$ and $|\vec{r}| = 3$, then $\vec{r}$ equals :
- (A) $\pm 3(\hat{i} + \hat{j} + \hat{k})$ (B) $\pm \frac{1}{3}(\hat{i} + \hat{j} + \hat{k})$ (C) $\pm \frac{1}{\sqrt{3}}(\hat{i} + \hat{j} + \hat{k})$ (D) $\pm \sqrt{3}(\hat{i} + \hat{j} + \hat{k})$
36. If I is the centre of a circle inscribed in a triangle ABC , then $|\vec{BC}| |\vec{IA}| + |\vec{CA}| |\vec{IB}| + |\vec{AB}| |\vec{IC}|$ is :
- (A) $\vec{0}$ (B) $\vec{IA} + \vec{IB} + \vec{IC}$ (C) $\frac{\vec{IA} + \vec{IB} + \vec{IC}}{3}$ (D) None of these
37. If $\vec{\alpha} = 2\hat{i} + 3\hat{j} - \hat{k}$, $\vec{\beta} = -\hat{i} + 2\hat{j} - 4\hat{k}$ and $\vec{\gamma} = \hat{i} + \hat{j} + \hat{k}$, then $(\vec{\alpha} \times \vec{\beta}) \cdot (\vec{\alpha} \times \vec{\gamma})$ is :
- (A) 60 (B) 64 (C) 74 (D) -74
38. The scalar $\vec{A} \cdot (\vec{B} + \vec{C}) \times (\vec{A} + \vec{B} + \vec{C})$ is :
- (A) 0 (B) $[\vec{ABC}] + [\vec{BCA}]$ (C) $2[\vec{ABC}]$ (D) $[\vec{ABC}]$
39. For any three vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$, $(\vec{a} - \vec{b}) \cdot (\vec{b} + \vec{c}) \times (\vec{c} + \vec{a})$ is equal to :
- (A) $2\vec{a} \cdot (\vec{b} \times \vec{c})$ (B) $[\vec{a} \vec{b} \vec{c}]$ (C) $[\vec{a} \vec{b} \vec{c}]^2$ (D) 0
40. If D , E and F are respectively the mid-points of AB , AC and BC in $\triangle ABC$, then $\vec{BE} + \vec{AF}$ is equal to :
- (A) $\vec{DC}$ (B) $\frac{1}{2}\vec{BF}$ (C) $2\vec{BF}$ (D) $\frac{3}{2}\vec{BF}$

41. If C is the mid-point of AB and P is any point outside AB , then :
 (A) $\overrightarrow{PA} + \overrightarrow{PB} = \overrightarrow{PC}$ (B) $\overrightarrow{PA} + \overrightarrow{PB} = 2\overrightarrow{PC}$
 (C) $\overrightarrow{PA} + \overrightarrow{PB} + \overrightarrow{PC} = \vec{0}$ (D) $\overrightarrow{PA} + \overrightarrow{PB} + 2\overrightarrow{PC} = \vec{0}$
42. If $\vec{a}, \vec{b}, \vec{c}$ are mutually perpendicular vectors of equal magnitudes, then the angle between the vectors $\vec{a}$ and $\vec{a} + \vec{b} + \vec{c}$ is :
 (A) $\pi/3$ (B) $\pi/6$ (C) $\cos^{-1}(1/\sqrt{3})$ (D) $\pi/2$
43. A unit vector in the XY plane that makes an angle 45° with the vector $(\hat{i} + \hat{j})$ and an angle of 60° with the vector $(3\hat{i} - 4\hat{j})$ is :
 (A) $\hat{i}$ (B) $\frac{1}{\sqrt{2}}(\hat{i} - \hat{j})$ (C) $\frac{1}{\sqrt{2}}(\hat{i} + \hat{j})$ (D) None of these
44. The vectors $\vec{a}$ and $\vec{b}$ are not perpendicular and $\vec{c}$ and $\vec{d}$ are two vectors satisfying : $\vec{b} \times \vec{c} = \vec{b} \times \vec{d}$ and $\vec{a} \cdot \vec{d} = 0$. Then the vector $\vec{d}$ is equal to :
 (A) $\vec{c} + \left(\frac{\vec{a} \cdot \vec{c}}{\vec{a} \cdot \vec{b}}\right)\vec{b}$ (B) $\vec{b} + \left(\frac{\vec{b} \cdot \vec{c}}{\vec{a} \cdot \vec{b}}\right)\vec{c}$ (C) $\vec{c} - \left(\frac{\vec{a} \cdot \vec{c}}{\vec{a} \cdot \vec{b}}\right)\vec{b}$ (D) $\vec{b} - \left(\frac{\vec{b} \cdot \vec{c}}{\vec{a} \cdot \vec{b}}\right)\vec{c}$
45. Let $ABCD$ be a parallelogram such that $\overrightarrow{AB} = \vec{q}$, $\overrightarrow{AD} = \vec{p}$ and $\angle BAD$ be an acute angle. If $\vec{r}$ is the vector that coincides with the altitude directed from the vertex B to the side AD , then $\vec{r}$ is given by :
 (A) $\vec{r} = 3\vec{q} - \frac{3(\vec{p} \cdot \vec{q})}{(\vec{p} \cdot \vec{p})}\vec{p}$ (B) $\vec{r} = -\vec{q} + \frac{(\vec{p} \cdot \vec{q})}{(\vec{p} \cdot \vec{p})}\vec{p}$
 (C) $\vec{r} = \vec{q} - \frac{(\vec{p} \cdot \vec{q})}{(\vec{p} \cdot \vec{p})}\vec{p}$ (D) $\vec{r} = -3\vec{q} + \frac{3(\vec{p} \cdot \vec{q})}{(\vec{p} \cdot \vec{p})}\vec{p}$

Integer Answer Type Questions

The Answer to the following questions are positive integers of 1/2/3 digits and zero

46. Let ABC be a triangle whose circumcentre is at P . If the position vectors of A, B, C and P are $\vec{a}, \vec{b}, \vec{c}$ and $\frac{\vec{a} + \vec{b} + \vec{c}}{4}$ respectively, then the position vector of the orthocentre of this triangle, is $\frac{\vec{a} + \vec{b} + \vec{c}}{k}$. The value of k is ____.
47. Let $\vec{a}$ and $\vec{b}$ be two unit vectors such that $|\vec{a} + \vec{b}| = \sqrt{3}$. If $\vec{c} = \vec{a} + 2\vec{b} + 3(\vec{a} \times \vec{b})$, then $2|\vec{c}|$ is equal to $\sqrt{11k}$. The value of k is:
48. In a parallelogram $ABCD$, $|\overrightarrow{AB}| = a$, $|\overrightarrow{AD}| = b$ and $|\overrightarrow{AC}| = c$, then $\overrightarrow{DB} \cdot \overrightarrow{AB}$ has the value $\frac{ka^2 + b^2 - c^2}{2}$. The value of k is:
49. If $|\vec{a}| = 2$, $|\vec{b}| = 3$ and $|2\vec{a} - \vec{b}| = 5$, then $|2\vec{a} + \vec{b}|$ equals ____.

50. If $|\vec{c}|^2 = 60$ and $\vec{c} \times (\hat{i} + 2\hat{j} + 5\hat{k}) = \vec{0}$, then the value of $\vec{c} \cdot (-7\hat{i} + 2\hat{j} + 3\hat{k})$ is $4k\sqrt{2}$. The value of k is ____.
51. If $\hat{x}$, $\hat{y}$ and $\hat{z}$ are three unit vectors in three-dimensional space, then the minimum value of $|\hat{x} + \hat{y}|^2 + |\hat{y} + \hat{z}|^2 + |\hat{z} + \hat{x}|^2$ is ____.
52. If $\vec{x} = 3\hat{i} - 6\hat{j} - \hat{k}$, $\vec{y} = \hat{i} + 4\hat{j} - 3\hat{k}$ and $\vec{z} = 3\hat{i} - 4\hat{j} - 12\hat{k}$, then the magnitude of the projection of $\vec{x} \times \vec{y}$ on $\vec{z}$ is $2k$. The value of k is ____.
53. If $\hat{a}$, $\hat{b}$ and $\hat{c}$ are unit vectors satisfying $\hat{a} - \sqrt{3}\hat{b} + \hat{c} = \vec{0}$, then the angle between the vectors $\hat{a}$ and $\hat{c}$ is $\frac{\pi}{k}$. The value of k is ____.
54. If $\vec{a}$ and $\vec{b}$ are non-collinear vectors, then the value of α for which the vectors $\vec{u} = (\alpha - 2)\vec{a} + \vec{b}$ and $\vec{v} = (2 + 3\alpha)\vec{a} - 3\vec{b}$ are collinear is $\frac{l}{m}$ (where l and m are co-prime). The value of $(l + m)$ is ____.
55. Let $\vec{a}, \vec{b}, \vec{c}$ be three non-coplanar vectors such that $\vec{r}_1 = \vec{a} - \vec{b} + \vec{c}$, $\vec{r}_2 = \vec{b} + \vec{c} - \vec{a}$, $\vec{r}_3 = \vec{c} + \vec{a} + \vec{b}$, $\vec{r} = 2\vec{a} - 3\vec{b} + 4\vec{c}$. If $\vec{r} = \lambda_1\vec{r}_1 + \lambda_2\vec{r}_2 + \lambda_3\vec{r}_3$, then the value of $\lambda_1 + \lambda_3$ is ____.
56. If $\hat{a}, \hat{b}, \hat{c}$ are unit vectors such that $\hat{a} + \hat{b} + \hat{c} = \vec{0}$, then the value of $\hat{a} \cdot \hat{b} + \hat{b} \cdot \hat{c} + \hat{c} \cdot \hat{a}$ is $-\left(\frac{l}{m}\right)$ (where l and m are co-prime). The value of $(l + m)$ is ____.
57. If $\vec{a} = \frac{1}{\sqrt{10}}(3\hat{i} + \hat{k})$ and $\vec{b} = \frac{1}{7}(2\hat{i} + 3\hat{j} - 6\hat{k})$, then the absolute value of $(2\vec{a} - \vec{b}) \cdot [(\vec{a} \times \vec{b}) \times (\vec{a} + 2\vec{b})]$ is ____.
58. Let $\hat{a}$ and $\hat{b}$ be two unit vectors. If the vectors $\vec{c} = \hat{a} + 2\hat{b}$ and $\vec{d} = 5\hat{a} - 4\hat{b}$ are perpendicular to each other, then the angle between $\hat{a}$ and $\hat{b}$ is $\frac{\pi}{k}$. The value of k is: ____.
59. If $[\vec{a} \times \vec{b} \vec{b} \times \vec{c} \vec{c} \times \vec{a}] = \lambda [\vec{a} \vec{b} \vec{c}]^2$ then λ is equal to ____.
60. If the vector $\overrightarrow{AB} = 3\hat{i} + 4\hat{k}$ and $\overrightarrow{AC} = 5\hat{i} - 2\hat{j} + 4\hat{k}$ are the sides of a triangle ABC , then the length of the median through A is $\sqrt{11k}$. The value of k is: ____.

Three Dimensional Geometry

CHOOSE THE CORRECT ALTERNATIVE. ONLY ONE CHOICE IS CORRECT.

- If a straight line makes an angle of $\frac{\pi}{4}$ with each of X-axis and Y-axis, then the angle made by the line with Z-axis is:
 (A) $\frac{\pi}{4}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{2}$ (D) None of these
- A vector $\vec{r}$ is equally inclined with the coordinate axes. If the tip of $\vec{r}$ is in the positive octant and $|\vec{r}| = 6$, then $\vec{r}$ is :
 (A) $2\sqrt{3}(\hat{i} - \hat{j} + \hat{k})$ (B) $2\sqrt{3}(-\hat{i} + \hat{j} + \hat{k})$ (C) $2\sqrt{3}(\hat{i} + \hat{j} - \hat{k})$ (D) $2\sqrt{3}(\hat{i} + \hat{j} + \hat{k})$
- If the direction cosines of a line are $\left(\frac{1}{c}, \frac{1}{c}, \frac{1}{c}\right)$ then :
 (A) $0 < c < 1$ (B) $c > 2$ (C) $c > 0$ (D) $c = \pm\sqrt{3}$
- The value of k so that $\frac{x-1}{-3} = \frac{y-2}{2k} = \frac{z-4}{2}$ and $\frac{x-1}{3k} = \frac{y-1}{1} = \frac{z-6}{-5}$ may be perpendicular is given by :
 (A) $-\frac{7}{10}$ (B) $-\frac{10}{7}$ (C) -10 (D) $\frac{10}{7}$
- The points $(5, 2, 4)$, $(6, -1, 2)$ and $(8, -7, k)$ are collinear if k is equal to :
 (A) -1 (B) 3 (C) 2 (D) -2
- The angle between the lines $x = 1, y = 2$ and $y = -1, z = 0$ is :
 (A) 30° (B) 60° (C) 90° (D) 0°
- The ratio in which the line joining the points (a, b, c) and $(-a, -c, -b)$ is divided by the XY-plane is :
 (A) $a : b$ (B) $b : c$ (C) $c : a$ (D) $c : b$
- The equation of straight line passing through the point (a, b, c) and parallel to Z-axis, is :
 (A) $\frac{x-a}{1} = \frac{y-b}{1} = \frac{z-c}{0}$ (B) $\frac{x-a}{0} = \frac{y-b}{1} = \frac{z-c}{1}$
 (C) $\frac{x-a}{1} = \frac{y-b}{0} = \frac{z-c}{0}$ (D) $\frac{x-a}{0} = \frac{y-b}{0} = \frac{z-c}{1}$
- If l_1, m_1, n_1 and l_2, m_2, n_2 be the direction cosines of two mutually perpendicular lines, then the direction ratios of the line perpendicular to both the lines are :
 (A) l_1l_2, m_1m_2, n_1n_2 (B) $m_1n_2 - m_2n_1, n_1l_2 - n_2l_1, l_1m_2 - l_2m_1$
 (C) $\frac{l_1}{l_2}, \frac{m_1}{m_2}, \frac{n_1}{n_2}$ (D) $m_1n_2 + m_2n_1, n_1l_2 + n_2l_1, l_1m_2 + l_2m_1$
- If l_1, m_1, n_1 and l_2, m_2, n_2 be the direction cosines of two concurrent lines, then the direction cosines of the line bisecting the angles between them are proportional to :
 (A) l_1l_2, m_1m_2, n_1n_2 (B) $\frac{l_1}{l_2}, \frac{m_1}{m_2}, \frac{n_1}{n_2}$
 (C) $l_1 + l_2, m_1 + m_2, n_1 + n_2$ (D) $m_1n_2 + m_2n_1, n_1l_2 + n_2l_1, l_1m_2 + l_2m_1$

11. The shortest distance between the lines $\vec{r} = \vec{a} + t\vec{b}$ and $\vec{r} = \vec{a} + s\vec{c}$ ($\vec{b}, \vec{c}$ are non-collinear) is :
 (A) 0 (B) $\vec{b} \cdot \vec{c}$ (C) $\frac{\vec{b} \cdot \vec{c}}{|\vec{a}|}$ (D) None of these
12. The equation of the plane containing the line $\vec{r} = \hat{i} + \hat{j} + \lambda(2\hat{i} + \hat{j} + 4\hat{k})$ and origin is :
 (A) $\vec{r} \cdot (-\hat{i} - 2\hat{j} + \hat{k}) = 3$ (B) $\vec{r} \cdot (4\hat{i} - 4\hat{j} - \hat{k}) = 0$
 (C) $\vec{r} \cdot (\hat{i} + 2\hat{j} - \hat{k}) = 3$ (D) None of these
13. The equation of the plane through $(2, 3, 4)$ and parallel to the plane $x + 2y + 4z = 5$ is :
 (A) $x + 2y + 4z = 24$ (B) $x + y + 2z = 2$ (C) $x + 2y + 4z = 3$ (D) $x + 2y + 4z = 10$
14. ABC is a triangle in which $A = (2, 3, 5)$, $B = (-1, 3, 2)$ and $C = (\lambda, 5, \mu)$. If the median through A is equally inclined to the axes then :
 (A) $\lambda = \mu = 5$ (B) $\lambda = 5, \mu = 7$ (C) $\lambda = 7, \mu = 10$ (D) $\lambda = 0, \mu = 0$
15. The perpendicular distance of a corner of a unit cube from a diagonal not passing through it is :
 (A) $\frac{1}{\sqrt{3}}$ (B) $\frac{2}{3}$ (C) $\sqrt{\frac{2}{3}}$ (D) $\frac{2}{\sqrt{3}}$
16. Given the line $L : \frac{x-1}{3} = \frac{y+1}{2} = \frac{z+3}{1}$ and the plane $\pi : x - 2y + z = 0$. Of the following assertions, the only one that is always true is :
 (A) L is $\perp$ to π (B) L lies in π (C) L is parallel to π (D) None of these
17. If the point of the intersection of the line $\vec{r} = (\hat{i} + 2\hat{j} + 3\hat{k}) + \lambda(2\hat{i} + \hat{j} + 2\hat{k})$ and the plane $\vec{r} \cdot (2\hat{i} - 6\hat{j} + 3\hat{k}) + 5 = 0$ lies on the plane $\vec{r} \cdot (\hat{i} + 75\hat{j} + 60\hat{k}) - \alpha = 0$, then $19\alpha + 17$ is equal to :
 (A) 1555 (B) 1550 (C) 2563 (D) 2000
18. The equation of the plane through the line of intersection of planes $ax + by + cz + d = 0$, $a'x + b'y + c'z + d' = 0$ and parallel to the line $y = 0, z = 0$ is :
 (A) $(ab' - a'b)x + (bc' - b'c)y + (ad' - a'd) = 0$ (B) $(ab' - a'b)x + (bc' - b'c)y + (ad' - a'd)z = 0$
 (C) $(ab' - a'b)y + (ac' - a'c)z + (ad' - a'd) = 0$ (D) None of these
19. If a variable plane forms a tetrahedron of constant volume $64k^3$ with the co-ordinate planes, then the locus of the centroid of the tetrahedron is :
 (A) $x^3 + y^3 + z^3 = 6k^3$ (B) $xyz = 6k^3$
 (C) $x^2 + y^2 + z^2 = 4k^2$ (D) $x^{-2} + y^{-2} + z^{-2} = 4k^{-2}$
20. The line $\frac{x-2}{3} = \frac{y+1}{2} = \frac{z-1}{-1}$ intersects the curve $xy = c^2, z = 0$ if c is equal to :
 (A) ± 1 (B) $\pm \frac{1}{3}$ (C) $\pm \sqrt{5}$ (D) None of these
21. A point moves so that the sum of the square of its distances from the six faces of a cube given by $x = \pm 1, y = \pm 1$ and $z = \pm 1$ is 10 units. The locus of the point is :
 (A) $x^2 + y^2 + z^2 = 1$ (B) $x^2 + y^2 + z^2 = 2$ (C) $x + y + z = 1$ (D) $x + y + z = 2$

Paragraph For Questions 22 - 24

Given : $P(2, 3, -4)$, $\vec{b} = 2\hat{i} - \hat{j} + 2\hat{k}$

22. Vector equation of a plane passing through the point P perpendicular to the vector $\vec{b}$ is :
- (A) $\vec{r} \cdot (2\hat{i} - \hat{j} + 2\hat{k}) = -7$ (B) $\vec{r} \cdot (2\hat{i} - \hat{j} + 2\hat{k}) = 7$
- (C) $\vec{r} \cdot (2\hat{i} + 3\hat{j} + 4\hat{k}) = -7$ (D) $\vec{r} \cdot (2\hat{i} + 3\hat{j} + 4\hat{k}) = 7$
23. Cartesian equation of a plane π passing through the point with position vector $\vec{b}$ and perpendicular to the vector $\vec{OP}$, O being the origin is :
- (A) $2x - y + 2z + 7 = 0$ (B) $2x - y + 2z - 7 = 0$
- (C) $2x + 3y - 4z + 7 = 0$ (D) $2x + 3y - 4z - 7 = 0$
24. Sum of the lengths of the intercepts made by the above plane on the coordinate axes is :
- (A) 14 (B) $91/12$ (C) $9/7$ (D) $5/7$
25. If θ is the angle between the line $\frac{x+1}{3} = \frac{y-1}{2} = \frac{z-2}{4}$ and the plane $2x + y - 3z + 4 = 0$, then $64 \cos^2 \theta$ is equal to:
- (A) 1246 (B) 5625 (C) 2500 (D) 1624
26. P, Q, R, S are the points $(1, 2, -2), (8, 10, 11), (1, 2, 3)$ and $(3, 5, 7)$ respectively. If s denotes the projection of PQ on RS , then $29s^2 + 29$ is equal to :
- (A) 2195 (B) 8129 (C) 3100 (D) 2500
27. The number of lines which are equally inclined to the axes is :
- (A) 2 (B) 4 (C) 6 (D) 8
28. The value of k such that $\frac{x-4}{1} = \frac{y-2}{1} = \frac{z-k}{2}$ lies in the plane $2x - 4y + z = 7$, is :
- (A) 7 (B) 6 (C) all values of k (D) -7
29. If $P(\vec{p}), Q(\vec{q}), R(\vec{r})$ and $S(\vec{s})$ be four points such that $3\vec{p} + 8\vec{q} = 6\vec{r} + 5\vec{s}$, then the lines PQ and RS are :
- (A) Skew (B) Intersecting (C) Parallel (D) None of these
30. The line $\frac{x}{1} = \frac{y}{2} = \frac{z}{3}$ and the plane $x - 2y + z = 0$:
- (A) Meet in a unique point (B) Do not meet at all
- (C) Meet in infinitely points (D) None of these
31. The coordinates of the foot of the perpendicular drawn from the point $(2, 5, 7)$ on the X axis are given by :
- (A) $(2, 0, 0)$ (B) $(0, 5, 0)$ (C) $(0, 0, 7)$ (D) $(0, 5, 7)$
32. The reflection of the point (α, β, γ) in the XY plane is :
- (A) $(\alpha, \beta, 0)$ (B) $(0, 0, \gamma)$ (C) $(-\alpha, -\beta, \gamma)$ (D) $(\alpha, \beta, -\gamma)$
33. The plane $2x - 3y + 6z - 11 = 0$ makes an angle $\sin^{-1}(\alpha)$ with X axis. The value of α is equal to :
- (A) $\frac{\sqrt{3}}{2}$ (B) $\frac{\sqrt{2}}{3}$ (C) $\frac{2}{7}$ (D) $\frac{3}{7}$
34. Under what condition is a straight line $\frac{x-x_0}{l} = \frac{y-y_0}{m} = \frac{z-z_0}{n}$ parallel to the XY plane?
- (A) $l = 0$ (B) $m = 0$ (C) $n = 0$ (D) $l = 0, m = 0$

35. The equation of the plane bisecting the obtuse angle between the planes $2x - y + 2z + 3 = 0$ and $3x - 2y + 6z + 8 = 0$ is :
 (A) $23x - 13y + 32z + 45 = 0$ (B) $6x - 5y + 17z + 19 = 0$
 (C) $3x + 5y - 11z - 20 = 0$ (D) None of these
36. The image of the point $P(1, 3, 4)$ in the plane $2x - y + z + 3 = 0$ is :
 (A) $(3, 5, -2)$ (B) $(-3, 5, 2)$ (C) $(3, -5, 2)$ (D) $(-1, 4, 2)$
37. If points $P(6, 3, 2)$, $Q(5, 1, 4)$ and $R(3, 3, 5)$ are the vertices of a triangle PQR , then the value of $\angle PQR$ is :
 (A) 0 (B) $\pi/6$ (C) $\pi/3$ (D) $\pi/2$
38. The equation of the plane which bisects the line joining $(2, 3, 4)$ and $(6, 7, 8)$ is :
 (A) $x + y + z - 15 = 0$ (B) $x - y + z - 15 = 0$
 (C) $x - y - z - 15 = 0$ (D) $x + y + z + 15 = 0$
39. **Statement : 1** The point $A(1, 0, 7)$ is the mirror image of the point $B(1, 6, 3)$ in the line $\frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3}$.
Statement : 2 The line $\frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3}$ bisects the line segment joining $A(1, 0, 7)$ and $B(1, 6, 3)$.
 (A) Statement-1 is True, Statement-2 is True and Statement-2 is a correct explanation for Statement-1.
 (B) Statement-1 is True, Statement-2 is True and Statement-2 is NOT a correct explanation for Statement-1.
 (C) Statement-1 is True, Statement-2 is False.
 (D) Statement-1 is False, Statement-2 is True.
40. The shortest distance between the lines $\frac{x}{y} = \frac{y}{2} = \frac{z}{1}$ and $\frac{x+2}{-1} = \frac{y-4}{8} = \frac{z-5}{4}$ lies in the interval :
 (A) $[0, 1)$ (B) $[1, 2)$ (C) $(2, 3]$ (D) $(3, 4]$
41. If the points $(1, 1, \lambda)$ and $(-3, 0, 1)$ are equidistant from the plane, $3x + 4y - 12z + 13 = 0$, then λ satisfies the equation :
 (A) $3x^2 - 10x + 7 = 0$ (B) $3x^2 + 10x + 7 = 0$
 (C) $3x^2 + 10x - 13 = 0$ (D) $3x^2 - 10x + 21 = 0$
42. If the shortest distance between the lines $\frac{x-1}{\alpha} = \frac{y+1}{-1} = \frac{z}{1}$, $(\alpha \neq -1)$ and $x + y + z + 1 = 0 = 2x - y + z + 3$ is $\frac{1}{\sqrt{3}}$, then a value of α is :
 (A) $-\frac{16}{19}$ (B) $-\frac{19}{16}$ (C) $\frac{32}{19}$ (D) $\frac{19}{32}$
43. The shortest distance between the z -axis and the line $x + y + 2z - 3 = 0 = 2x + 3y + 4z - 4$, is :
 (A) 1 (B) 2 (C) 3 (D) 4
44. A plane containing the point $(3, 2, 0)$ and the line $\frac{x-1}{1} = \frac{y-2}{5} = \frac{z-3}{4}$ also contains the point :
 (A) $(0, -3, 1)$ (B) $(0, 7, 10)$ (C) $(0, 7, -10)$ (D) $(0, 3, 1)$
45. A line in the 3-dimensional space makes an angle θ ($0 < \theta \leq \frac{\pi}{2}$) with both the x and y axes. Then the set of all values of θ is the interval :
 (A) $\left(0, \frac{\pi}{4}\right]$ (B) $\left[\frac{\pi}{6}, \frac{\pi}{3}\right]$ (C) $\left[\frac{\pi}{4}, \frac{\pi}{2}\right]$ (D) $\left(\frac{\pi}{3}, \frac{\pi}{2}\right]$

Integer Answer Type Questions

The Answer to the following questions are positive integers of 1/2/3 digits and zero

46. The perimeter of the triangle with vertices at $(1, 0, 0)$, $(0, 1, 0)$ and $(0, 0, 1)$ is $k\sqrt{2}$. The value of k is:
47. The equation of the plane containing the lines $\frac{x-2}{4} = \frac{y+1}{3} = \frac{z-3}{5}$, $x+2y+3z-9=0$ and $2x-y+2z-11=0$ is $7x-11y+z=7k$. The value of k is:
48. If a line makes angles α, β, γ with the coordinate axes, then $\cos 2\alpha + \cos 2\beta + \cos 2\gamma + k = 0$. The value of k is:
49. The ratio in which the YZ plane divides the line segment joining $(-3, 4, -2)$ and $(2, 1, 3)$ is $k:2$. The value of k is:
50. If the angle between the line $x = \frac{y-1}{2} = \frac{z-3}{\lambda}$ and the plane $x+2y+3z=4$ is $\cos^{-1}\left(\sqrt{\frac{5}{14}}\right)$, then λ equals $\left(\frac{l}{m}\right)$ (Where l and m are co-prime). The value of $(m-l)$ is ____.
51. The equation of a plane parallel to the plane $x-2y+2z-5=0$ and at a unit distance from the origin is $x-2y+2z-k=0$. The value of k is ____.
52. If the line $\frac{x-1}{2} = \frac{y+1}{3} = \frac{z-1}{4}$ and $\frac{x-3}{1} = \frac{y-k}{2} = \frac{z}{1}$ intersect, then k is equal to $\frac{\alpha}{2}$. The value of α is ____.
53. The image of the line $\frac{x-1}{3} = \frac{y-3}{1} = \frac{z-4}{-5}$ in the plane $2x-y+z+3=0$ is the line $\frac{x+3}{3} = \frac{y-k}{1} = \frac{z-2}{-5}$. The value of k is ____.
54. The angle between the lines whose direction cosines satisfy the equations $l+m+n=0$ and $l^2=m^2+n^2$ is $\frac{2\pi}{k}$. The value of k is ____.
55. Distance between two parallel planes $2x+y+2z=8$ and $4x+2y+4z+5=0$ is $\frac{k}{2}$. The value of k is ____.
56. If the lines $\frac{x-2}{1} = \frac{y-3}{1} = \frac{z-4}{-k}$ and $\frac{x-1}{k} = \frac{y-4}{2} = \frac{z-5}{1}$ are coplanar, then k can have how many values ____.
57. The distance of the point $(1, -2, 4)$ from the plane through the point $(1, 2, 2)$ and perpendicular to the planes $x-y+2z=3$ and $2x-2y+z+12=0$, is $k\sqrt{2}$. The value of k is ____.
58. Equation of the plane which passes through the point of intersection of lines $\frac{x-1}{3} = \frac{y-2}{1} = \frac{z-3}{2}$ and $\frac{x-3}{1} = \frac{y-1}{2} = \frac{z-2}{3}$ and has the largest distance from the origin is $4x+3y+5z=10k$. The value of k is ____.
59. A line is drawn from the line of intersection of the planes $x=ay+b$ and $z=cy+d$ perpendicular to the plane $\frac{x-b-a}{a} = \frac{y-1}{k} = \frac{z-d-c}{c}$. The value of k is ____.
60. Equation of the line of the shortest distance between the lines $\frac{x}{1} = \frac{y}{-1} = \frac{z}{1}$ and $\frac{x-1}{0} = \frac{y+1}{-2} = \frac{z}{1}$ is $\frac{x-1}{k} = \frac{y+1}{-1} = \frac{z}{-2}$. The value of k is ____.

Probability

CHOOSE THE CORRECT ALTERNATIVE. ONLY ONE CHOICE IS CORRECT.

- The chance of an event happening is the square of the chance of happening of second event but the odds against the first are the cube of the odds against the second. The chance of the events:
 (A) $\frac{1}{9}, \frac{1}{3}$ (B) $\frac{1}{16}, \frac{1}{4}$ (C) $\frac{1}{4}, \frac{1}{2}$ (D) None of these
- The odds against a certain event are 5 : 2 and the odds in favour of another independent event are 6 : 5. The probability that at least one of the events will happen is:
 (A) $25/77$ (B) $52/77$ (C) $12/77$ (D) $65/77$
- The probability that at least one of the events A and B occurs is 0.6. If A and B occur simultaneously with probability 0.2, then $P(\bar{A}) + P(\bar{B})$ is:
 (A) 0.4 (B) 0.8 (C) 1.2 (D) 1.4
- Odds in favour of an event A is 2 to 1 and odds in favour of $A \cup B$ are 3 to 1. Consistent with information the smallest and largest values for the probability of event B are given by :
 (A) $\frac{1}{6} \leq P(B) \leq \frac{1}{3}$ (B) $\frac{1}{3} \leq P(B) \leq \frac{1}{3}$ (C) $\frac{1}{12} \leq P(B) \leq \frac{3}{4}$ (D) None of these
- An integer is chosen at random from the first 200 positive integers. The probability that the integer chosen is divisible by 6 or 8 is :
 (A) $1/3$ (B) $1/4$ (C) $1/5$ (D) None of these
- If ten different objects are distributed at random among ten persons, the probability that at least one of them will not get any object is :
 (A) $\frac{10^{10} - 10}{10^{10}}$ (B) $\frac{10^{10} - 10!}{10^{10}}$ (C) $\frac{10^{10} - 1}{10^{10}}$ (D) None of these
- If two events A and B are such that $P(\bar{A}) = 0.3$, $P(B) = 0.4$, $P(A \cap \bar{B}) = 0.5$, then $P(B/A \cup \bar{B})$ is :
 (A) 0.20 (B) 0.25 (C) 0.30 (D) 0.35
- One ticket is selected at random from 100 tickets numbered 00, 01, 02, ..., 99. Suppose A and B are the sum and product of the digit found on the ticket. Then $P(A = 7/B = 0)$ is given by :
 (A) $2/13$ (B) $2/19$ (C) $1/50$ (D) None of these
- A box contains 100 tickets numbered 1, 2, ..., 100. Two tickets are chosen at random. It is given that the greater number on the two chosen tickets is not more than 10. The probability that the smaller number is 5 is :
 (A) $13/15$ (B) $1/330$ (C) $1/3$ (D) $1/9$
- A bag contains four tickets marked with numbers 112, 121, 211, 222. One ticket is drawn at random from the bag. Let E_i ($i = 1, 2, 3$) denote the event that i^{th} digit on the ticket is 2. then which of the following is incorrect :
 (A) E_1 and E_2 are independent (B) E_2 and E_3 are independent
 (C) E_3 and E_1 are independent (D) E_1, E_2, E_3 are independent
- Two dice are rolled one after the other. The probability that the number on the first is smaller than the number on the second is:
 (A) $1/2$ (B) $7/18$ (C) $3/4$ (D) $5/12$

12. If A and B are two independent events, the probability that both A and B occur is $1/8$ and the probability that neither of them occurs is $3/8$. The probability of the occurrence of A is :
 (A) $1/2$ (B) $1/3$ (C) $1/6$ (D) $1/5$
13. There are 20 cards. 10 of these cards have the letter 'T' printed on them and the other 10 have the letter 'I' printed on them. If three cards are drawn without replacement and kept in the same order, the probability of making word IIT is :
 (A) $4/27$ (B) $5/38$ (C) $1/8$ (D) $9/80$
14. A bag contains ' m ' white and ' n ' black balls. Two players A and B alternately draw a ball from the bag, replacing the ball each time after draw. A begins the game. If the probability of A winning (that is drawing a white ball) is twice the probability of B winning, then the ratio $m : n$ is equal to :
 (A) $1 : 2$ (B) $2 : 1$ (C) $1 : 1$ (D) None of these
15. A bag contains 4 tickets numbered 1, 2, 3, 4 and another bag contains 6 tickets numbered 2, 4, 6, 7, 8, 9. One bag is chosen and a ticket is drawn. The probability that the ticket bears the number 4 is:
 (A) $1/48$ (B) $1/8$ (C) $5/24$ (D) None of these
16. In an entrance test, there are multiple choice questions. There are four possible answers to each question, of which one is correct. The probability that a student knows the answer to the question is 90%. If he gets the correct answer to the question, then the probability that he was guessing is :
 (A) $37/40$ (B) $1/37$ (C) $36/37$ (D) $1/9$
17. There are 3 bags which are known to contain 2 white and 3 black, 4 white and 1 black and 3 white and 7 black balls respectively. A ball is drawn at random from one of the bags and found to be a black ball. Then the probability that it was drawn from the bag containing the most black balls is :
 (A) $7/15$ (B) $5/19$ (C) $3/4$ (D) None of these
18. A pack of playing cards was found to contain only 51 cards. If the first 13 cards which are examined are all red, then the probability that the missing cards is black, is :
 (A) $1/3$ (B) $2/3$ (C) $1/2$ (D) $\frac{{}^{25}C_{13}}{{}^{51}C_{13}}$
19. If m rupee coins and n ten paise coins are placed in a line, then the probability that the extreme coins are ten paise coins is : (Assume coins of same type are identical)
 (A) ${}^{m+n}C_m$ (B) $\frac{n(n-1)}{(m+n)(m+n-1)}$ (C) ${}^{m+n}P_m$ (D) ${}^{m+n}P_n$
20. Twelve different balls are distributed among three boxes. Then probability that the first box contains 3 balls is :
 (A) $\frac{110}{9} \left(\frac{2}{3}\right)^{10}$ (B) $\frac{9}{110} \left(\frac{2}{3}\right)^{10}$ (C) $\frac{{}^{12}C_3}{12^3} \cdot 2^9$ (D) $\frac{{}^{12}C_3}{3^{12}}$
21. A committee of five is to be chosen from a group of 9 people. The probability that a certain married couple will either serve together or not at all is :
 (A) $1/2$ (B) $5/9$ (C) $4/9$ (D) $2/3$
22. A cricket club has 15 members, of whom only 5 can bowl. If the names of 15 members are put into a box and 11 are drawn at random, then the probability of obtaining an eleven containing at least 3 bowlers is :
 (A) $7/13$ (B) $6/13$ (C) $11/15$ (D) $12/13$
23. An ordinary cube has four blank faces, one face marked 2 and another marked 3. Then the probability of obtaining 9 in 5 throws is :
 (A) $3/777$ (B) $5/259$ (C) $5/194$ (D) $5/162$

24. An bat contains 50 tickets numbered 1, 2, 3, 50 of which five are drawn at random and arranged in a ascending order of magnitude ($x_1 < x_2 < x_3 < x_4 < x_5$). The probability that $x_3 = 30$ is :
- (A) $\frac{{}^{20}C_2}{{}^{20}C_5}$ (B) $\frac{{}^{29}C_2}{{}^{50}C_5}$ (C) $\frac{{}^{20}C_2 \times {}^{29}C_2}{{}^{50}C_5}$ (D) None of these
25. The probability that a marksman will hit a target is given as $1/5$. Then the probability that at least one hit in 10 shots is :
- (A) $1 - \left(\frac{4}{5}\right)^{10}$ (B) $\frac{1}{5^{10}}$ (C) $1 - \frac{1}{5^{10}}$ (D) None of these
26. A bag contains $(2n + 1)$ coins. It is known that n of these coins have a head on both sides, whereas the remaining $n + 1$ coins are fair. A coin is picked up at random from the bag and tossed. If the probability that the toss results in a head is $31/42$, then n is equal to :
- (A) 10 (B) 11 (C) 12 (D) 13
27. Two friends A and B have equal number of daughters. There are three cinema tickets which are to be distributed among the daughters of A and B. The probability that all the tickets go to daughters of A is $1/20$. The numbers of daughters each of them have is :
- (A) 4 (B) 5 (C) 6 (D) 3
28. Let $0 < P(A) < 1$, $0 < P(B) < 1$ and $P(A \cup B) = P(A) + P(B) - P(A)P(B)$. Then :
- (A) $P(B/A) = P(B) - P(A)$ (B) $P(A^c \cup B^c) = P(A^c) + P(B^c)$
 (C) $P(A \cup B)^c = P(A^c) + P(B^c)$ (D) $P(A/B) = P(A)$
29. For any two independent events E_1 and E_2 , $P\{(E_1 \cup E_2) \cap (\bar{E}_1 \cap \bar{E}_2)\}$
- (A) $< 1/4$ (B) $> 1/4$ (C) $\geq 1/2$ (D) None of these
30. Cards are drawn one-by-one at random from a well shuffled pack of 52 playing cards until 2 aces are obtained for the first time. The probability that 18 draws are required for this, is :
- (A) $3/34$ (B) $17/455$ (C) $561/15925$ (D) None of these
31. A box contains 24 identical balls of which 12 are white and 12 black. The balls are drawn at random from the box one at time with replacement. The probability that a white ball is drawn for the 4th time on 7th draw is :
- (A) $5/64$ (B) $27/32$ (C) $5/32$ (D) $1/2$
32. An elevator starts with m passengers and stops at n floors ($m \leq n$). The probability that no two passengers alight at same floor is :
- (A) $\frac{{}^nP_m}{m^n}$ (B) $\frac{{}^nP_m}{n^m}$ (C) $\frac{{}^nC_m}{m^n}$ (D) $\frac{{}^nC_m}{n^m}$
33. Three persons A, B and C are to speak at a function along with five others. If they all speak in random order, the probability that A speaks before B and B speaks before C is :
- (A) $3/8$ (B) $1/6$ (C) $3/5$ (D) None of these
34. A fair die is thrown twenty times. The probability that on the tenth throw the fourth six appears is :
- (A) $\frac{{}^{20}C_{10} \times 5^6}{6^{20}}$ (B) $\frac{120 \times 5^7}{6^{10}}$ (C) $\frac{84 \times 5^6}{6^{10}}$ (D) None of these

35. A biased coin with probability p , $0 < p < 1$ of heads is tossed until a head appears for the first time. If the probability that the number of tosses required is even is $2/5$, then p equals :
(A) $1/3$ **(B)** $2/3$ **(C)** $2/5$ **(D)** $3/5$
36. The probability that a teacher will give an unannounced test during any class meeting is $1/5$. If a student is absent twice, the probability that he will miss at least one test is :
(A) $7/25$ **(B)** $9/25$ **(C)** $16/25$ **(D)** $24/25$
37. There are 5 duplicate and 10 of original items in an automobile shop and 3 items are bought at random by a customer. The probability that none of the items is duplicate, is :
(A) $20/91$ **(B)** $22/91$ **(C)** $24/91$ **(D)** $89/91$
38. The probability that the 13th day of randomly chosen month is a second Saturday is :
(A) $1/7$ **(B)** $1/12$ **(C)** $1/84$ **(D)** $19/84$
39. Two persons each make a single throw with a pair of dice. The probability that the throws are unequal is given by :
(A) $\frac{1}{6^3}$ **(B)** $\frac{73}{6^3}$ **(C)** $\frac{51}{6^3}$ **(D)** None of these
40. Let X be a discrete random variable. The probability distribution of X is given below :

X	30	10	-10
$P(X)$	$1/5$	$3/10$	$1/2$

The $E(X)$ is equal to :
(A) 6 **(B)** 4 **(C)** 3 **(D)** -5
41. If $P(A) = \frac{3}{10}$, $P(B) = \frac{2}{5}$ and $P(A \cup B) = \frac{3}{5}$, then $P(B|A) + P(A|B)$ equal :
(A) $\frac{1}{4}$ **(B)** $\frac{1}{3}$ **(C)** $\frac{5}{12}$ **(D)** $\frac{7}{12}$
42. If $P(A) = \frac{2}{5}$, $P(B) = \frac{3}{10}$ and $P(A \cap B) = \frac{1}{5}$, then $P(A'|B') \cdot P(B'|A')$ is equal to :
(A) $\frac{5}{6}$ **(B)** $\frac{5}{7}$ **(C)** $\frac{25}{42}$ **(D)** 1
43. If A and B are such events that $P(A) > 0$ and $P(B) \neq 1$, then $P(A'|B')$ equals.
(A) $1 - P(P|B)$ **(B)** $1 - P(A'|B)$ **(C)** $\frac{1 - P(A \cup B)}{P(B')}$ **(D)** $P(A')/P(B')$
44. If two events are independent, then :
(A) They must be mutually exclusive **(B)** The sum of their probabilities must be equal to 1
(C) (A) and (B) both are correct **(D)** None of the above is correct
45. Three persons, A , B and C fire at a target in turn, starting with A . Their probability of hitting the target are 0.4, 0.3 and 0.2 respectively. The probability of two hits is :
(A) 0.024 **(B)** 0.188 **(C)** 0.336 **(D)** 0.452
46. Two dice are thrown. If it is known that the sum of numbers of the dice was less than 6, the probability of getting a sum 3, is :
(A) $1/18$ **(B)** $5/18$ **(C)** $1/5$ **(D)** $2/5$

47. Which one is not a requirement of a binomial distribution?
 (A) There are 2 outcomes for each trial (B) There is a fixed number of trials
 (C) The outcomes must be dependent on each other
 (D) The probability of success must be the same for all the trials
48. The probability that a person is not a swimmer is 0.3. The probability that out of 5 person 4 are swimmers is :
 (A) ${}^5C_4(0.7)^4(0.3)$ (B) ${}^5C_1(0.7)(0.3)^4$
 (C) ${}^5C_4(0.7)(0.3)^4 + {}^5C_4(0.7)^4(0.3)$ (D) $(0.7)^4(0.3)$
49. For the following probability distribution

X	1	2	3	4
$P(X)$	1/10	1/5	3/10	2/5

$E(X^2)$ is equal to :

- (A) 3 (B) 5 (C) 7 (D) 10
50. Suppose a random variable X follows the binomial distribution with parameters n and p , where $0 < p < 1$.
 If $P(X = r)/P(X = n - r)$ is independent of n and r , then p equals :
 (A) 1/2 (B) 1/3 (C) 1/5 (D) 1/7
51. A and B are two students. Their chances of solving a problem correctly are 1/3 and 1/4, respectively. If the probability of their making a common error is, 1/20 and the obtain the same answer, then the probability of their answer to be correct is :
 (A) 1/12 (B) 1/40 (C) 13/120 (D) 10/13
52. A box has 100 pens of which 10 are defective. What is the probability that out of a sample of 5 pens drawn one by one with replacement at most one is defective?
 (A) $\left(\frac{9}{10}\right)^5$ (B) $\frac{1}{2}\left(\frac{9}{10}\right)^4$ (C) $\frac{1}{2}\left(\frac{9}{10}\right)^5$ (D) $\left(\frac{9}{10}\right)^5 + \frac{1}{2}\left(\frac{9}{10}\right)^4$
53. Consider 5 independent Bernoulli's trials each with probability of success p . If the probability of at least one failure is greater than equal to $\frac{31}{32}$, then p lies in the interval.
 (A) $\left[\frac{3}{4}, \frac{11}{12}\right]$ (B) $\left[0, \frac{1}{2}\right]$ (C) $\left[\frac{11}{12}, 1\right]$ (D) $\left[\frac{1}{2}, \frac{3}{4}\right]$
54. If C and D are two events such that $C \subset D$ and $P(D) \neq 0$, then the correct statement among the following is :
 (A) $P(C | D) \geq P(C)$ (B) $P(C | D) < P(C)$ (C) $P(C | D) = \frac{P(D)}{P(C)}$ (D) $P(C | D) = P(C)$
55. Three numbers are chosen at random without replacement from $\{1, 2, 3, \dots, 8\}$. The probability that their minimum is 3, given that their maximum is 6, is :
 (A) 3/8 (B) 1/5 (C) 1/4 (D) 2/5
56. Let A and B be two events such that $P(\overline{A \cup B}) = \frac{1}{6}$, $P(A \cap B) = \frac{1}{4}$ and $P(\overline{A}) = \frac{1}{4}$, where $\overline{A}$ stands for the complement of the event A . Then the events A and B are :
 (A) Independent but not equally likely (B) Independent and equally likely
 (C) Mutually exclusive and independent (D) Equally likely but not independent

57. The variance of first 50 even natural numbers is :
- (A) 437 (B) $\frac{437}{4}$ (C) $\frac{833}{4}$ (D) 833
58. A multiple choice examination has 5 questions. Each question has three alternative answers of which exactly one is correct. The probability that a student will get 4 or more correct answers just by guessing is :
- (A) $\frac{17}{3^5}$ (B) $\frac{13}{3^5}$ (C) $\frac{11}{3^5}$ (D) $\frac{10}{3^5}$
59. If A and B are any two events such that $P(A) = \frac{2}{5}$ and $P(A \cap B) = \frac{3}{20}$, then the conditional probability $P(A|(A' \cup B))$, where A' denotes the complement of A , is equal to :
- (A) $\frac{1}{4}$ (B) $\frac{5}{17}$ (C) $\frac{8}{17}$ (D) $\frac{11}{20}$
60. An experiment succeeds twice as often as it fails. The probability of at least 5 successes in the six trials of this experiment is :
- (A) $\frac{240}{729}$ (B) $\frac{192}{729}$ (C) $\frac{256}{729}$ (D) $\frac{496}{729}$
61. Let x be set containing 10 elements and $p(x)$ be its power set. If A and B are picked up at random from $p(x)$, with replacement, then probability that A and B have equal number of elements, is :
- (A) $\frac{{}^{20}C_{10}}{2^{10}}$ (B) $\frac{2^{10}-1}{2^{20}}$ (C) $\frac{2^{10}-1}{2^{10}}$ (D) $\frac{{}^{20}C_{10}}{2^{20}}$
62. If the lengths of the sides of a triangle are decided by the three throws of a single fair die, then the probability that the triangle is of maximum area given that it is an isosceles triangle, is :
- (A) $\frac{1}{26}$ (B) $\frac{1}{27}$ (C) $\frac{1}{21}$ (D) $\frac{1}{15}$
63. A set S contains 7 elements. A non-empty subset A of S and an element x of S are chosen at random. Then the probability that $x \in A$ is :
- (A) $\frac{1}{2}$ (B) $\frac{64}{127}$ (C) $\frac{63}{128}$ (D) $\frac{31}{128}$
64. If X has a binomial distribution, $B(n, p)$ with parameters n and p such that $P(X = 2) = P(X = 3)$, then $E(X)$, the mean of variable X , is :
- (A) $2 - p$ (B) $3 - p$ (C) $p/2$ (D) $p/3$
65. A number x is chosen at random from the set $\{1, 2, 3, 4, \dots, 100\}$. Define the event : A = the chosen number x satisfies $\frac{(x-10)(x-50)}{(x-30)} \geq 0$. Then $P(A)$ is :
- (A) 0.51 (B) 0.71 (C) 0.70 (D) 0.20
66. Let A and E be any two events with positive probabilities :
- Statement I :** $P(E/A) \geq P(A/E)P(E)$
- Statement II :** $P(E/A) \geq P(A \cap E)$
- (A) Both the statements are true (B) Both the statements are false
- (C) Statement I is true, Statement II is false (D) Statement I is false, Statement II is true

67. The probability of a man hitting a target is $\frac{2}{5}$. He fires at the target k times (k , a given number). Then the minimum k , so that probability of hitting the target at least ones is more than $\frac{7}{10}$, is :
 (A) 3 (B) 5 (C) 2 (D) 4
68. Given two independent events, if the probability that exactly one of them occurs is $\frac{26}{49}$ and the probability that none of them occurs is $\frac{15}{49}$, then the probability of more probable of the two events is :
 (A) $\frac{4}{7}$ (B) $\frac{6}{7}$ (C) $\frac{3}{7}$ (D) $\frac{5}{7}$
69. A, B, C try to hit a target simultaneously but independently. Their respective probabilities of hitting the targets are $\frac{3}{4}$, $\frac{1}{2}$, $\frac{5}{8}$. The probability that the target is hit by A or B but not by C is :
 (A) $\frac{21}{64}$ (B) $\frac{7}{8}$ (C) $\frac{7}{32}$ (D) $\frac{9}{64}$
70. On the sides AB, BC, CA of $\triangle ABC$, 3, 4, 5 distinct points (excluding vertices A, B, C) are respectively chosen. The number of triangles that can be constructed using these chosen points as vertices are :
 (A) 210 (B) 205 (C) 215 (D) 220

Integer Answer Type Questions

The Answer to the following questions are positive integers of 1/2/3 digits and zero

71. A pair of fair dice is rolled together till a sum of either 5 or 7 is obtained. If p denotes the probability that 7 comes before 5, then the value of $5p$ is _____.
72. A bag A there are 5 white and 3 black balls. In bag B there are 3 white and 1 black balls. One ball is chosen at random from any bag and found to be white then the probability that it is from bag B is $\frac{p}{q}$ (in lowest form, $p, q \in N$) then $q - p$ is _____.
73. A and B are two matrices with 32 and 58 elements respectively then the probability that $(A \times B)$ is possible, is $\frac{p}{q}$ (in lowest form, $p, q \in N$) then $q - p$ is _____.
74. A four digit number is formed by the digits 1, 2, 3, 4, 5, 6 and 8. The probability that the number has odd digits at both ends is $\frac{p}{q}$ (in lowest form, $p, q \in N$) then $q - p$ is _____.
75. Five digits from the numbers 1, 2, 3, 4, 5, 6, 7 are written in random order. If the probability that this 5-digit number is divisible by 4 is λ then the value of 21λ is _____.
76. If A and B are events such that $P(A) = 0.42$, $P(B) = 0.48$ and $P(A \cap B) = 0.16$, then $100 \times P((A \cup B)')$ is _____.

77. The probability that 13th day of February is a Second Saturday is $\frac{p}{q}$ (in lowest form, $p, q \in N$) then $q - p$ is _____.
78. In throwing a pair of dice getting an even number on first dice and a total of 7 on both the dice is $\frac{p}{q}$ (in lowest form, $p, q \in N$) then $q - p$ is _____.
79. If A and B are two events such that $P(A) = \frac{1}{3}$, $P(B) = \frac{1}{4}$ and $P(A \cap B) = \frac{1}{5}$ then $P(B' / A')$ is $\frac{p}{q}$ (in lowest form, $p, q \in N$) then $q - p$ is _____.
80. If the odds in favour of an event be $\frac{4}{5}$, then the probability of non-occurrence of the event is $\frac{p}{q}$ (in lowest form, $p, q \in N$) then $q - p$ is _____.
81. A purse contain 4, 10 paise coin; 3, 25 paise coin; 2.50 paise coin. A coin is drawn at random. The probability of coin drawn is 25 paise or 50 paise is $\frac{p}{q}$ (in lowest form, $p, q \in N$) then $q - p$ is _____.
82. There are 4 white and 5 black ball in a box. In an another box there are 5 white and 4 black balls. An unbiased die is rolled. If it shows even number then a ball is drawn from the second box otherwise from first box. If the ball drawn is white then the probability that the ball was drawn from the first box is $\frac{p}{q}$ (in lowest form, $p, q \in N$) then $q - p$ is _____.
83. Three players A, B, C in this order. Draw a card from well shuffled pack of card and reshuffled after each drawn. If the winner is one who draws a red card, then C 's change of winning is $\frac{p}{q}$ (in lowest form, $p, q \in N$) then $q - p$ is _____.
84. There are 3 bags which are known to contain 2 white and 3 black balls, 4 white and 2 black balls and 3 white and 2 black balls. A ball is drawn randomly from one of the bag find the probability of ball being black is $\frac{p}{q}$ (in lowest form, $p, q \in N$) then $q - p$ is _____.
85. In an entrance test there are multiple choice questions. There are four possible answers to each question of which one is correct. The probability that a student knows the answer to questions is 75%. If he gets the correct answer to a question, the probability that he was guessing is $\frac{p}{q}$ (in lowest form, $p, q \in N$) then $q - p$ is _____.